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Scotchmer et al.

(10) Patent No.: US 12,409,504 B2
(45) Date of Patent: Sep. 9, 2025

(54) **WELDING APPLICATOR AND METHOD OF APPLICATION**

(71) Applicant: **HUYS INDUSTRIES LIMITED**, Weston (CA)

(72) Inventors: **Nigel Scotchmer**, Willowdale (CA); **Franklin Rementer**, Dover, DE (US); **Dominic Cheuk Hung Leung**, Thornhill (CA)

(73) Assignee: **HUYS INDUSTRIES LIMITED**, Weston (CA)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 377 days.

(21) Appl. No.: **16/781,638**

(22) Filed: **Feb. 4, 2020**

(65) **Prior Publication Data**

US 2020/0246900 A1 Aug. 6, 2020

Related U.S. Application Data

(60) Provisional application No. 62/801,437, filed on Feb. 5, 2019.

(51) **Int. Cl.**
B23K 9/04 (2006.01)
B23H 1/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***B23K 9/04*** (2013.01); ***B23H 1/02*** (2013.01); ***B23K 9/09*** (2013.01); ***B23K 9/1006*** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC B23K 9/09; B23K 9/124; B23K 9/1043; B23K 9/04; B23K 9/173; B23K 9/1006; B23H 1/02; B23H 1/00
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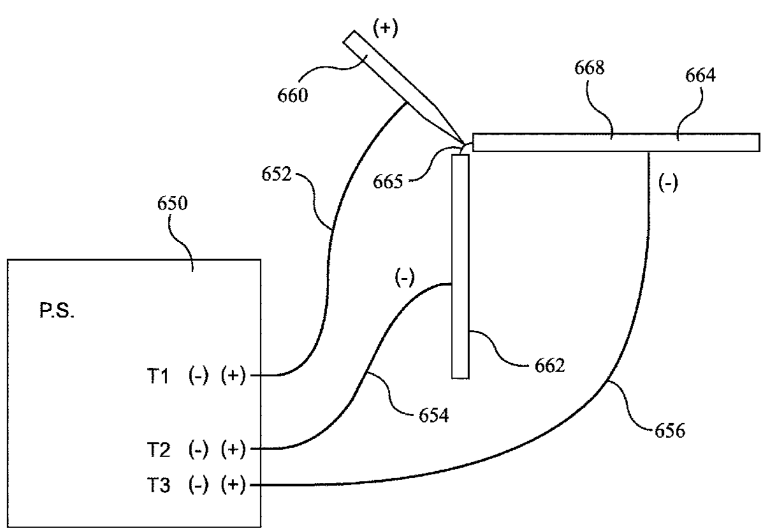
Assistant Examiner — Erwin J Wunderlich
(74) *Attorney, Agent, or Firm* — Vorys, Sater, Seymour and Pease LLP; Rex W. Miller, II

(57) **ABSTRACT**

A welding rod assembly has a seat, a chuck, and a welding rod. The chuck has a socket for the welding rod. The seat has a passageway having an inlet and an outlet, and a fluid conduit through the seat. The apparatus that has a power supply having a main control unit, and a welding electrode. It has synthetic discharge signal generator programmed to produce DC discharges as output electrical signals from the power supply. Each discharge has a voltage, charge, starting time and polarity. The discharges define a synthetic AC output. It has an output polarity reversing switch. The apparatus includes three output terminals, a first of the output terminals being connected to the welding electrode, a second of said output terminals and a third of said output terminals having respective workpiece connections.

20 Claims, 49 Drawing Sheets



(51) **Int. Cl.**

B23H 1/02	(2006.01)
B23K 9/09	(2006.01)
B23K 9/10	(2006.01)
B23K 9/12	(2006.01)
B23K 9/173	(2006.01)

(52) **U.S. Cl.**
CPC ***B23K 9/124*** (2013.01); ***B23K 9/173*** (2013.01); *B23H 1/00* (2013.01); *B23K 9/1043* (2013.01)

(58) **Field of Classification Search**
USPC 219/130.51, 70, 96, 113, 95, 76.13
See application file for complete search history.

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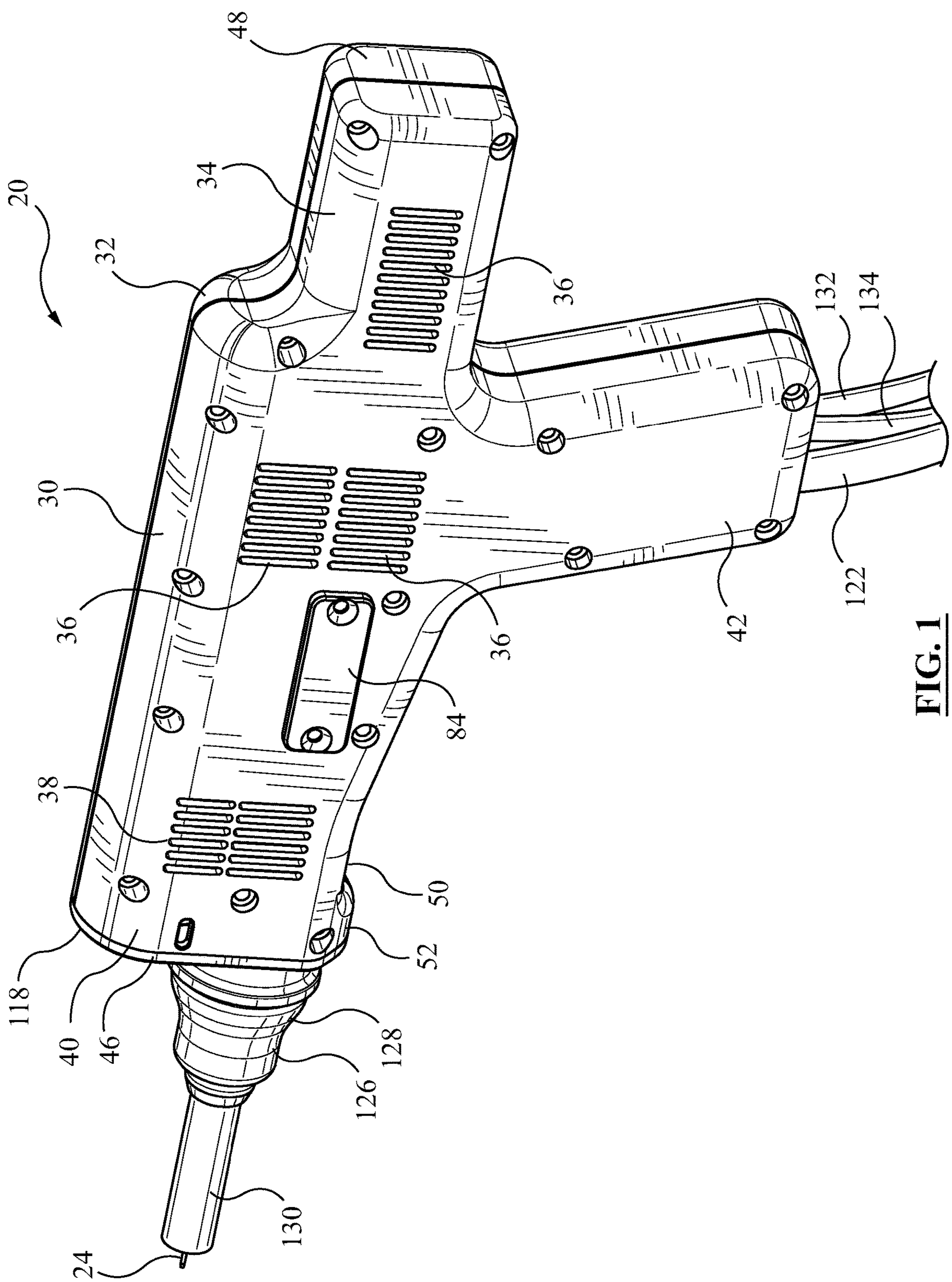
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FIG. 1

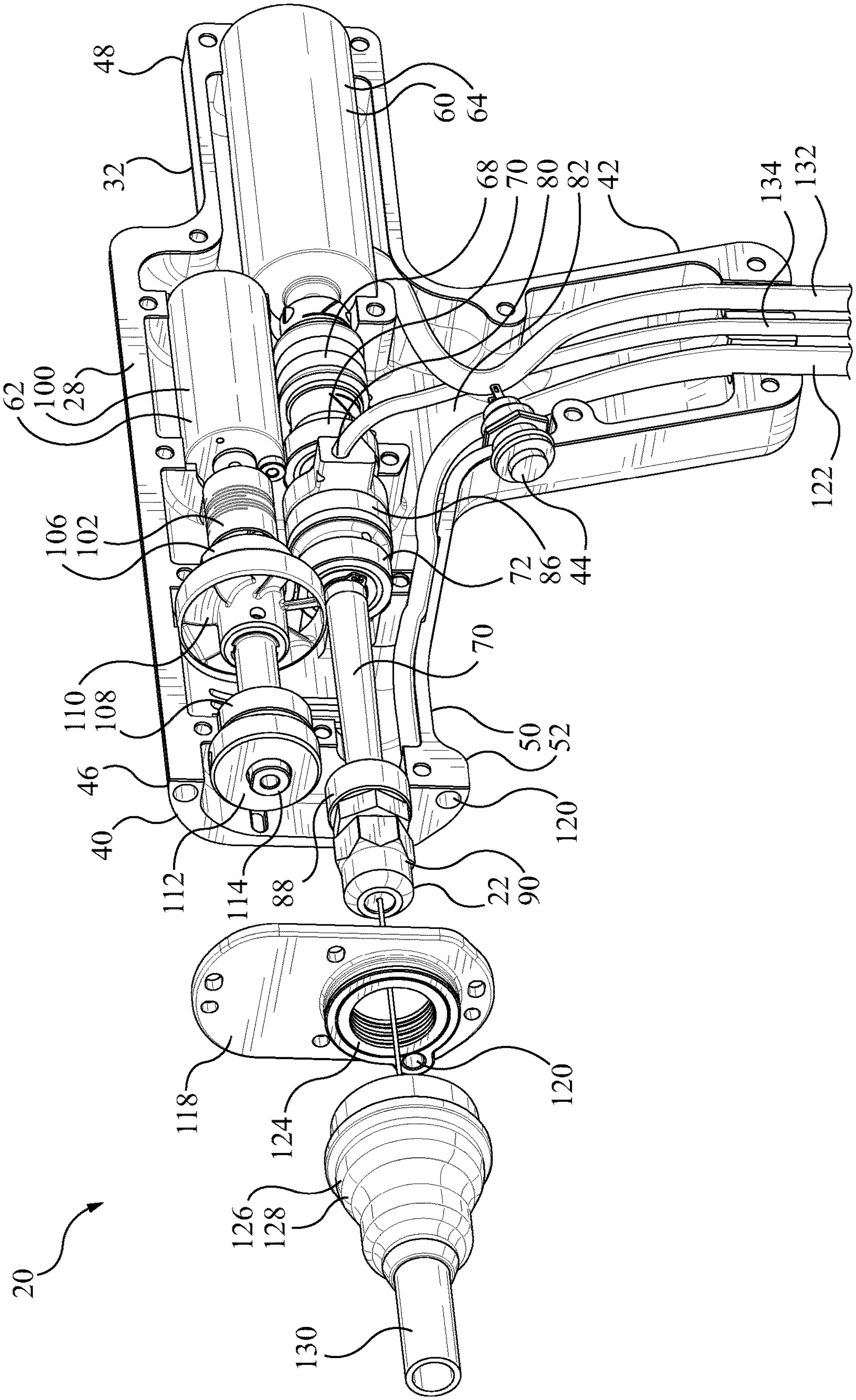
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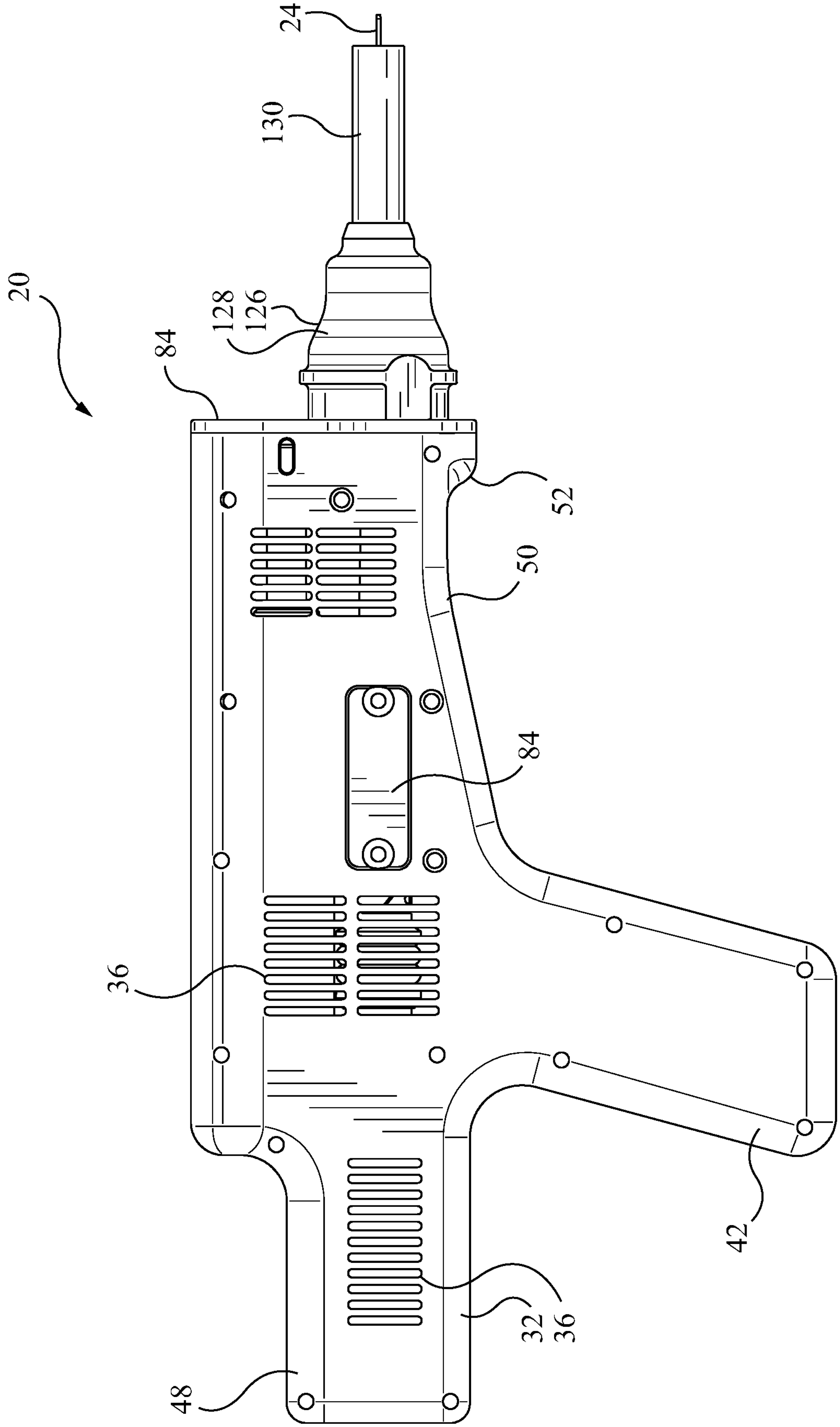
FIG. 2



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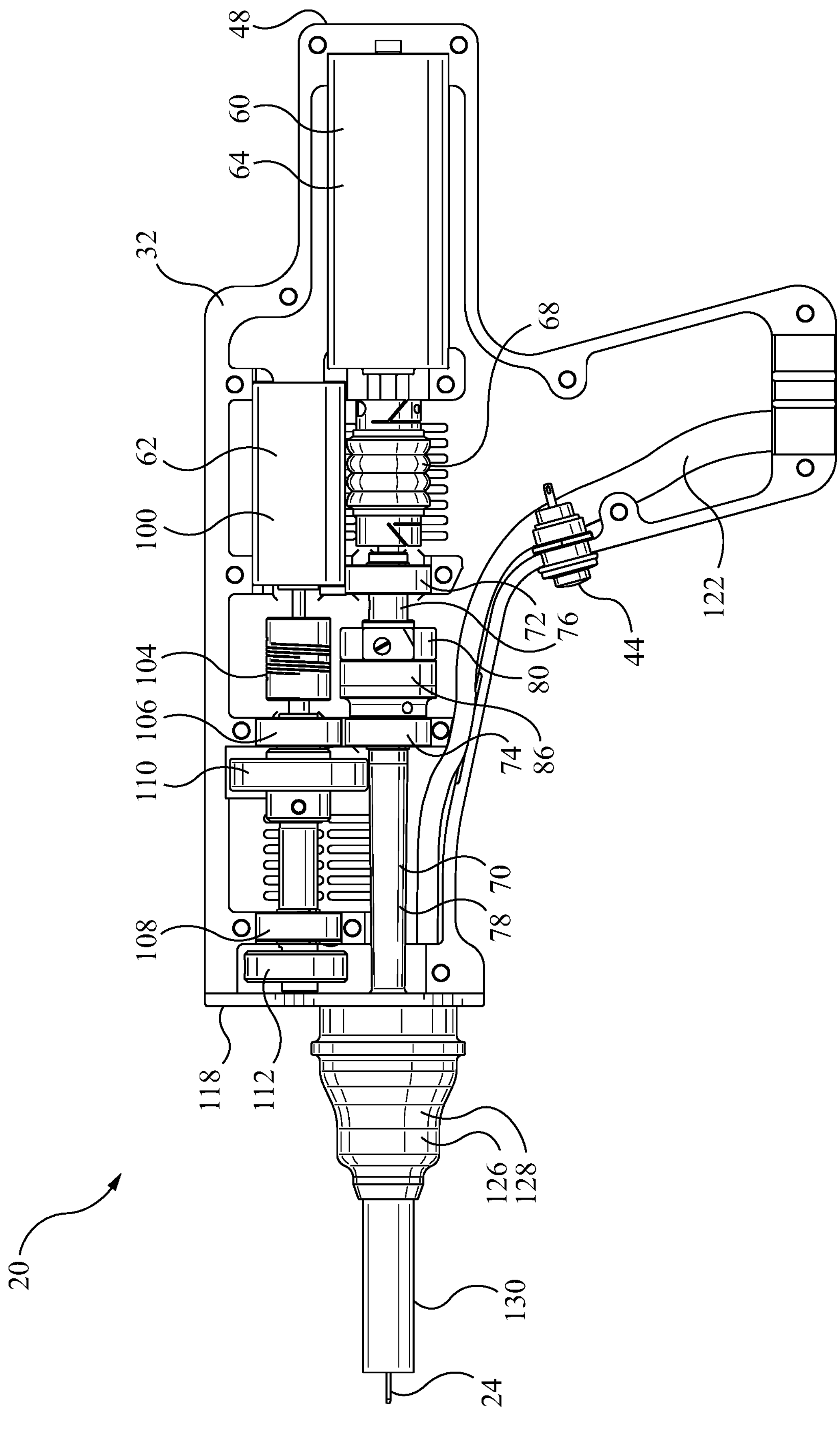
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FIG. 4

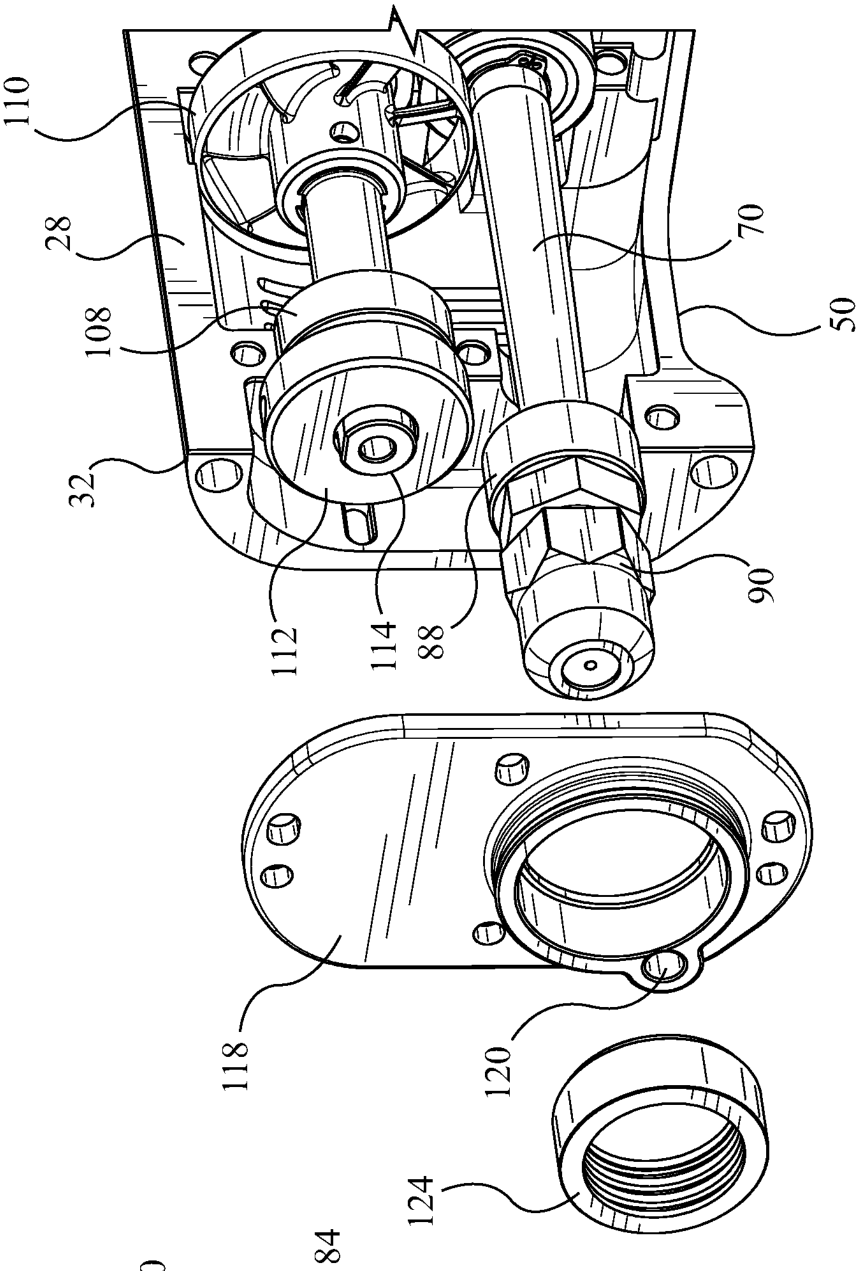
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FIG. 6

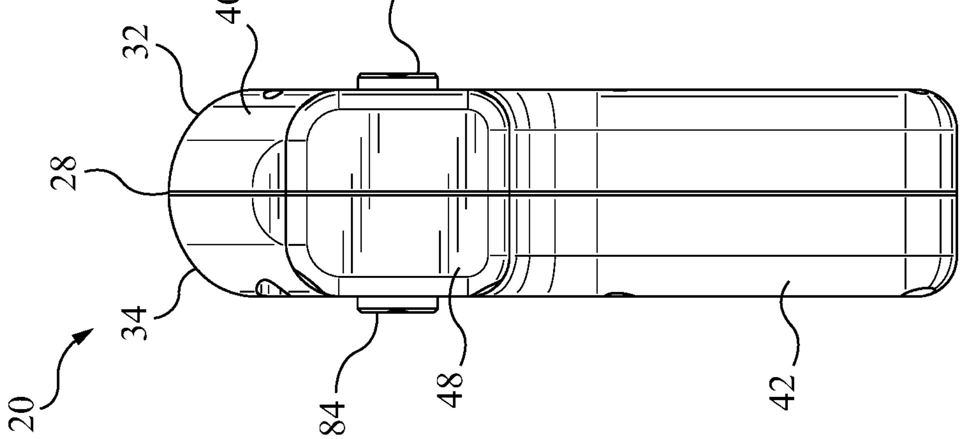
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FIG. 5

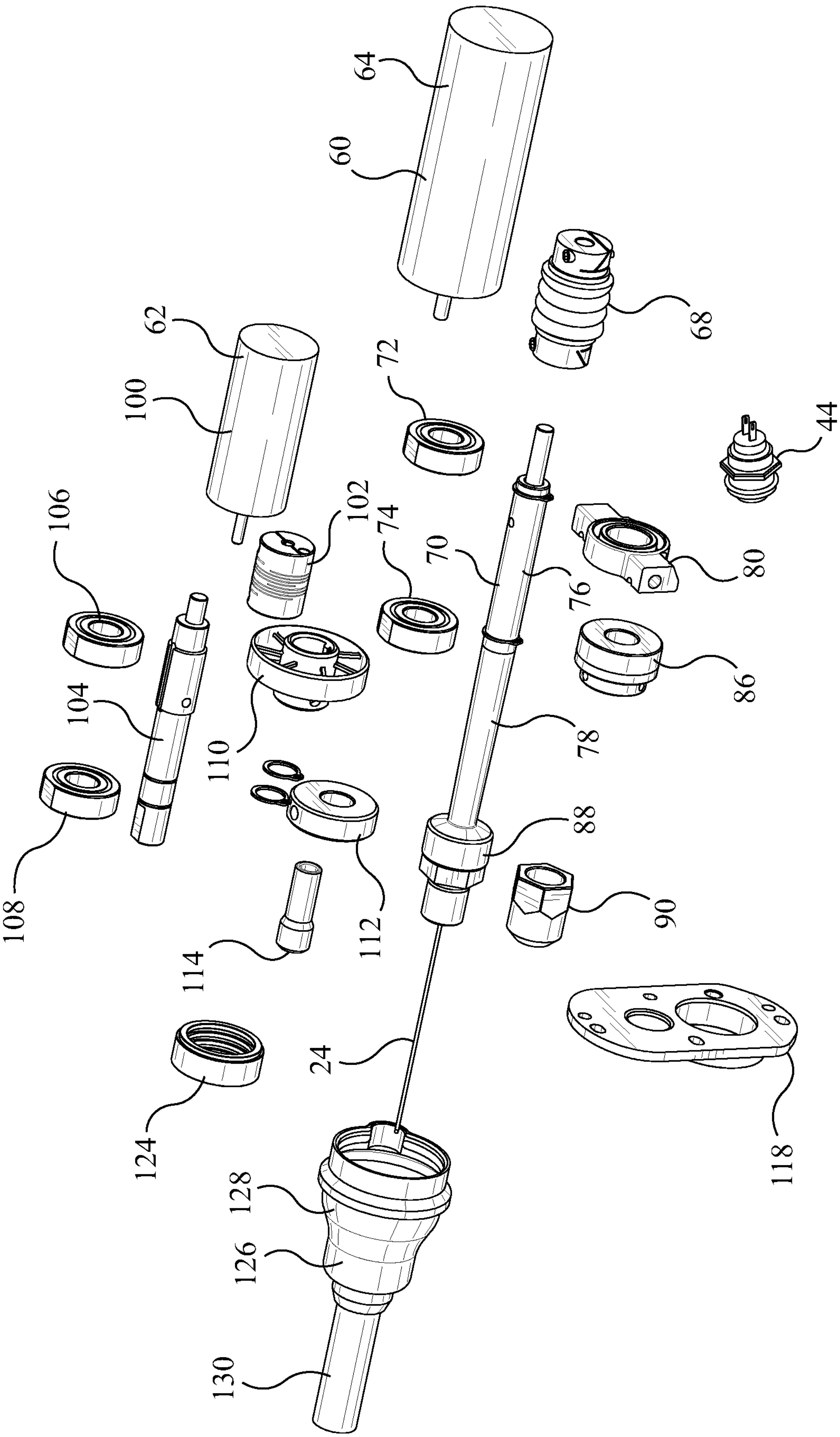
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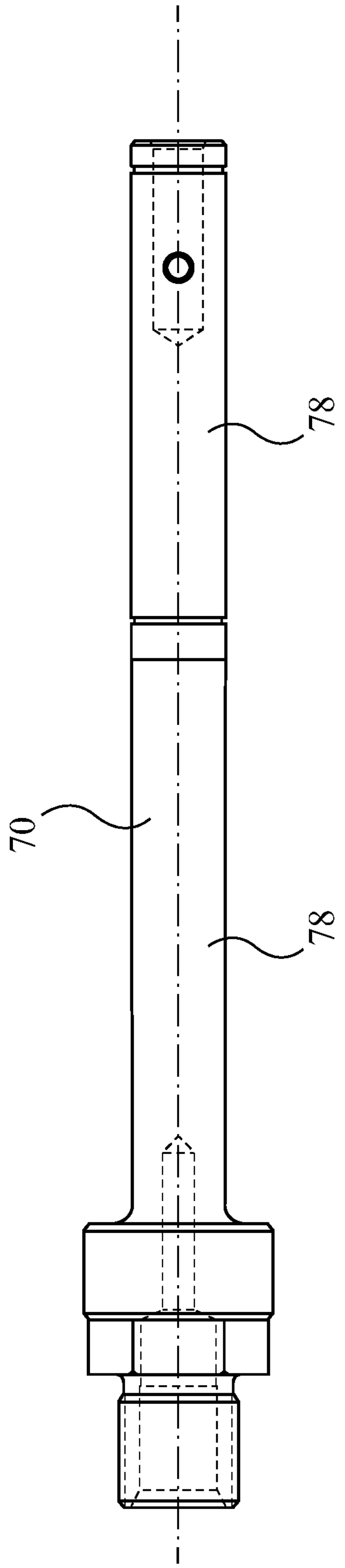
FIG. 7



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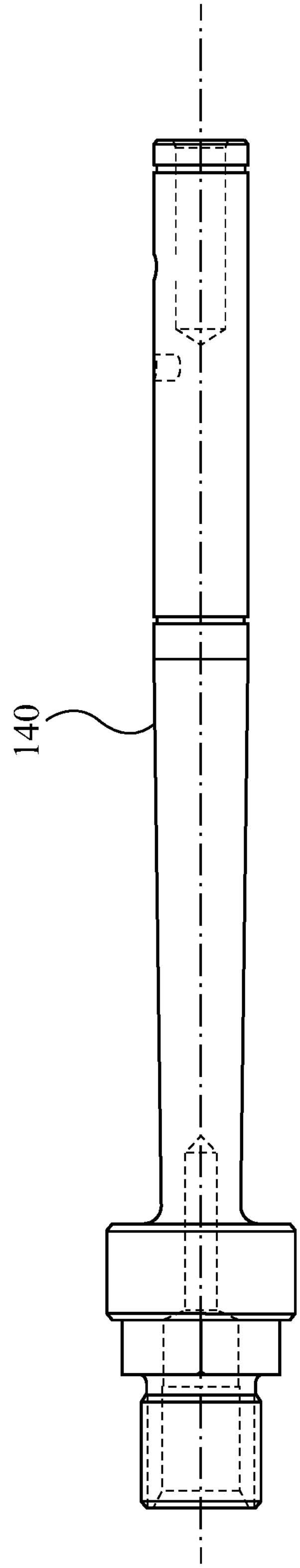

FIG. 8b

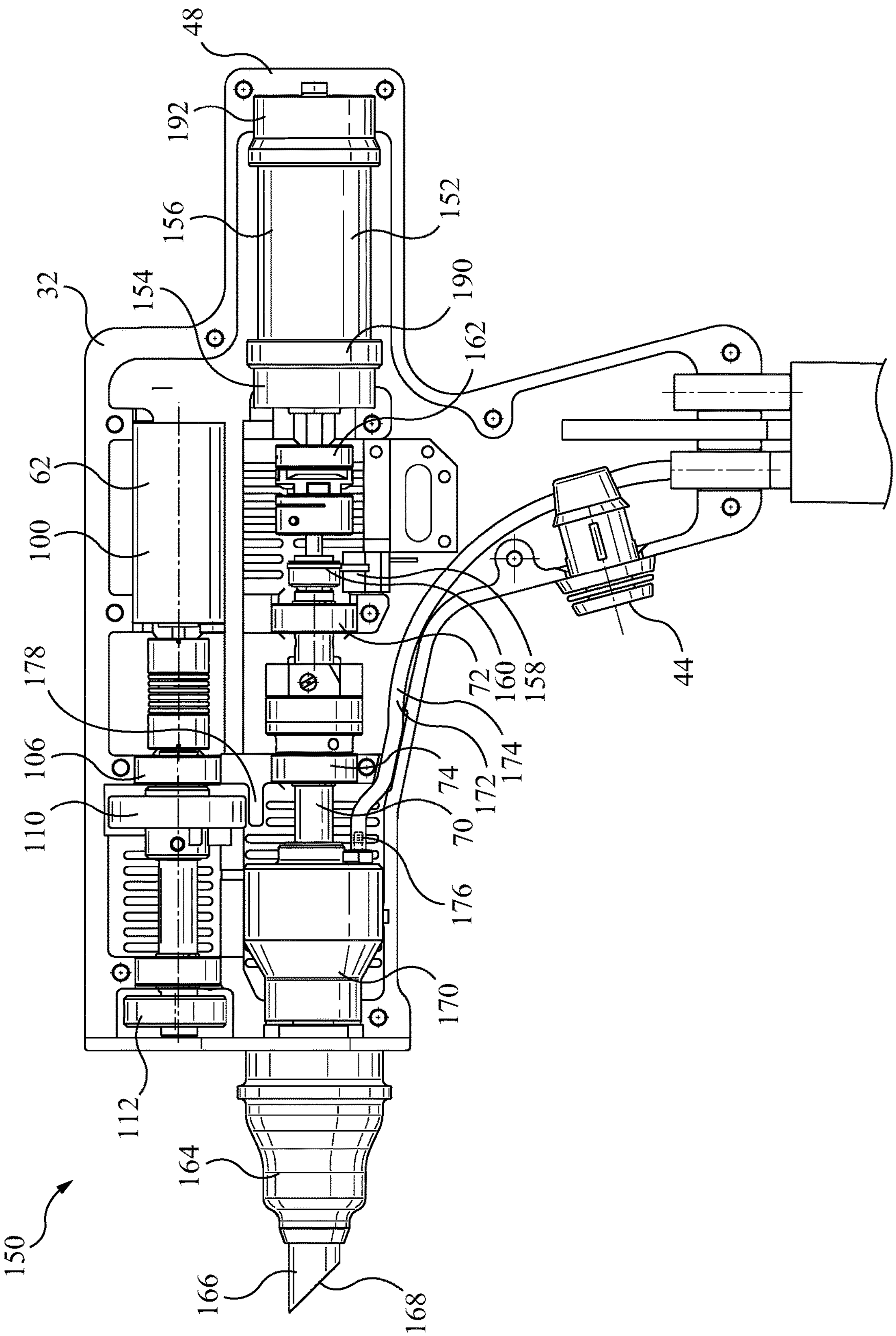
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FIG. 9a

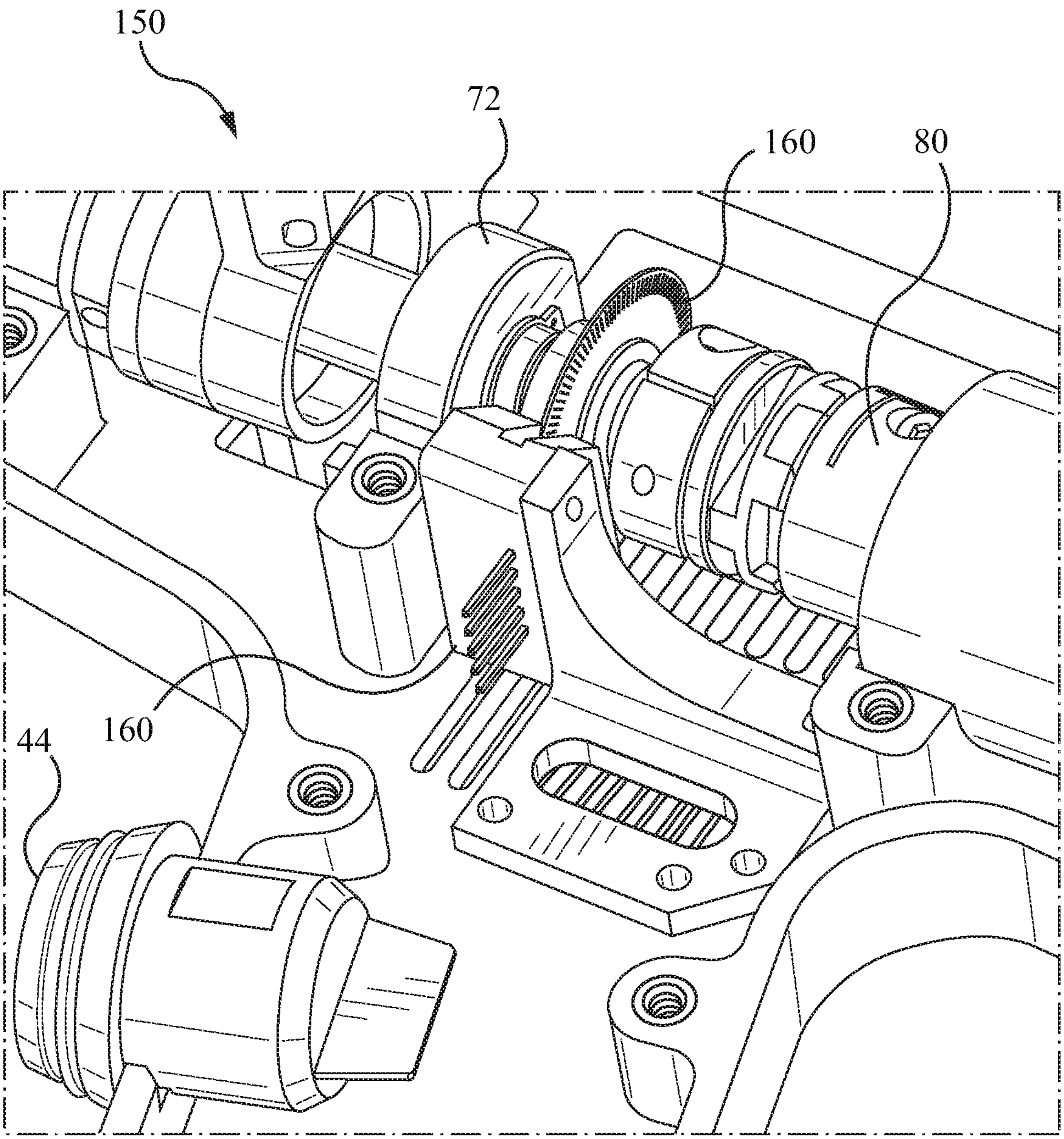
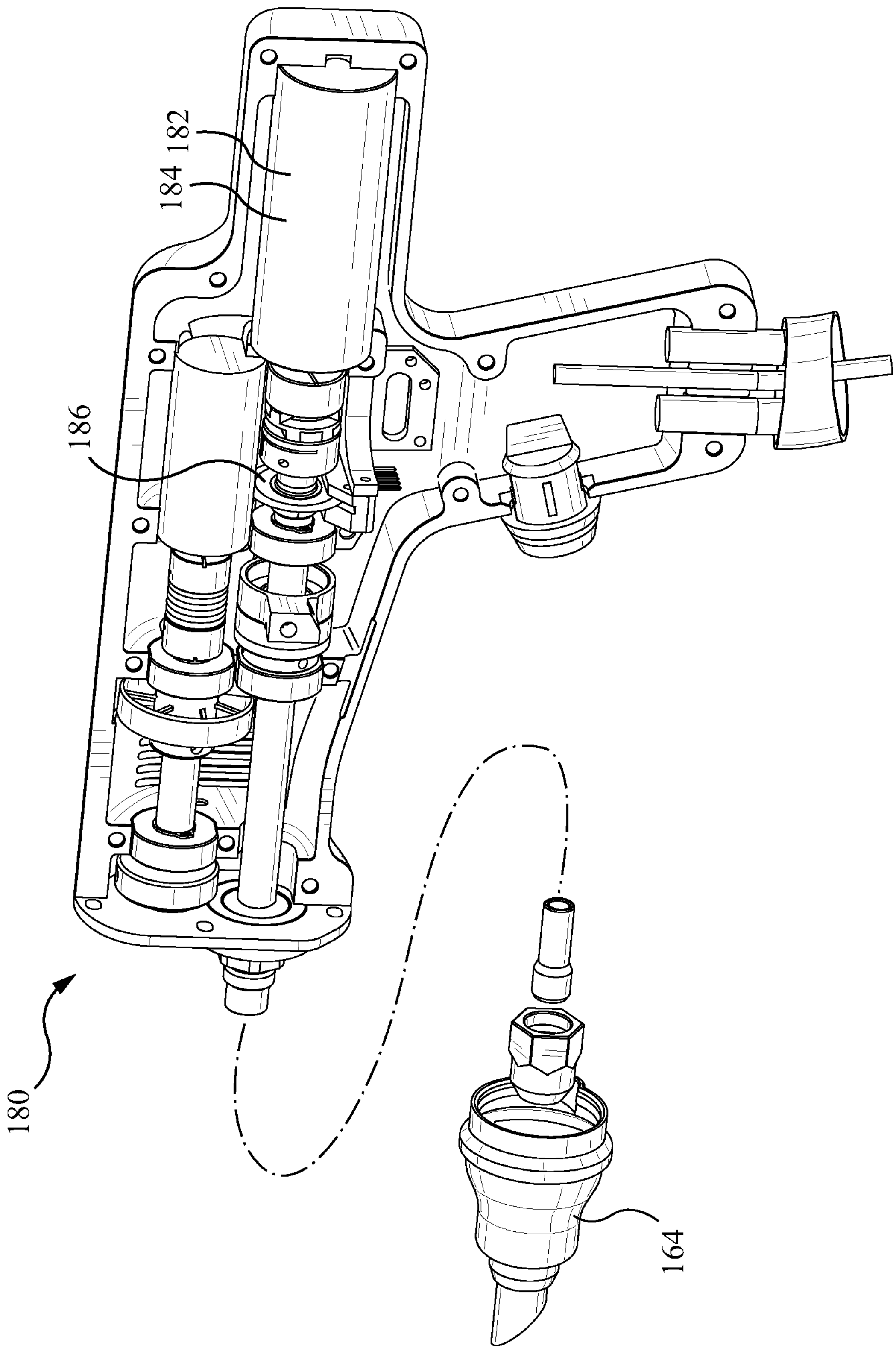


FIG. 9b



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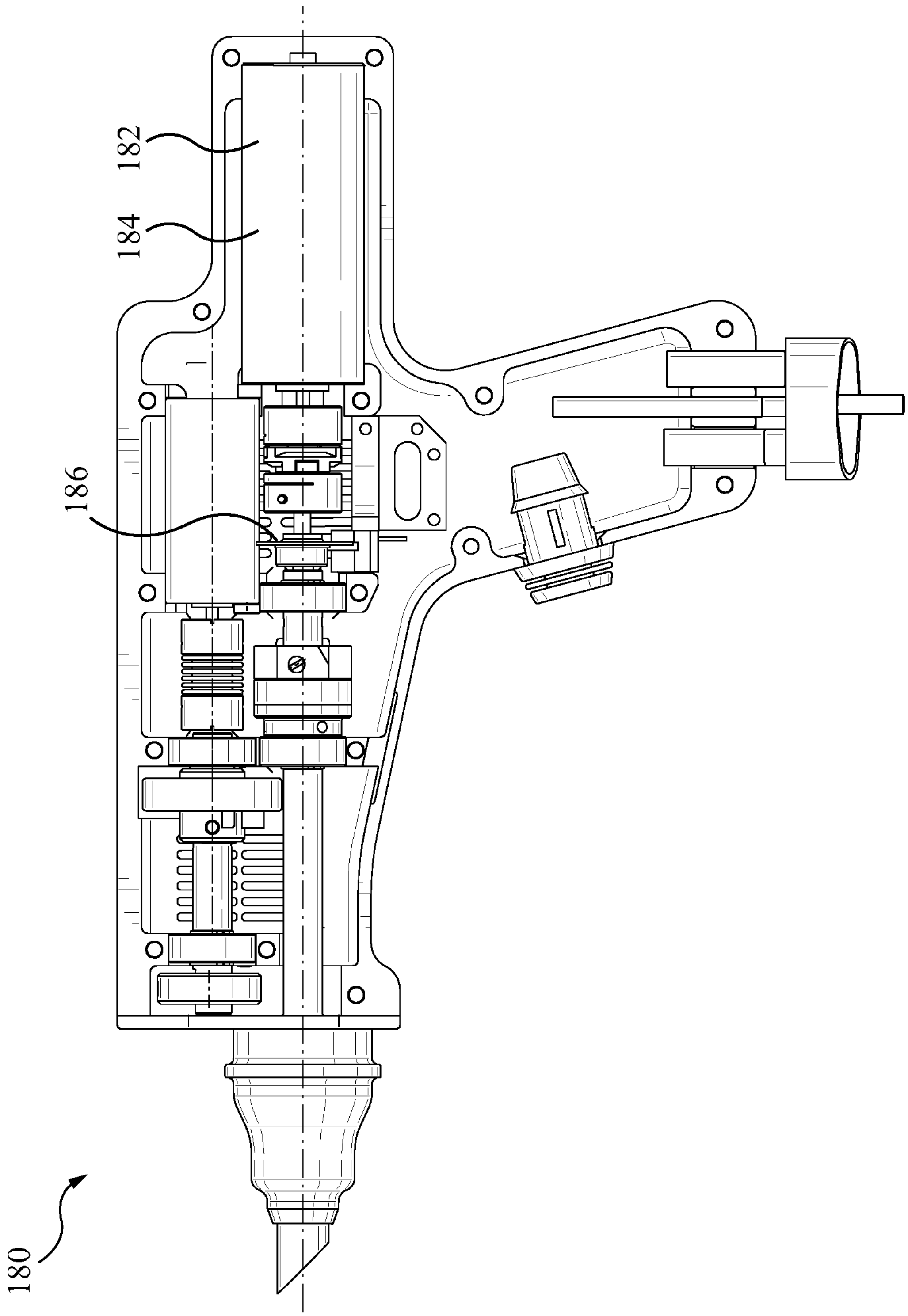

FIG. 10b

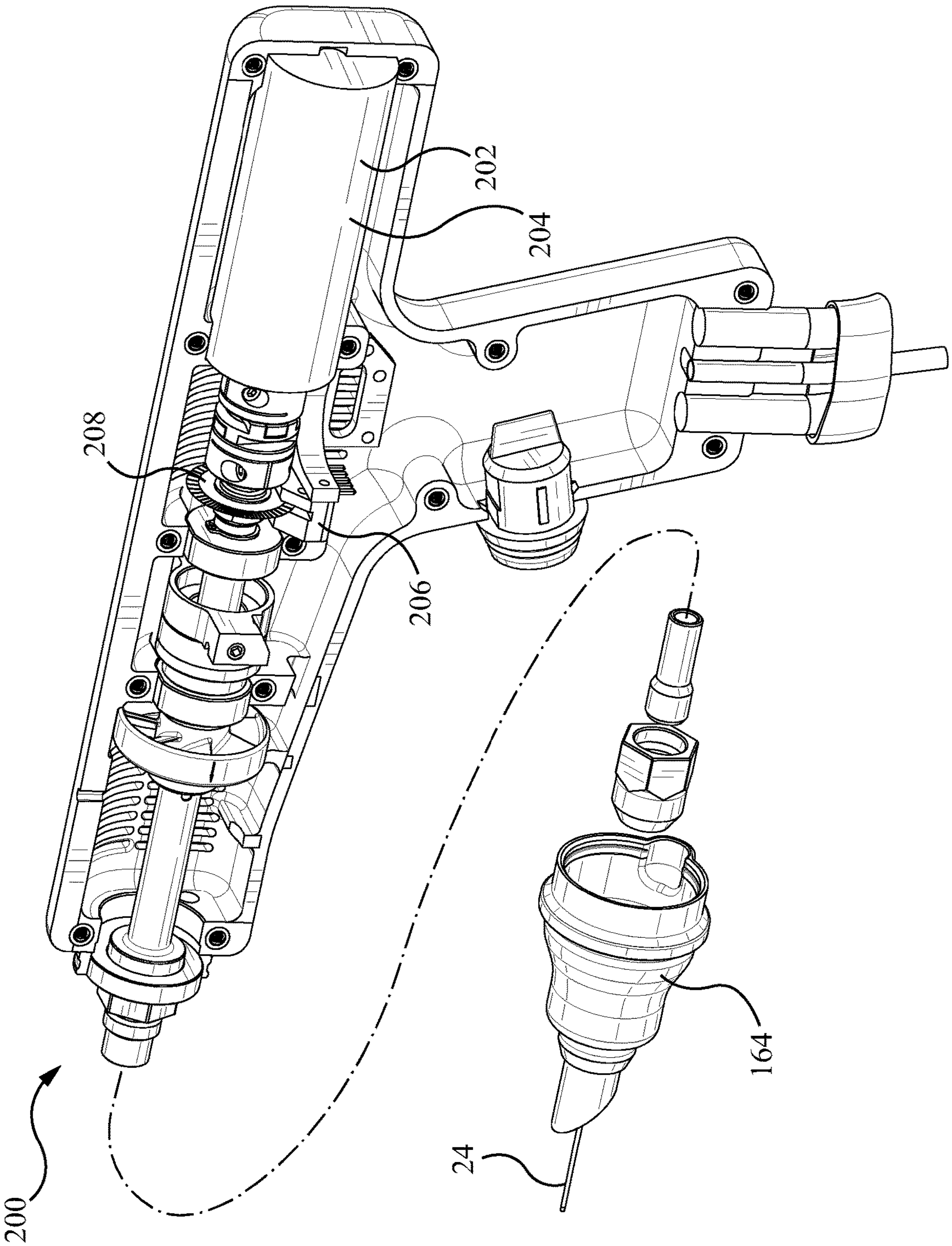

FIG. 11a

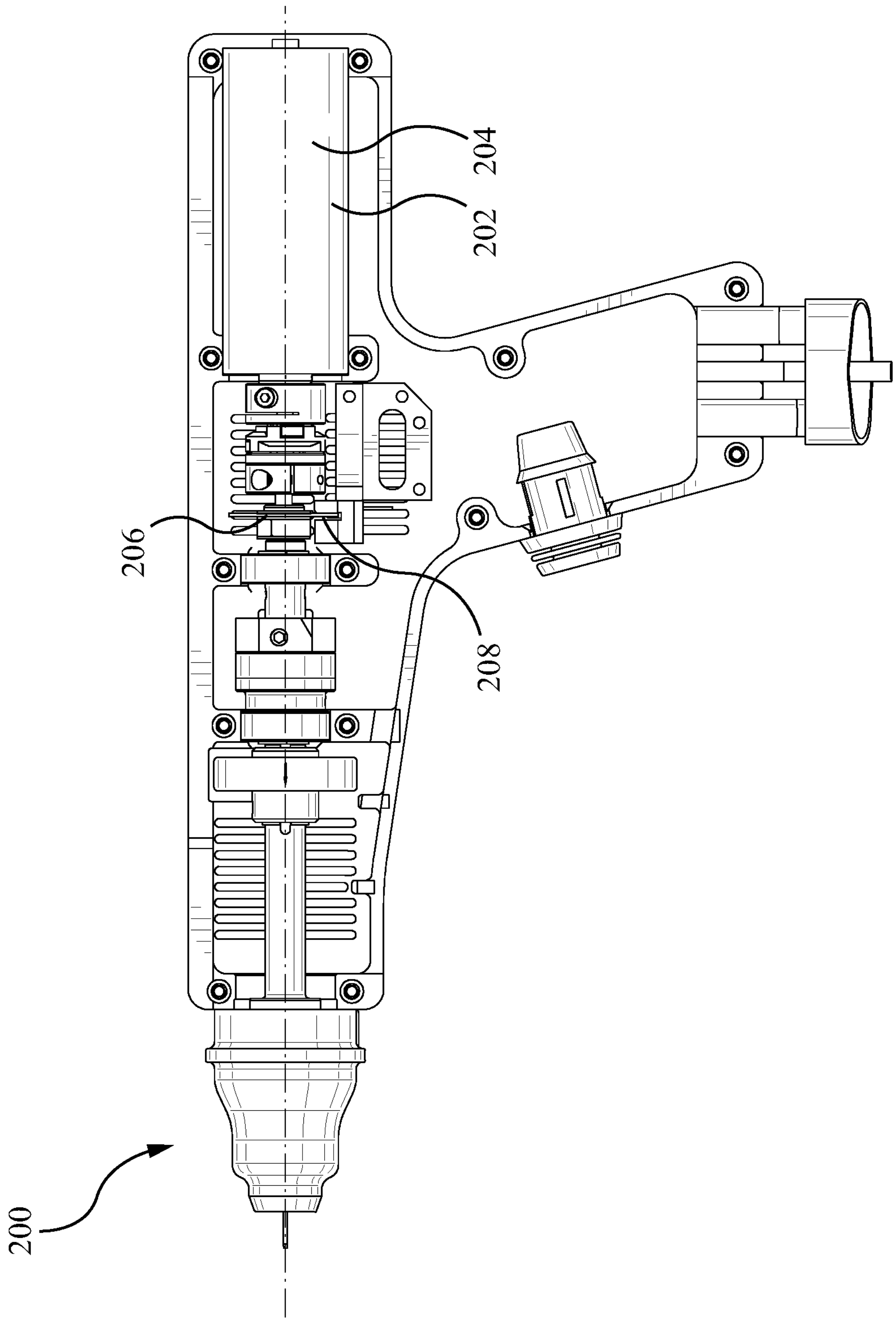

FIG. 11b

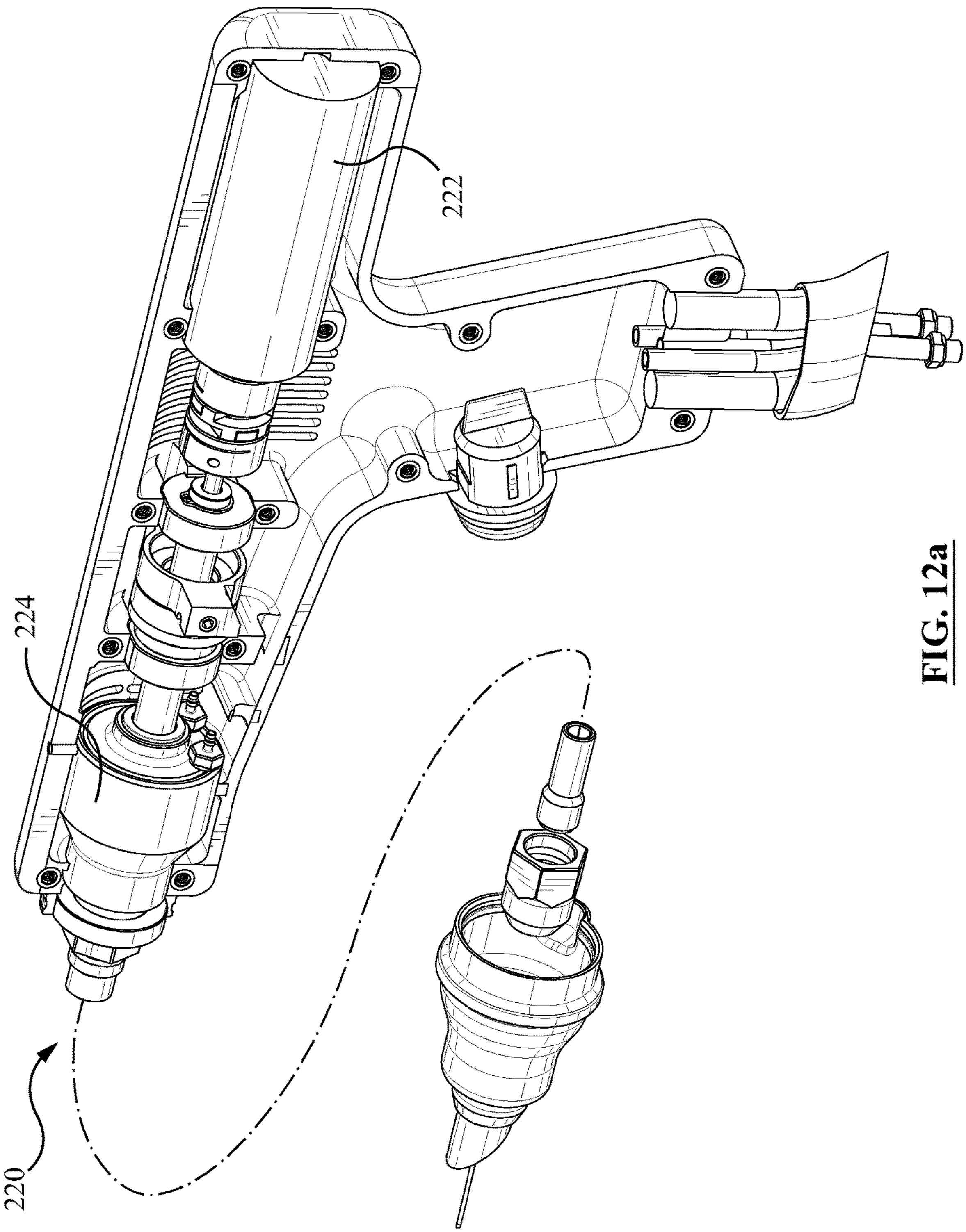

FIG. 12a

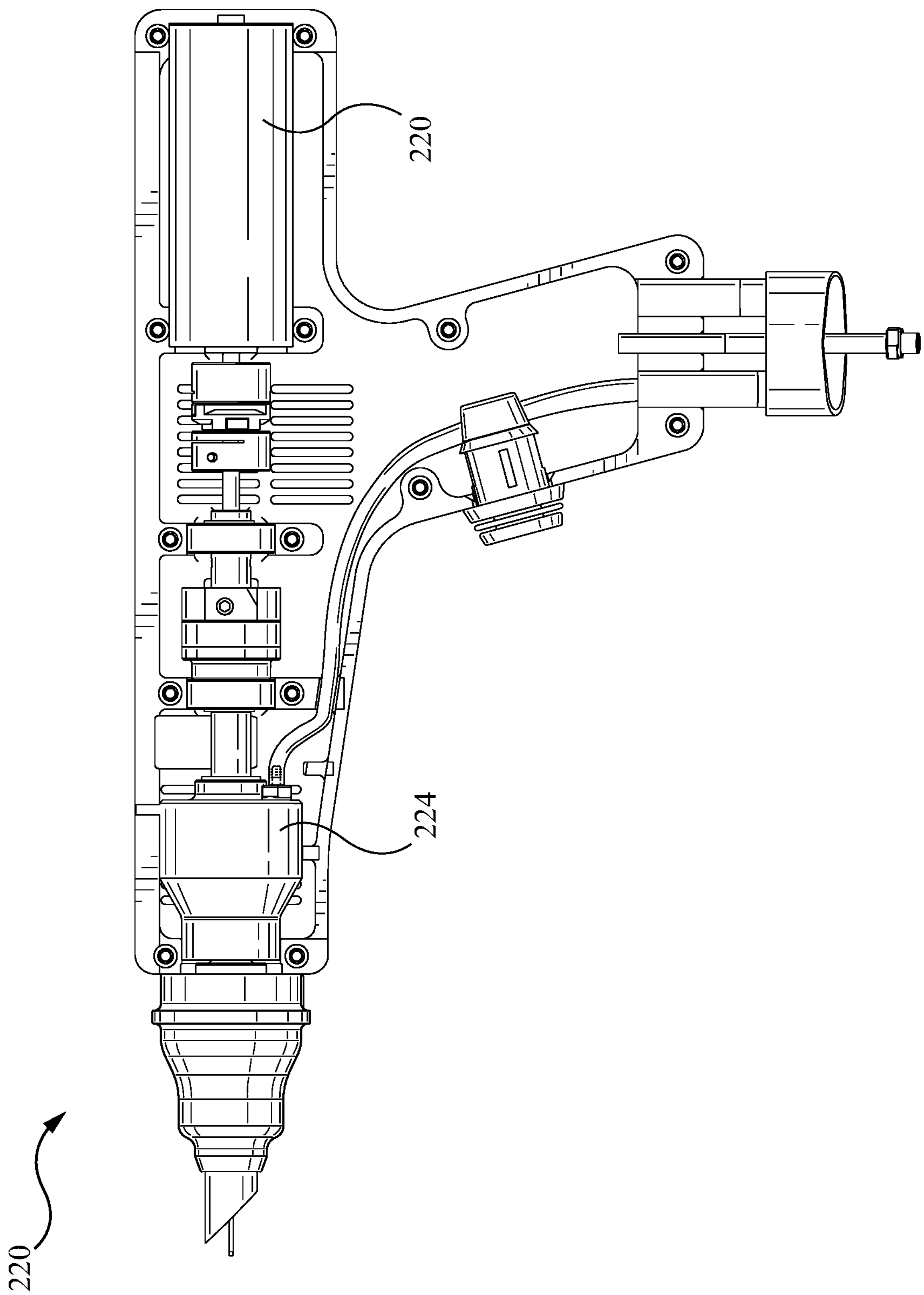

FIG. 12b

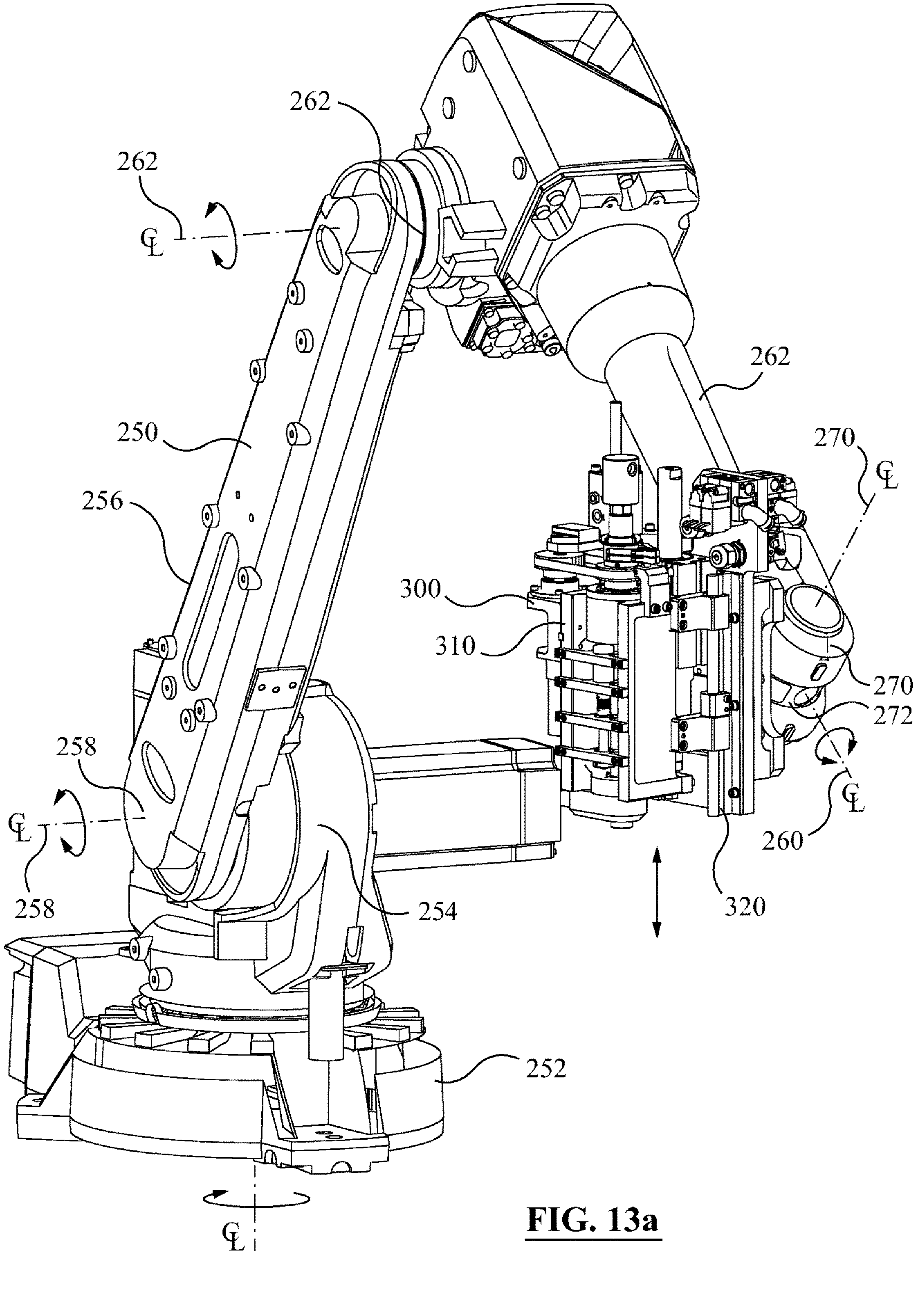
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FIG. 13a

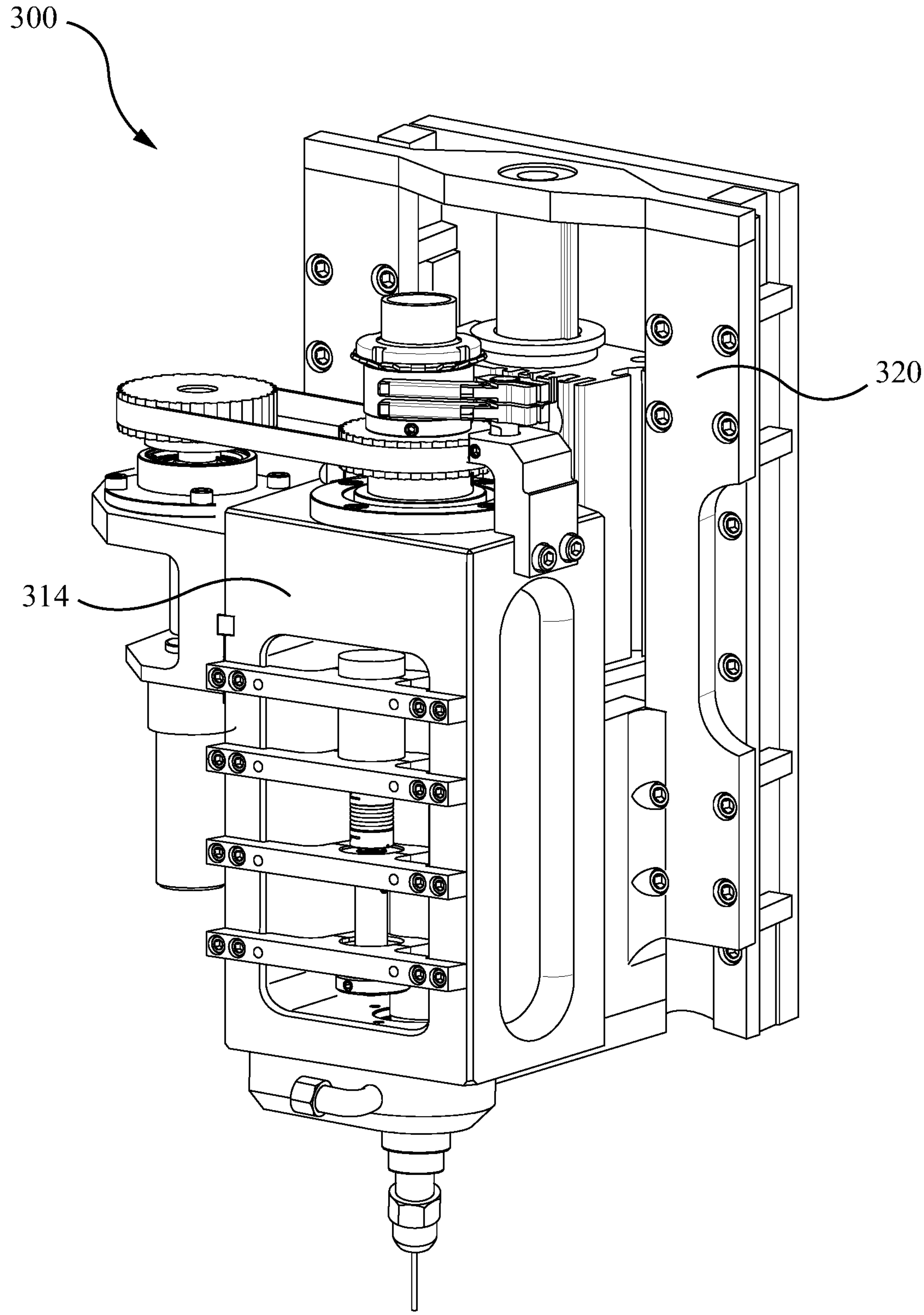


FIG. 13b

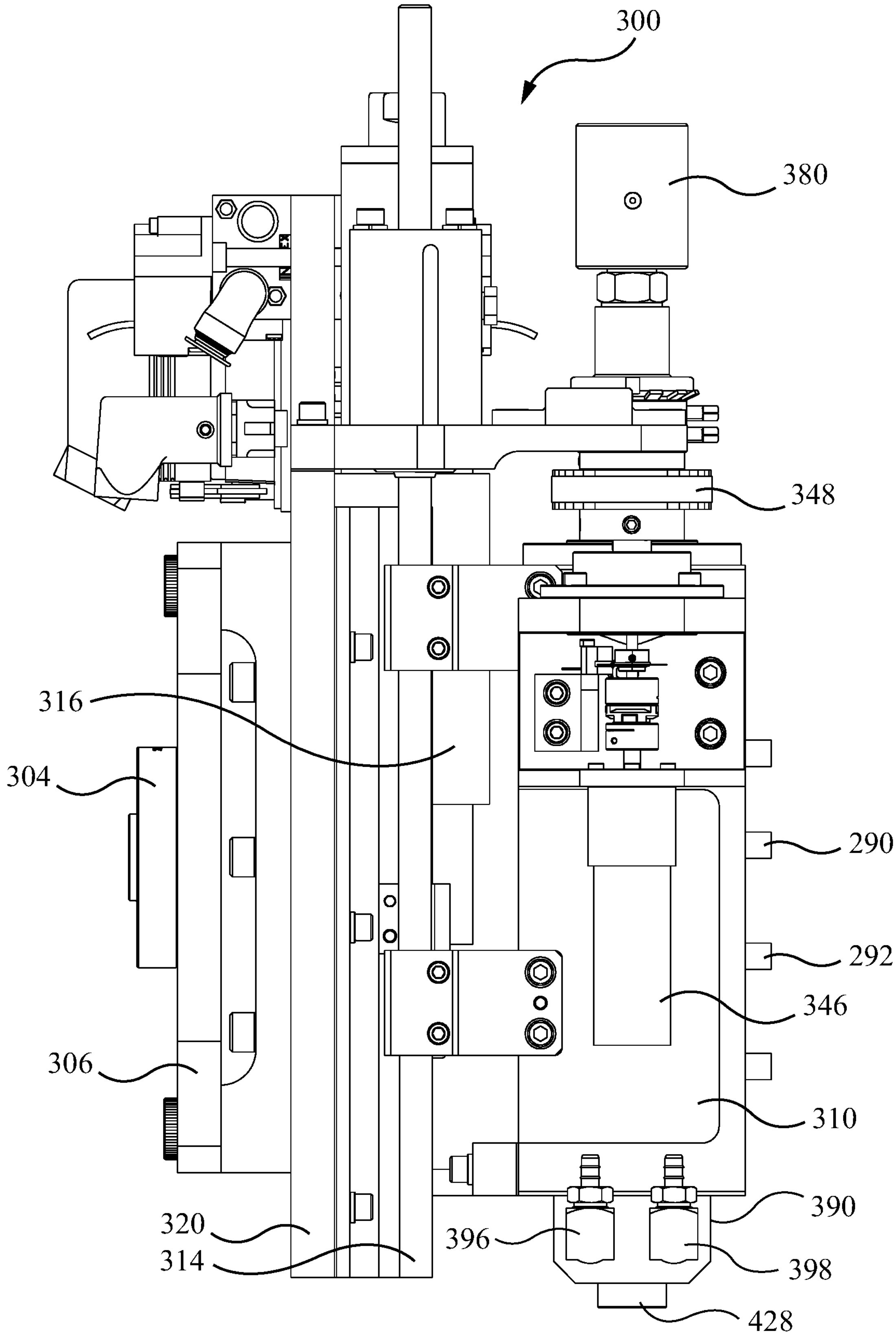


FIG. 14a

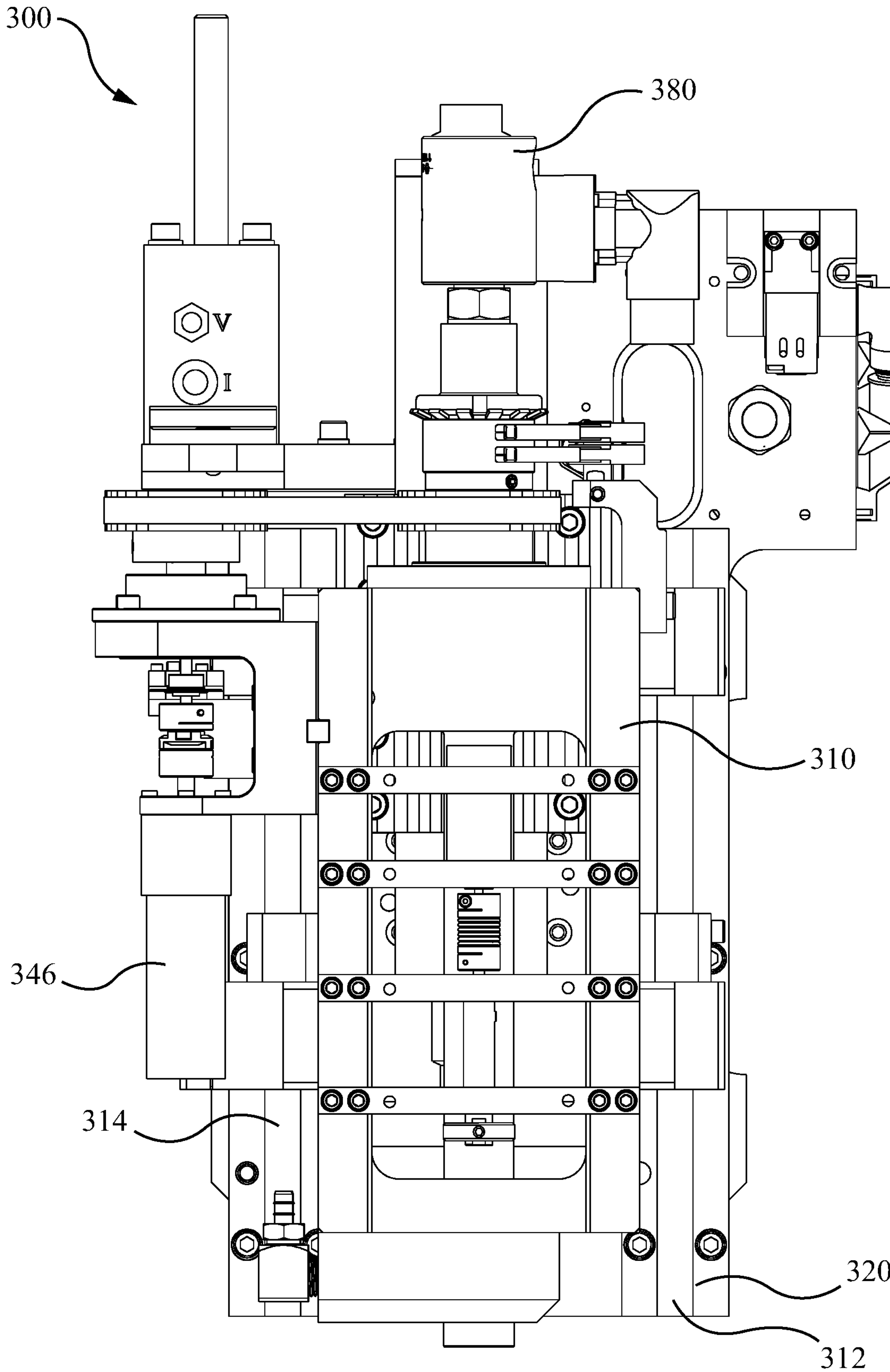


FIG. 14b

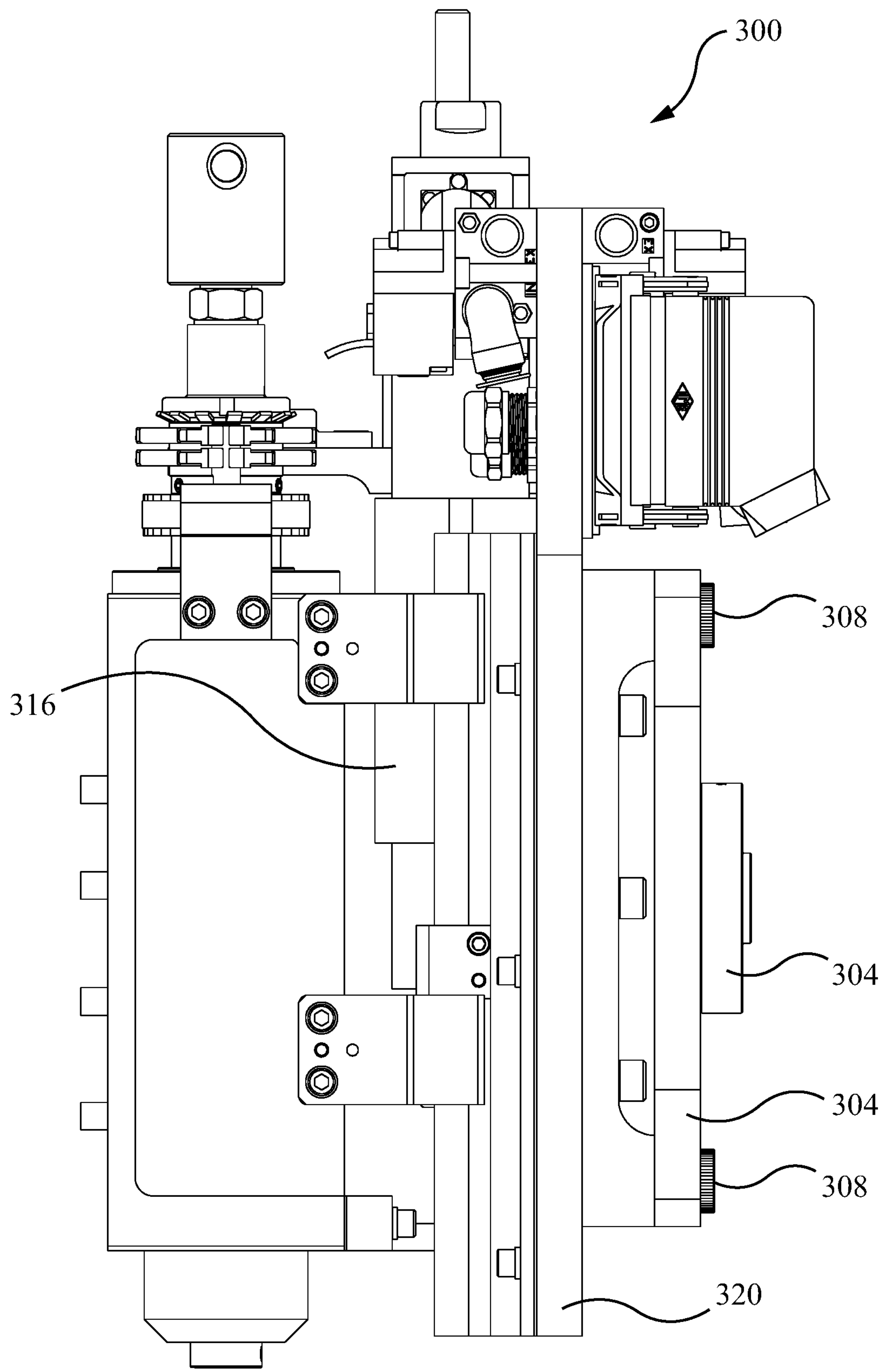


FIG. 14c

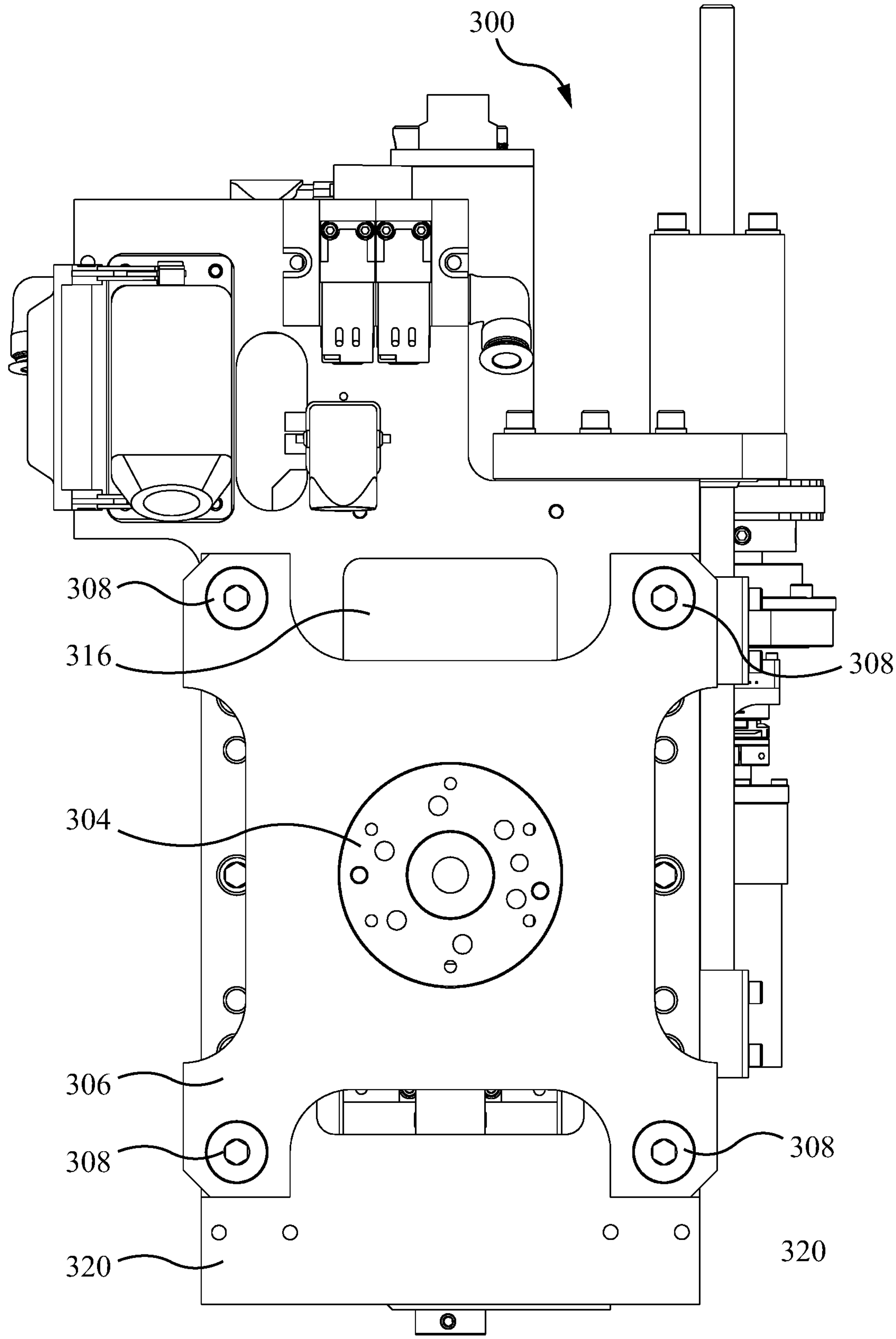


FIG. 14d

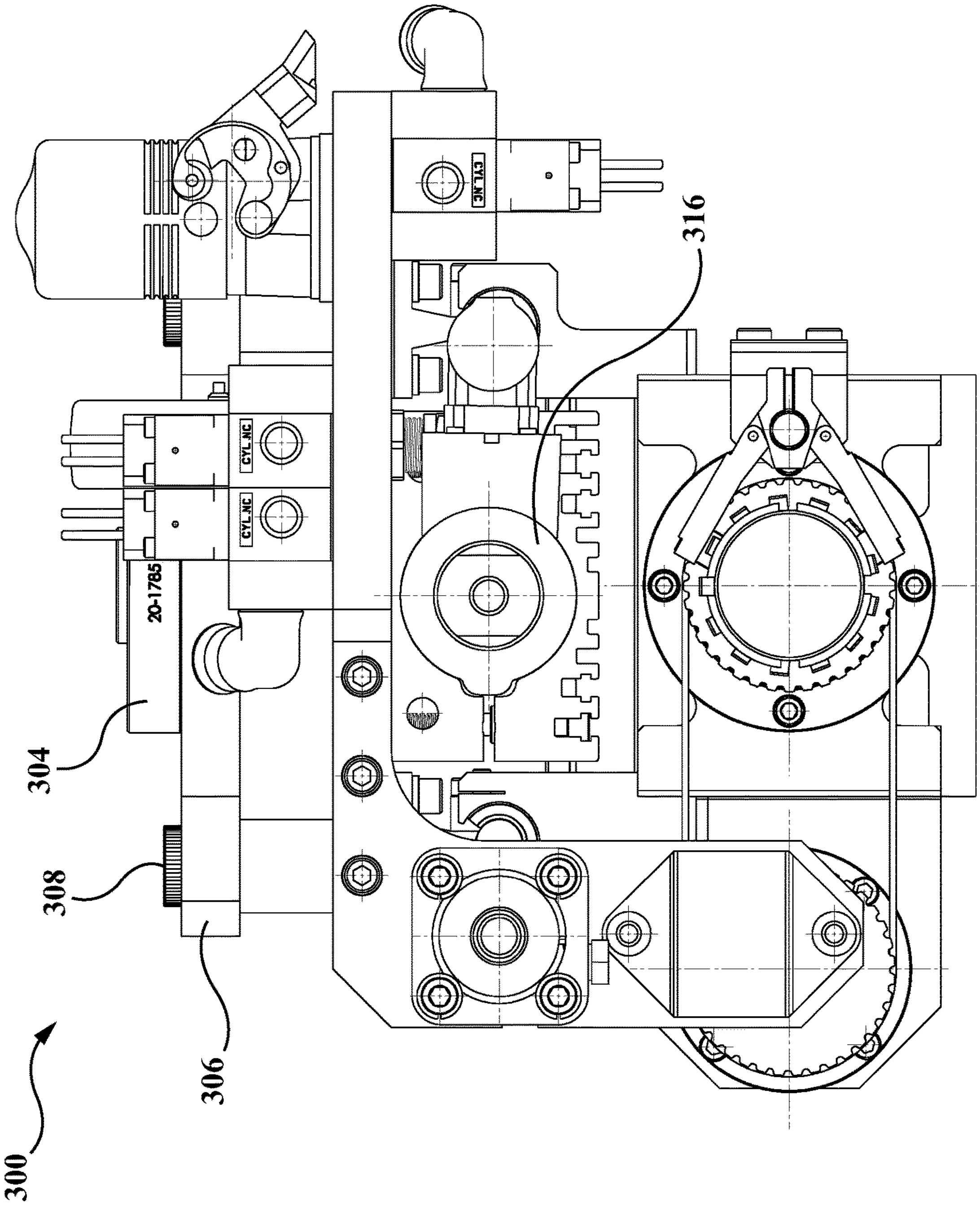
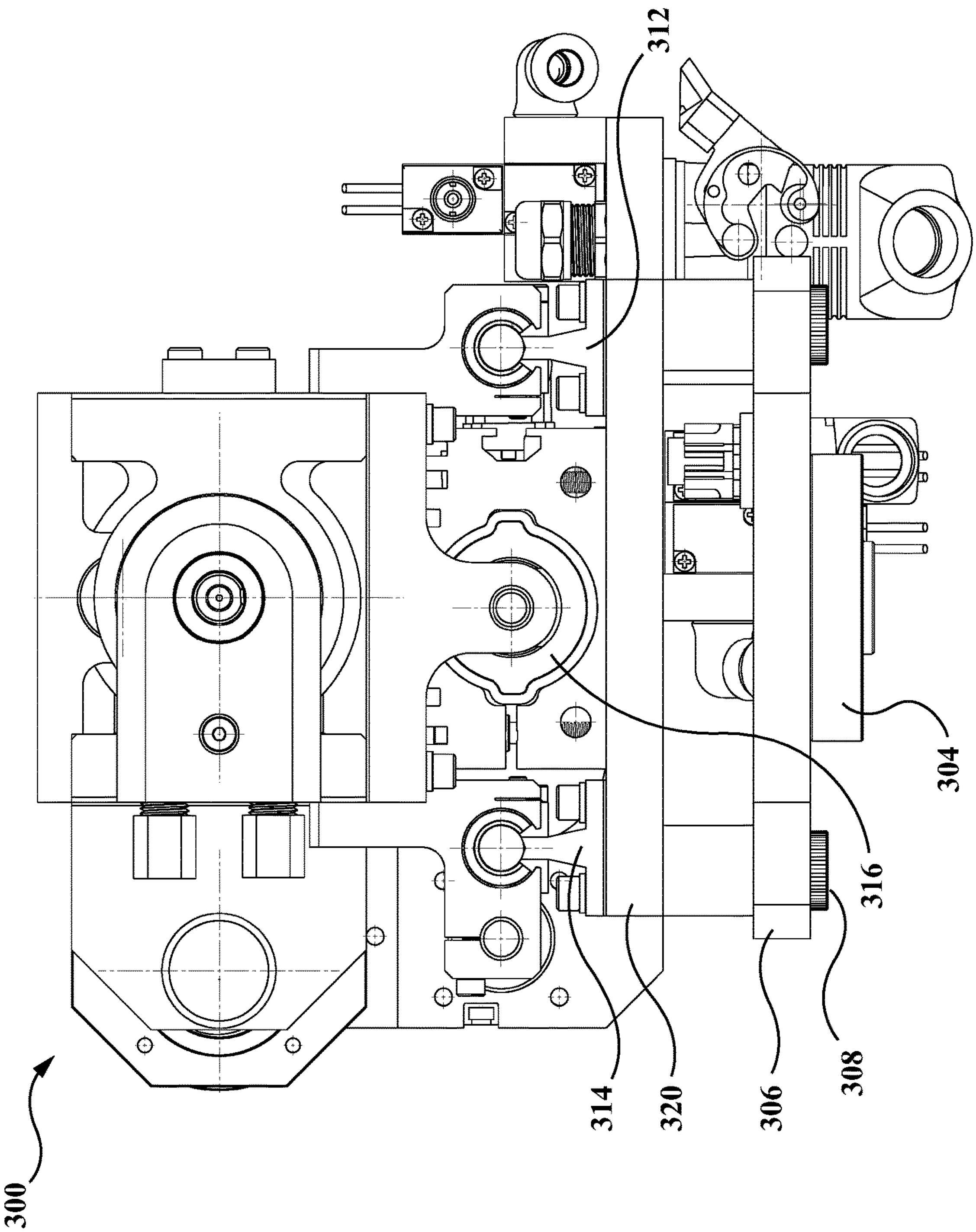


FIG. 14e



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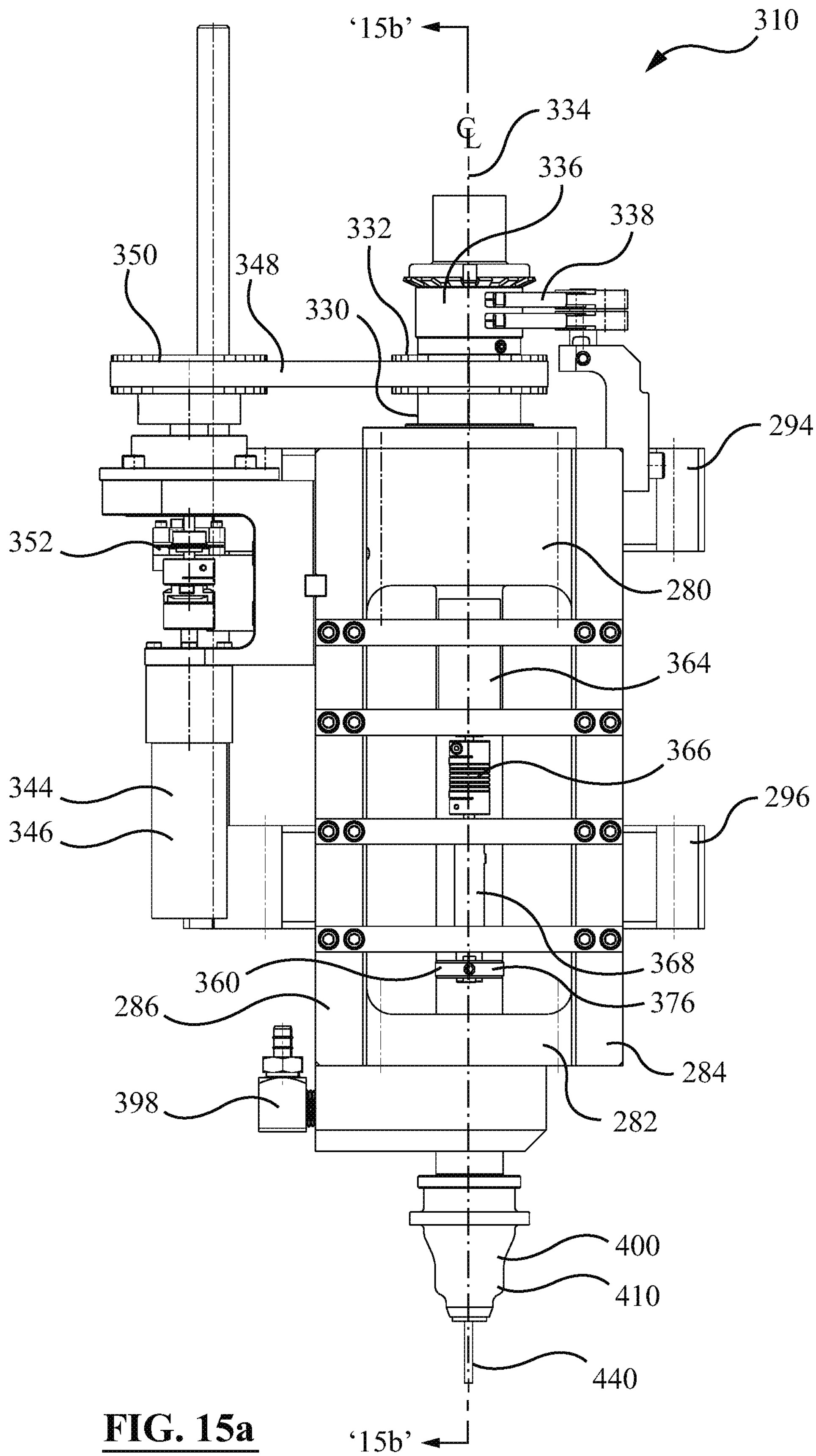
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FIG. 15a

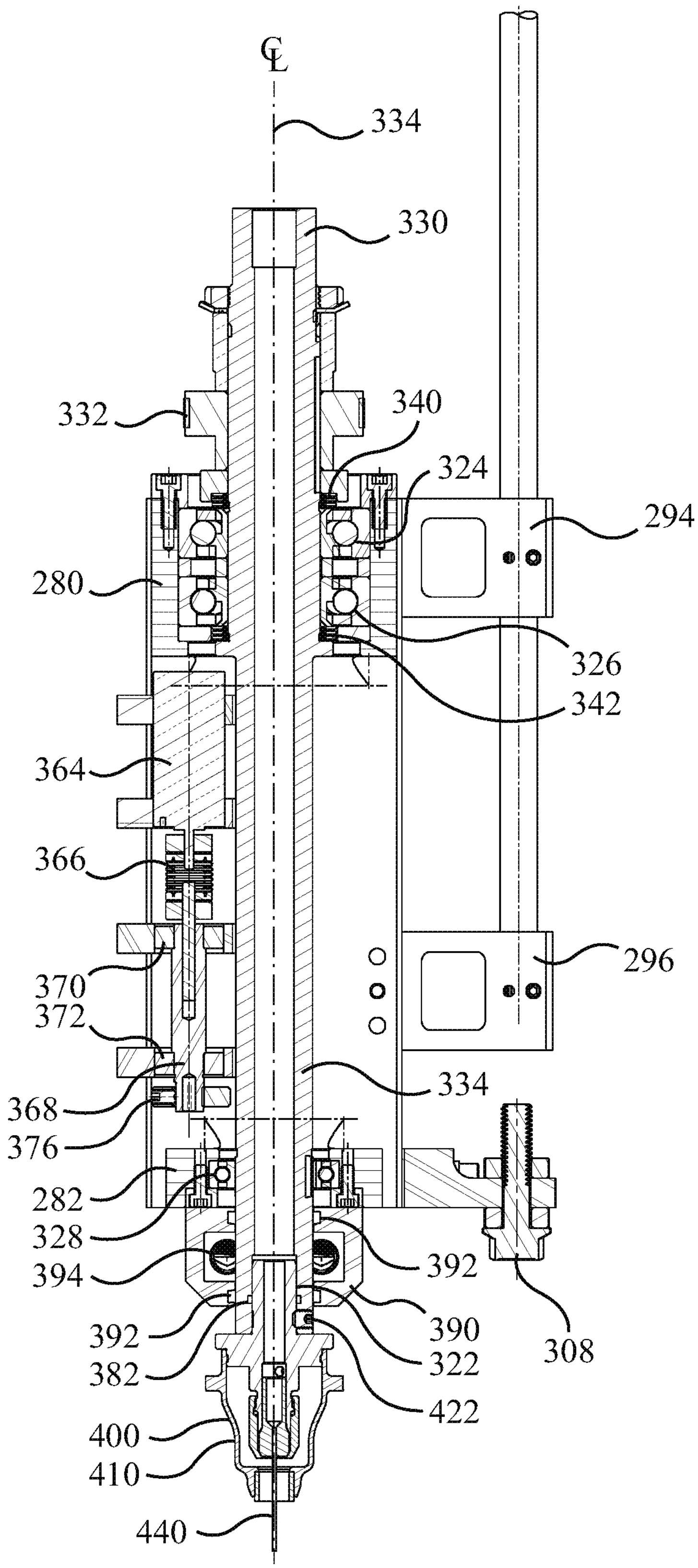
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FIG. 15b

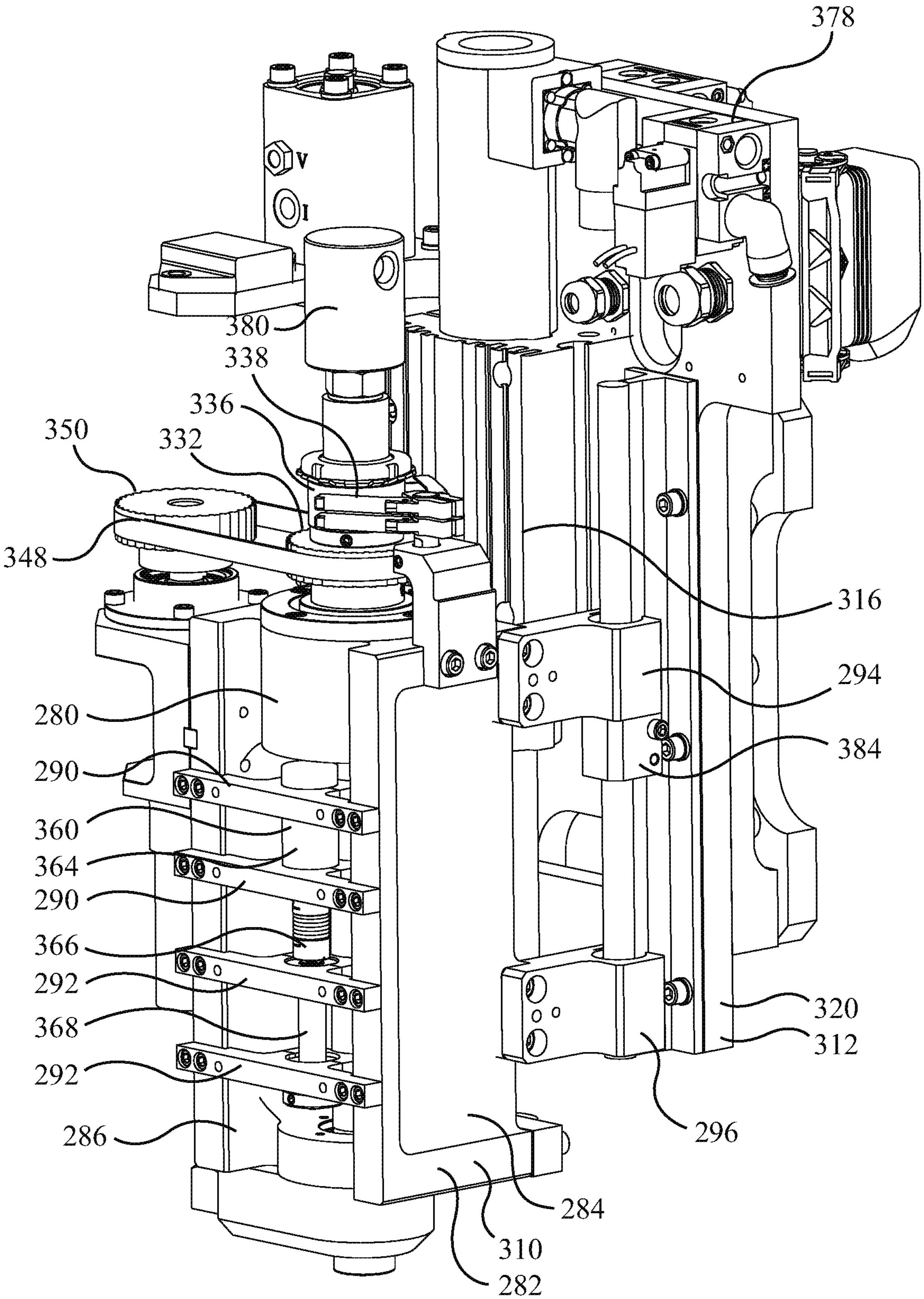
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FIG. 16a

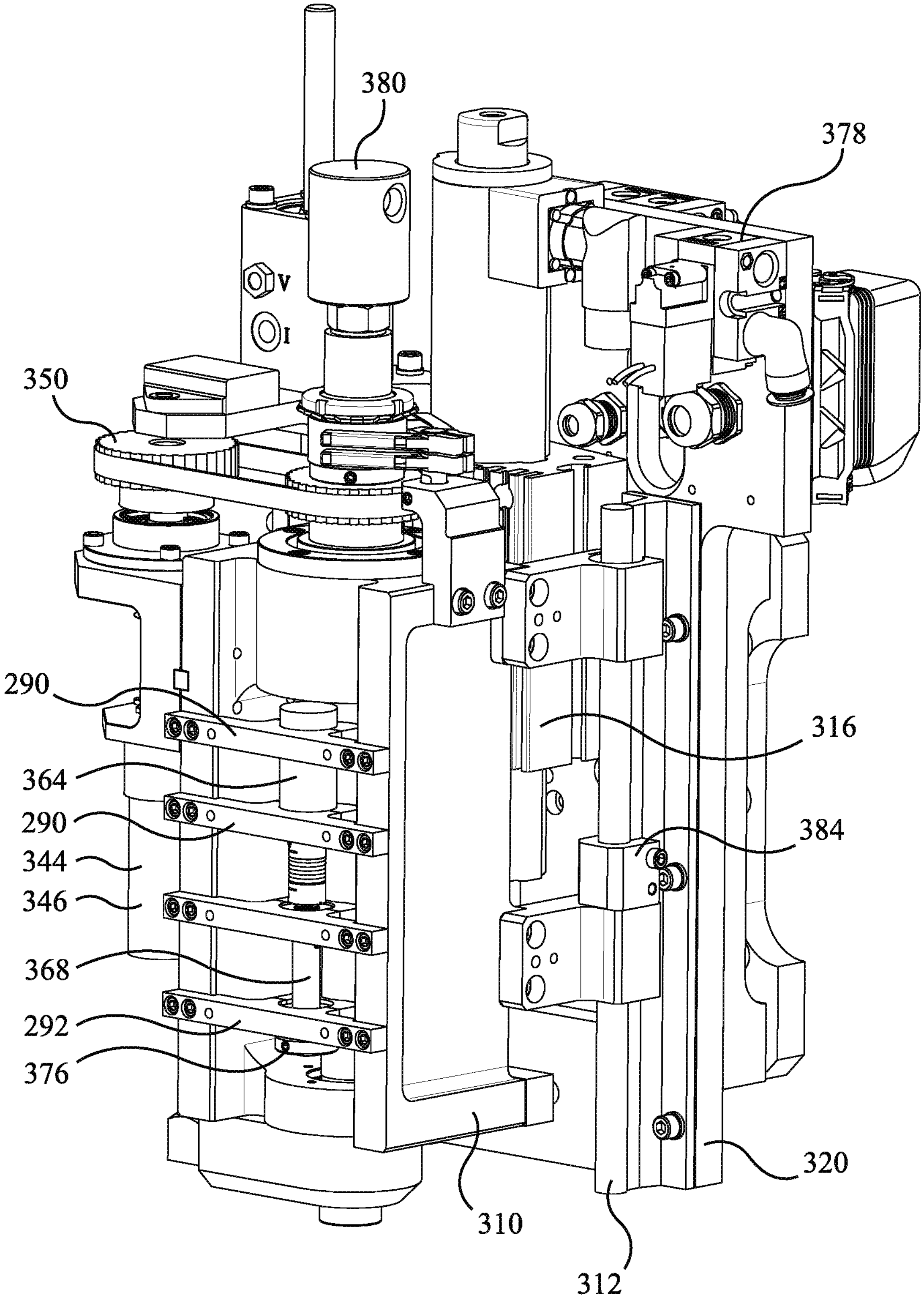
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FIG. 16b

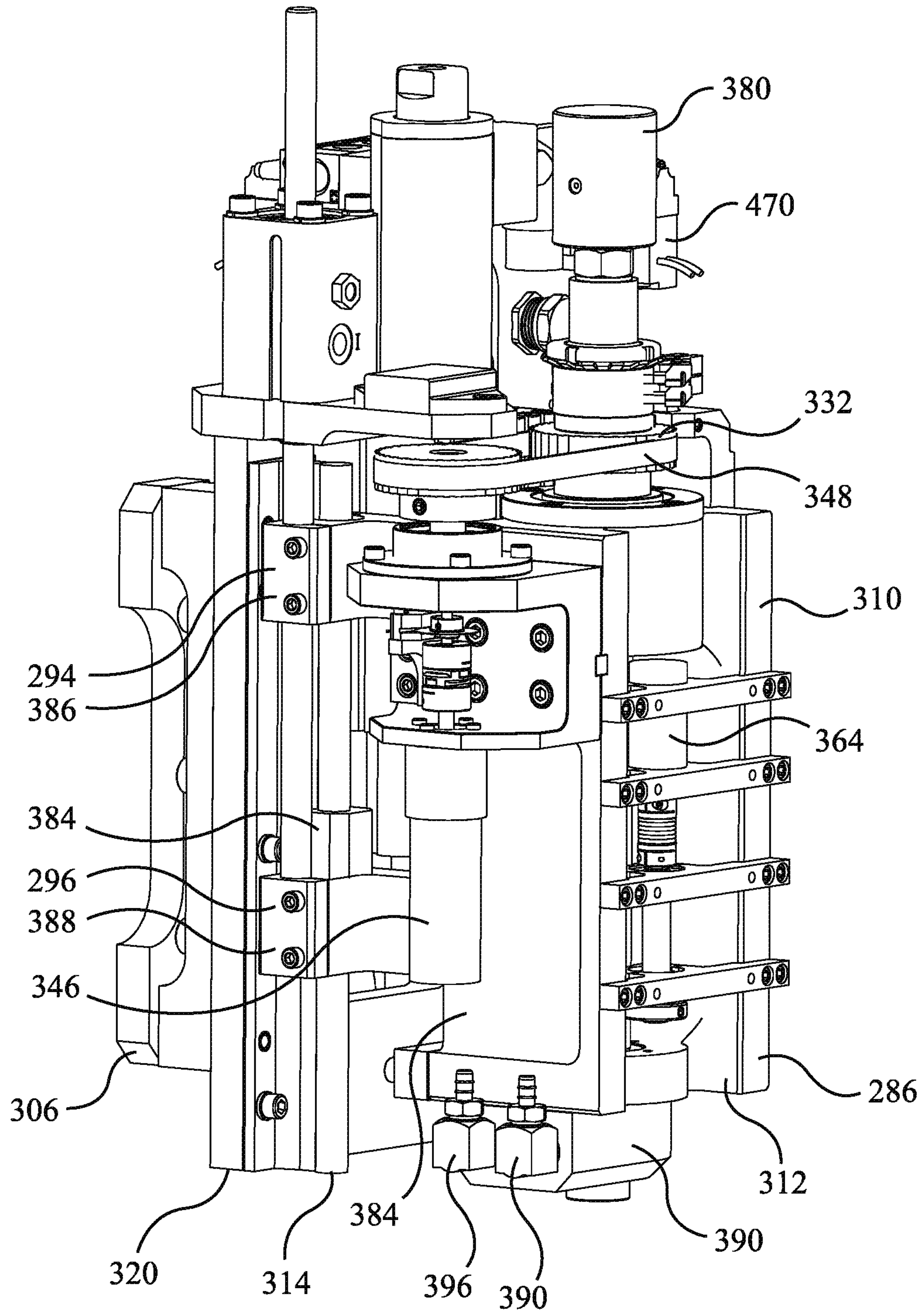
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FIG. 16c

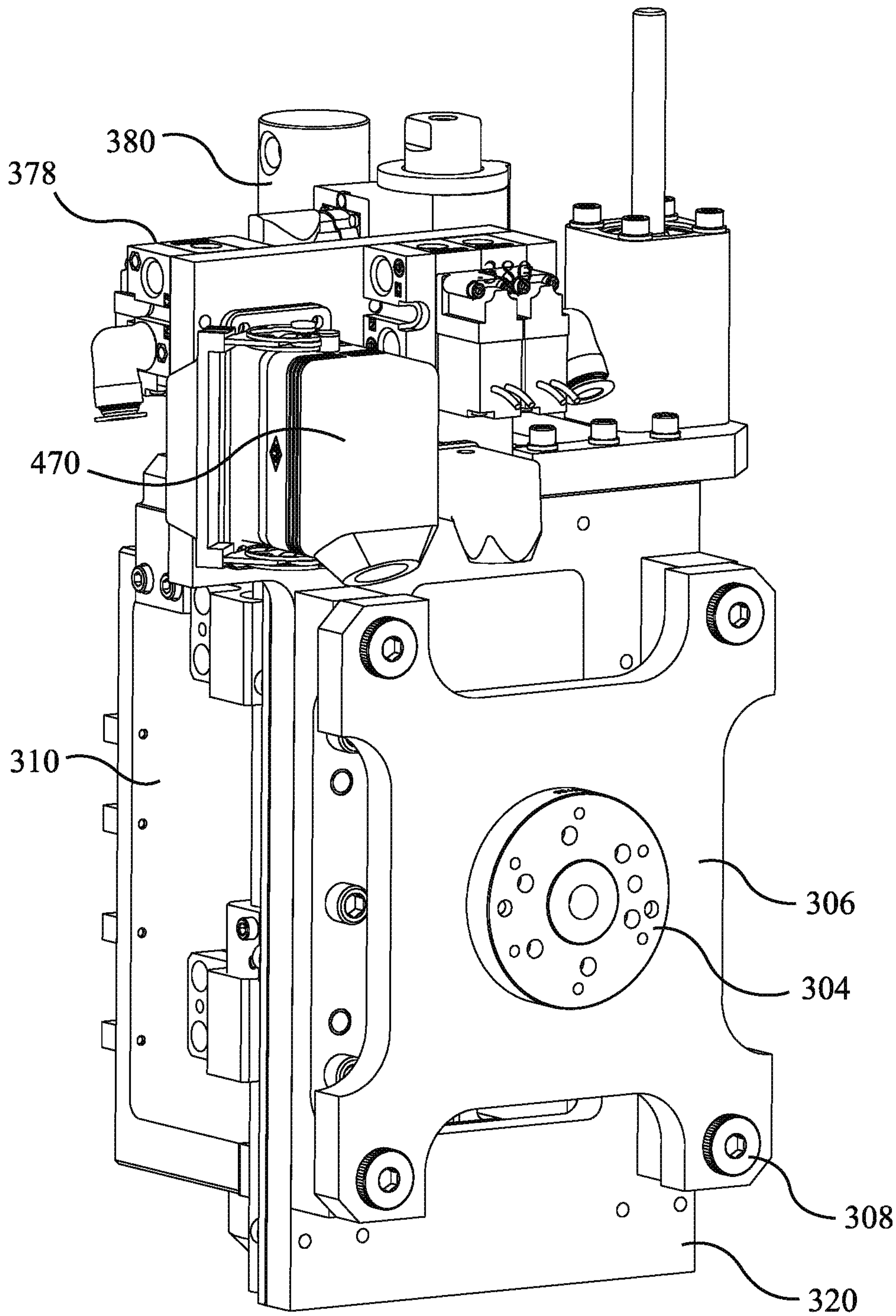
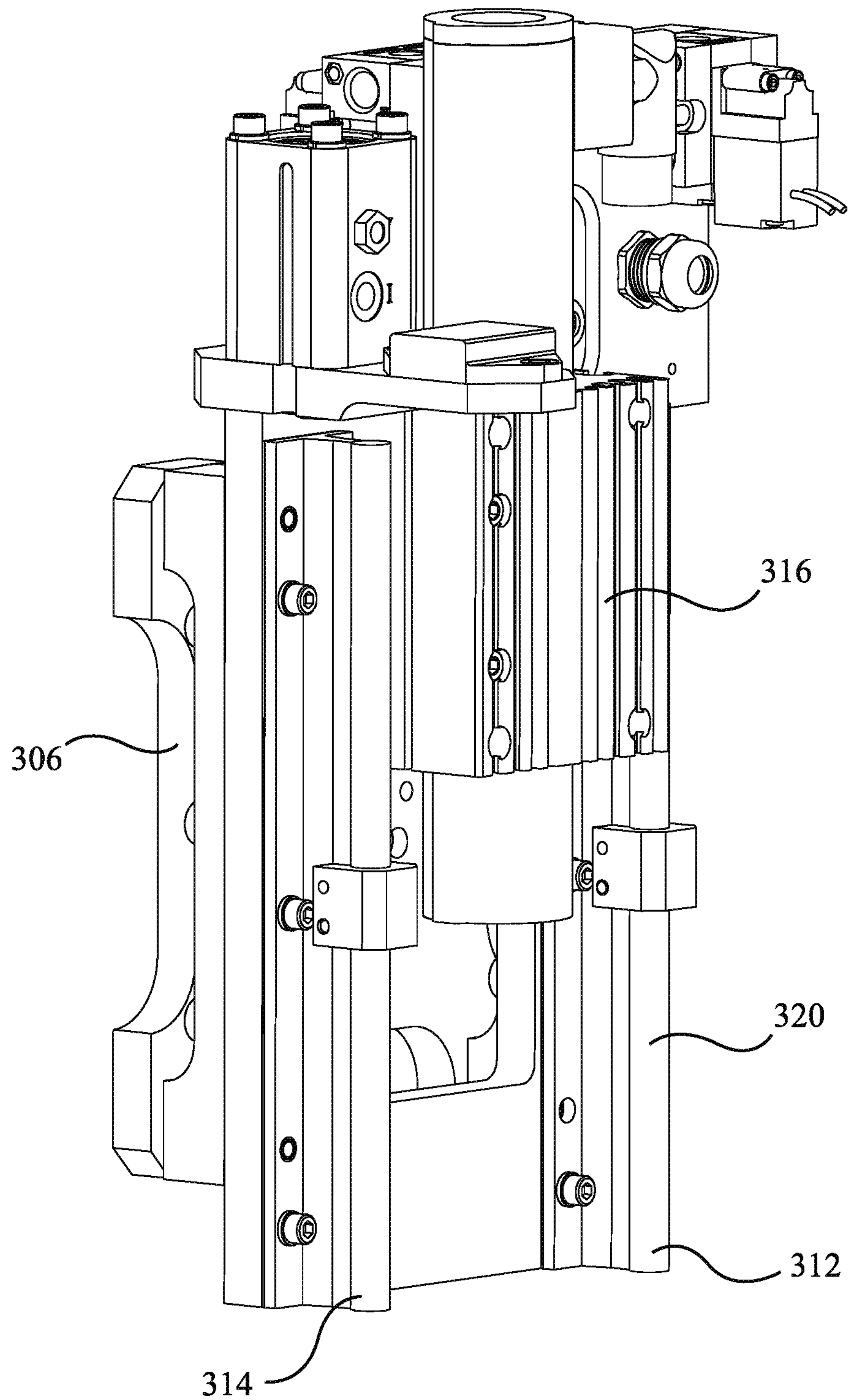


FIG. 16d



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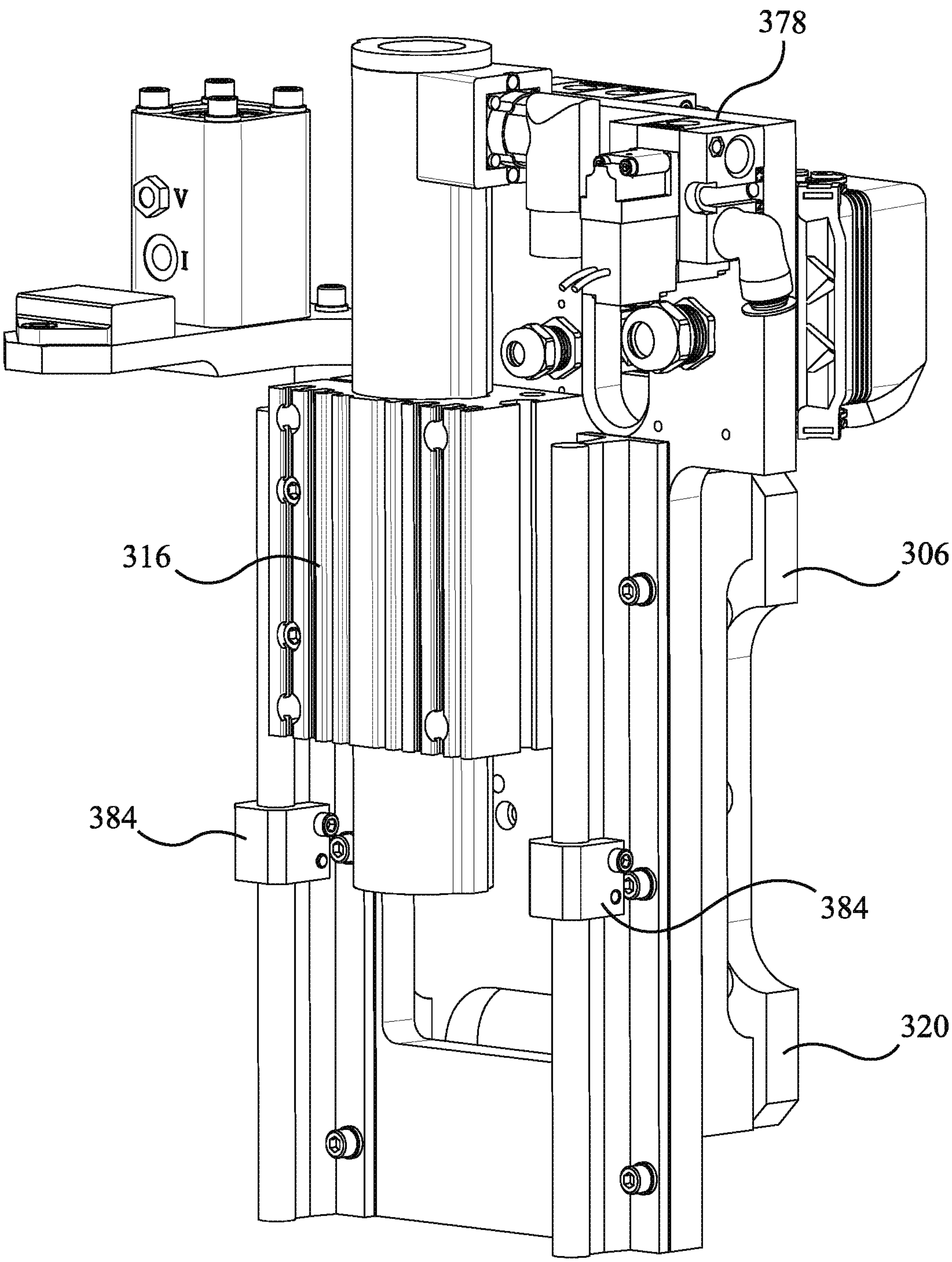


FIG. 17b

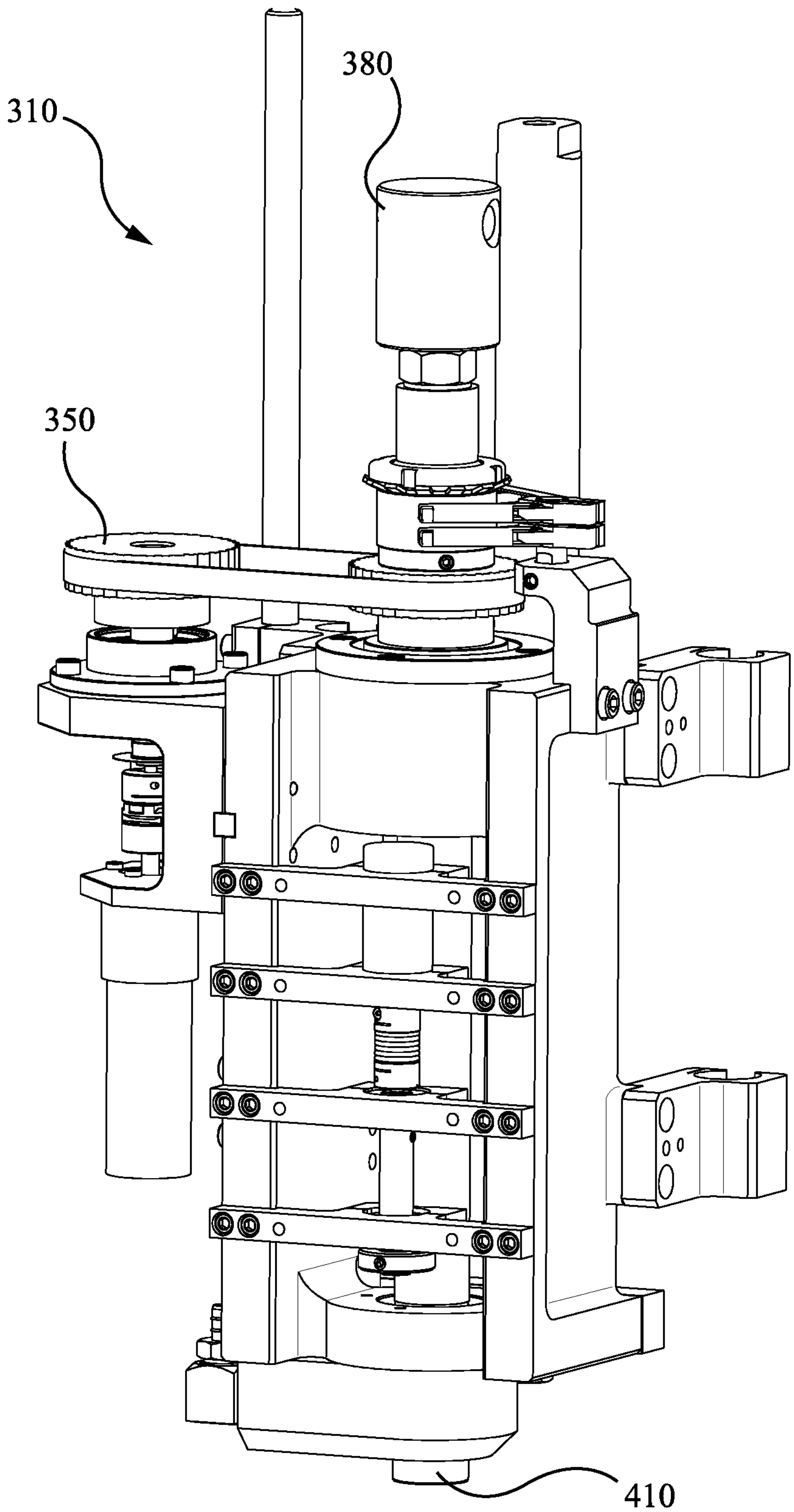


FIG. 18a

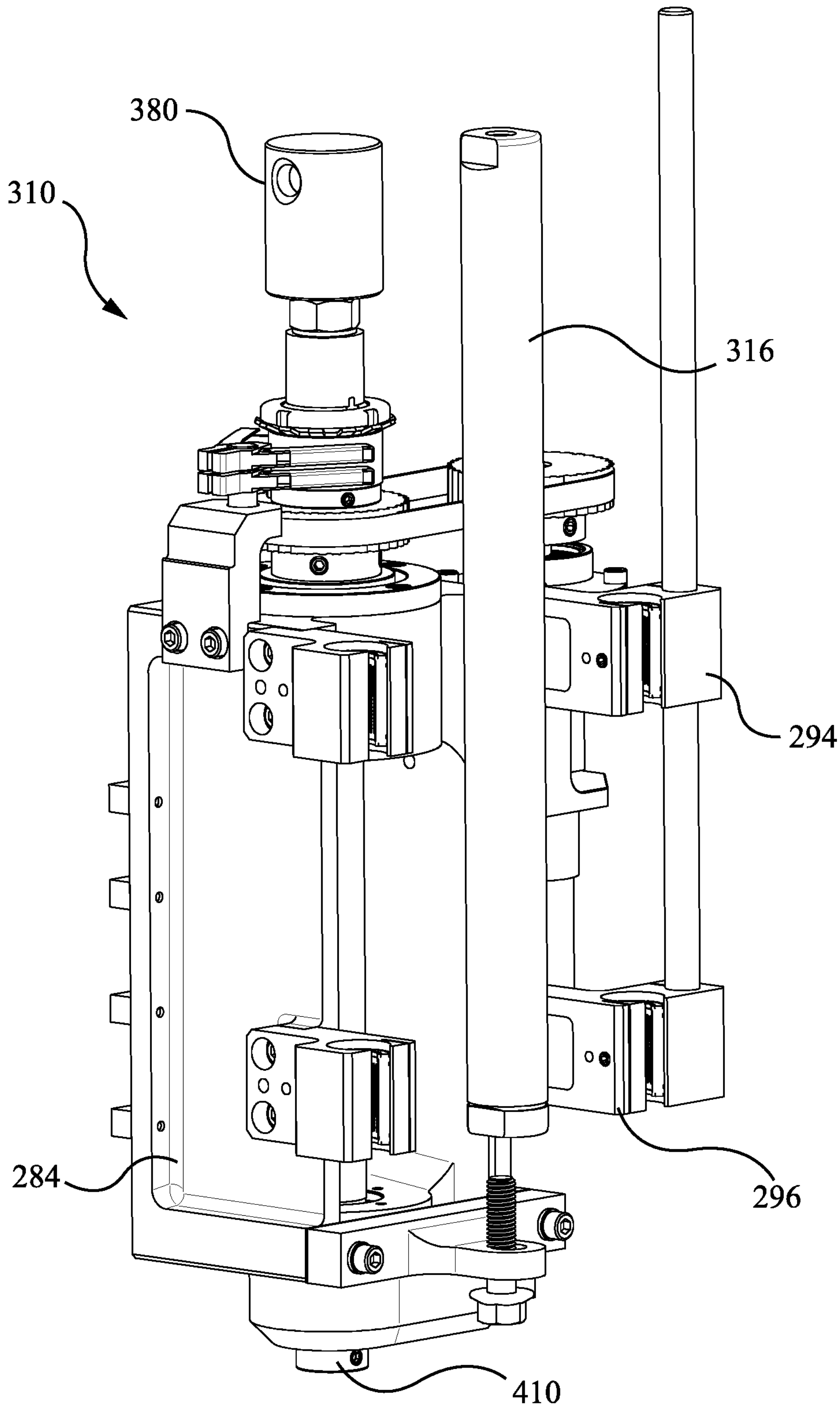


FIG. 18b

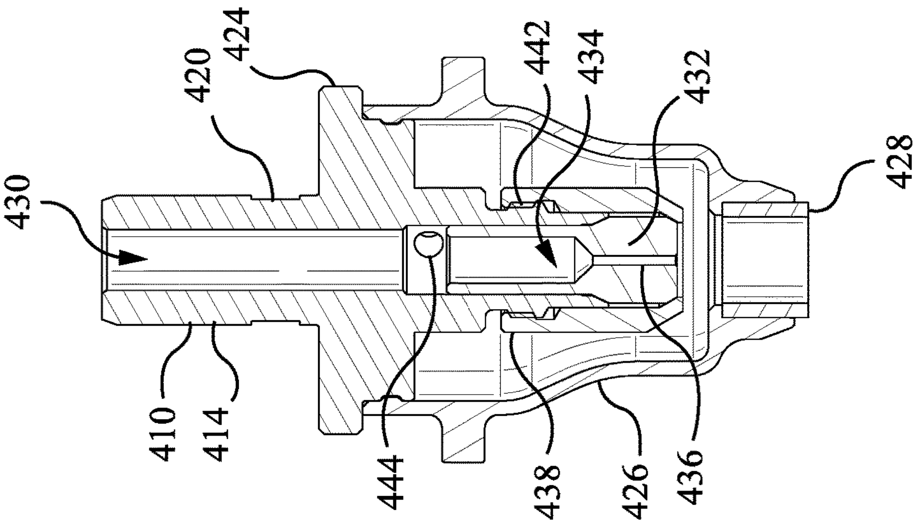


FIG. 19d

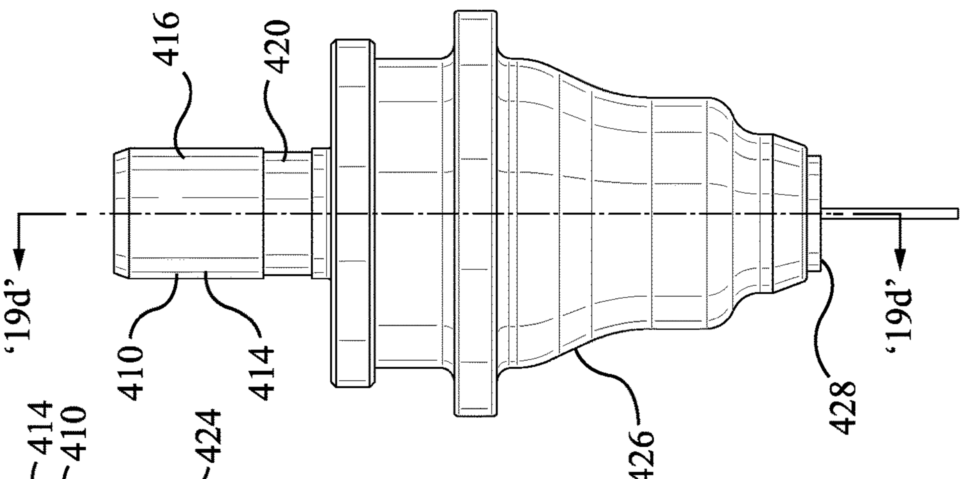


FIG. 19c

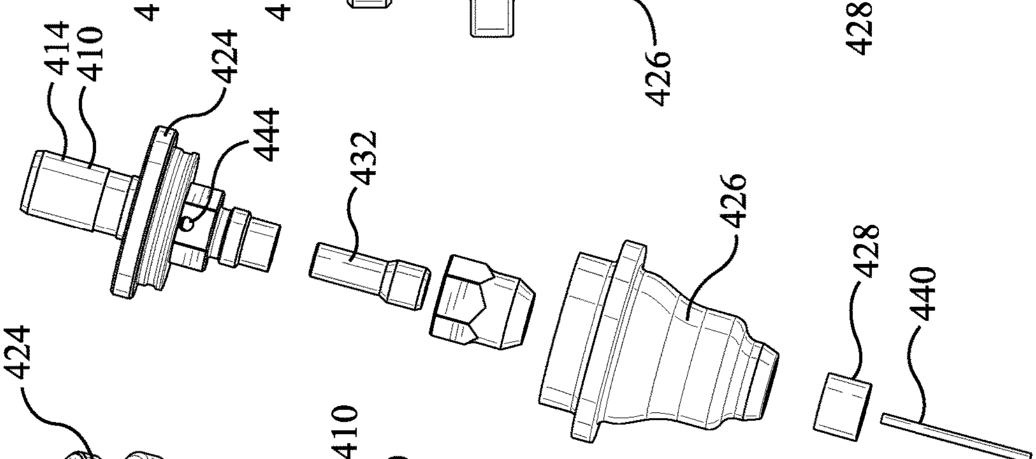


FIG. 19b

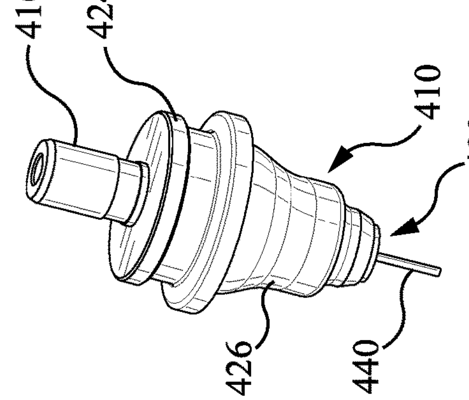


FIG. 19a

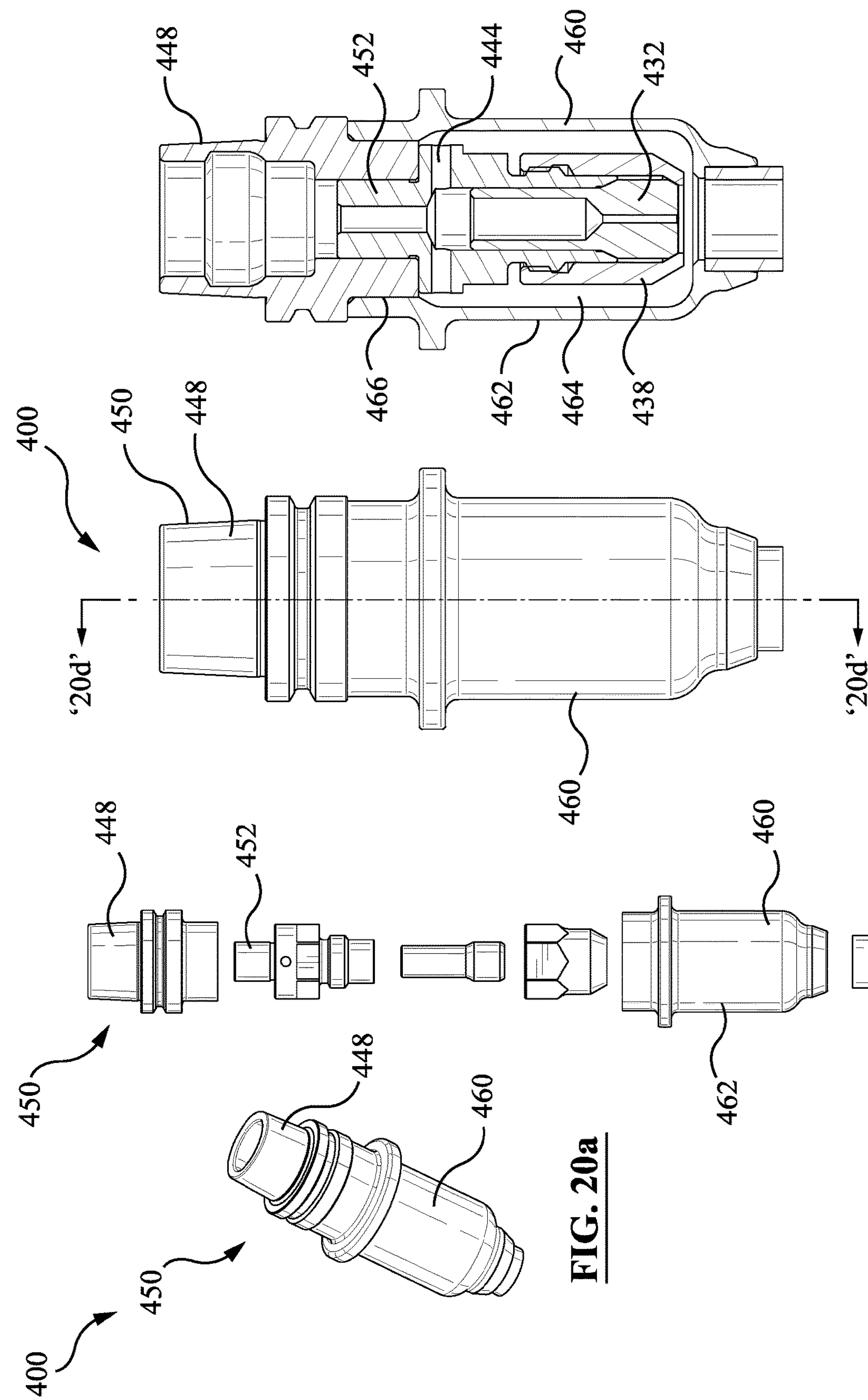


FIG. 20d

FIG. 20c

FIG. 20b

FIG. 20a

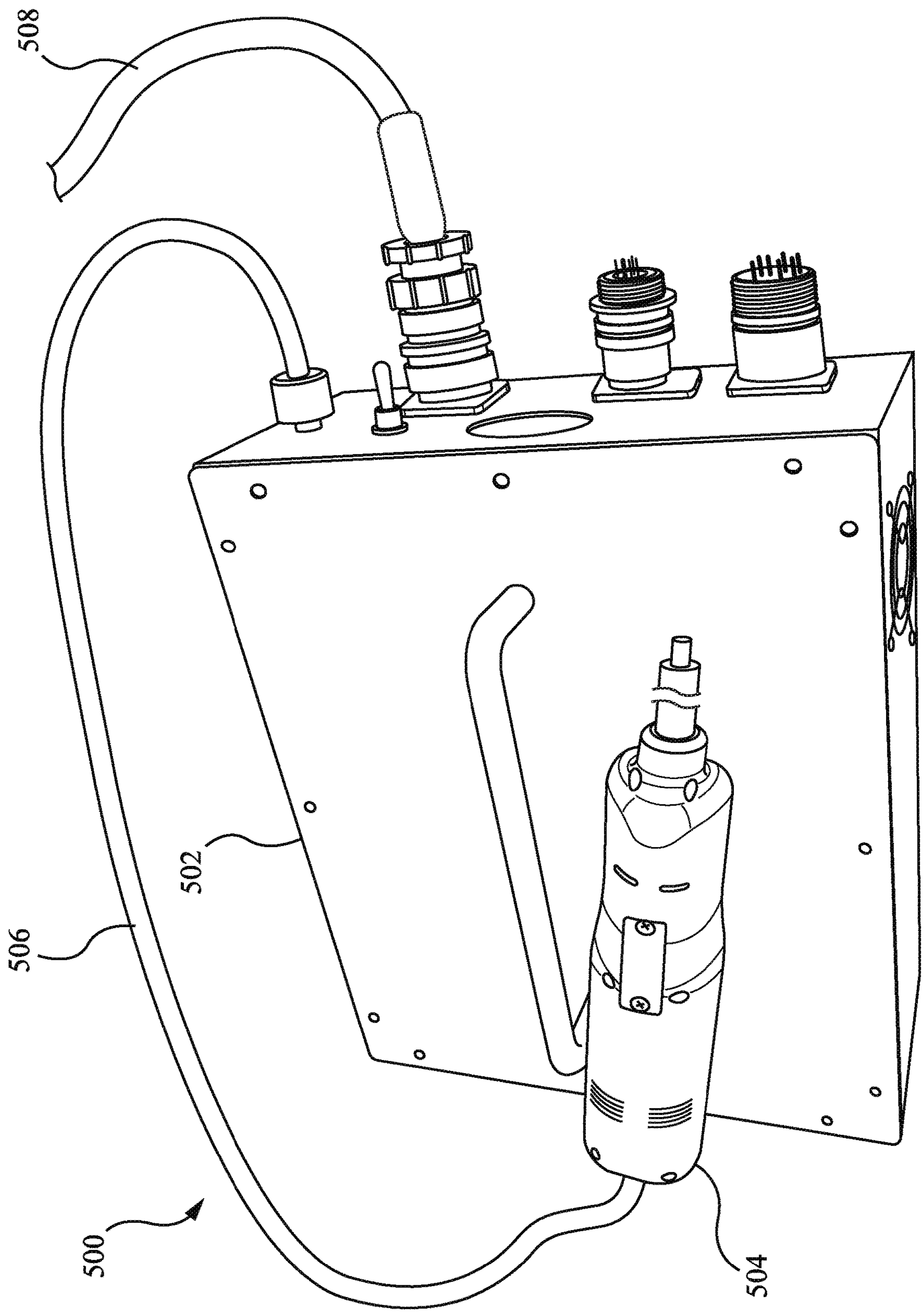


FIG. 21a

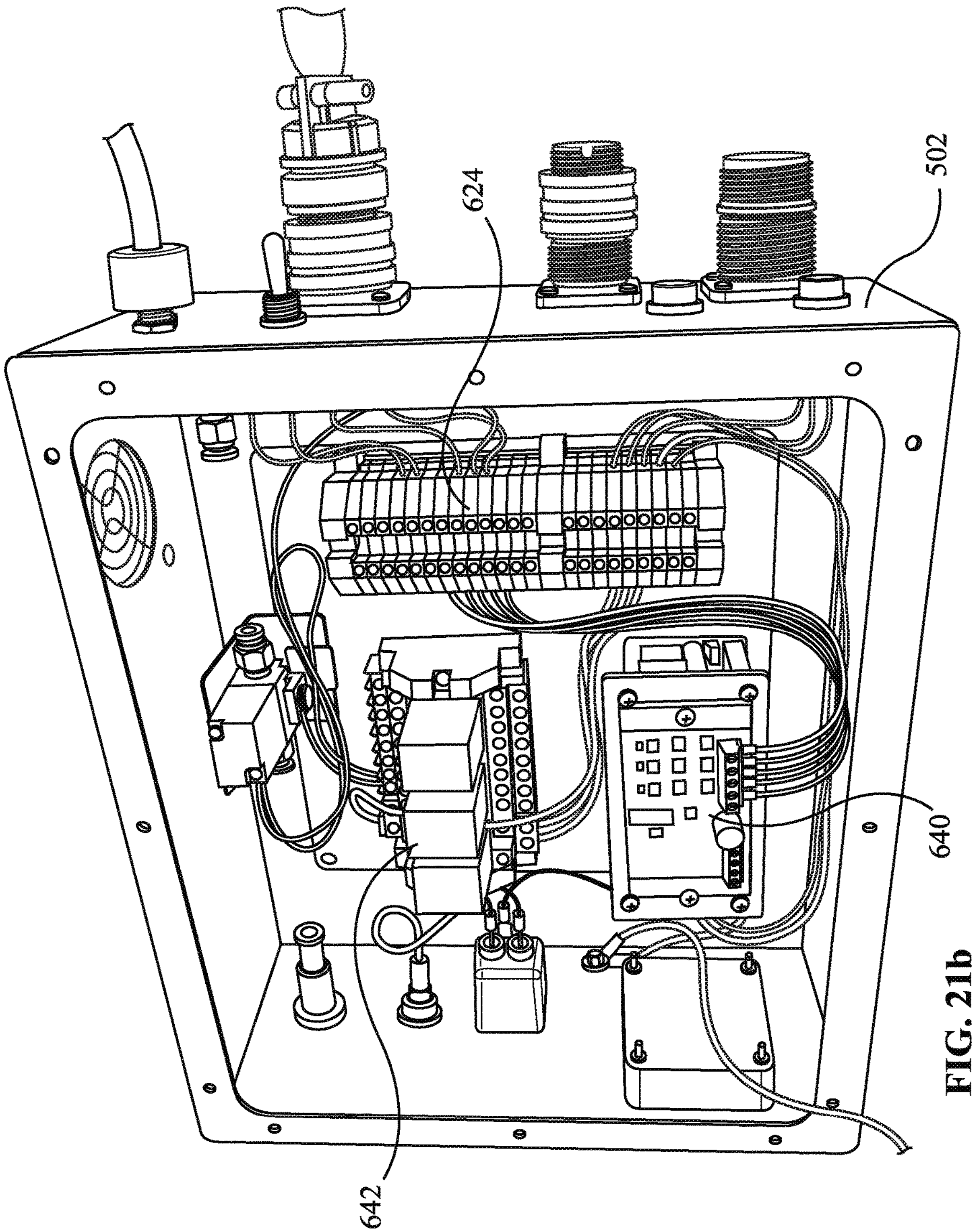


FIG. 21b

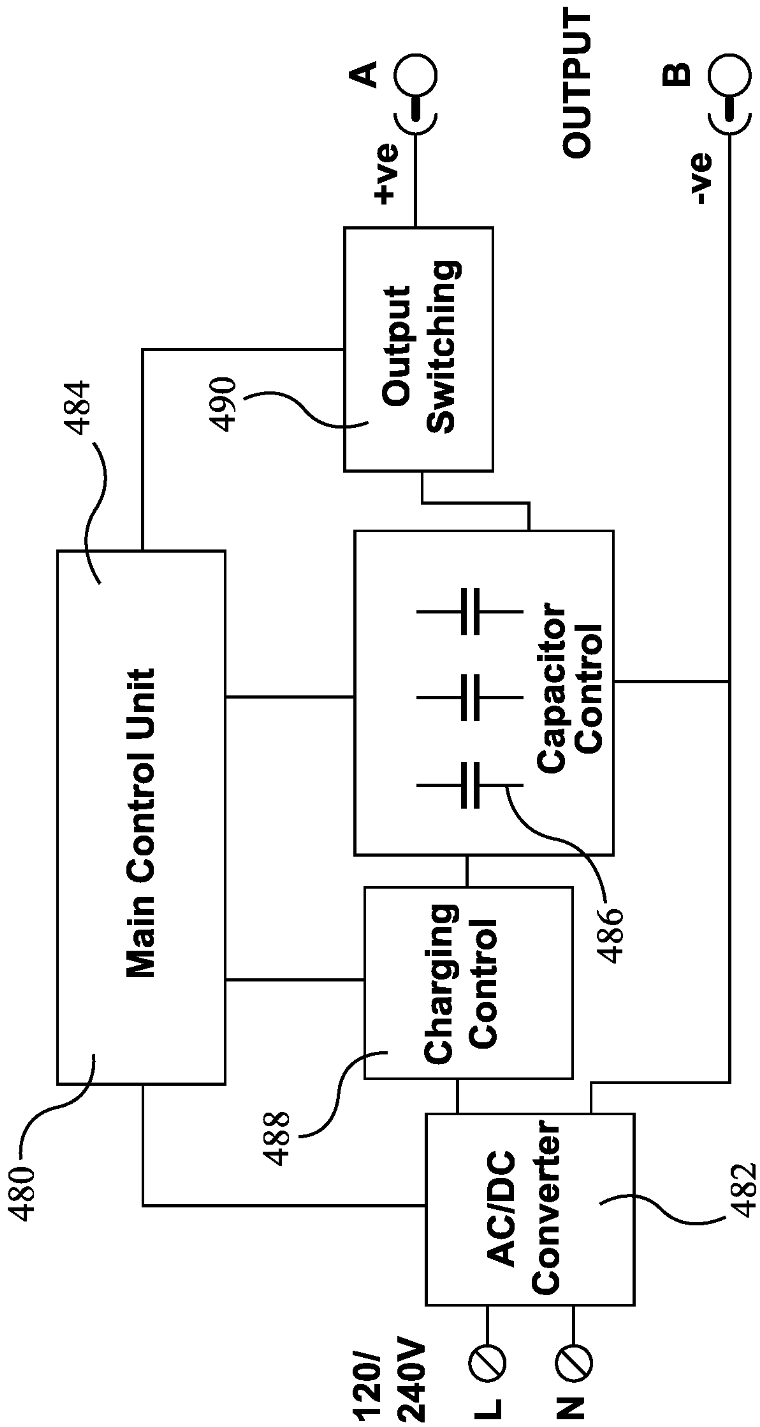

OUTPUT
A
B
+ve
-ve
Output Switching
490
484
Main Control Unit
Capacitor Control
486
Charging Control
488
480
AC/DC Converter
482
120/ 240V
L
N

FIG. 21c

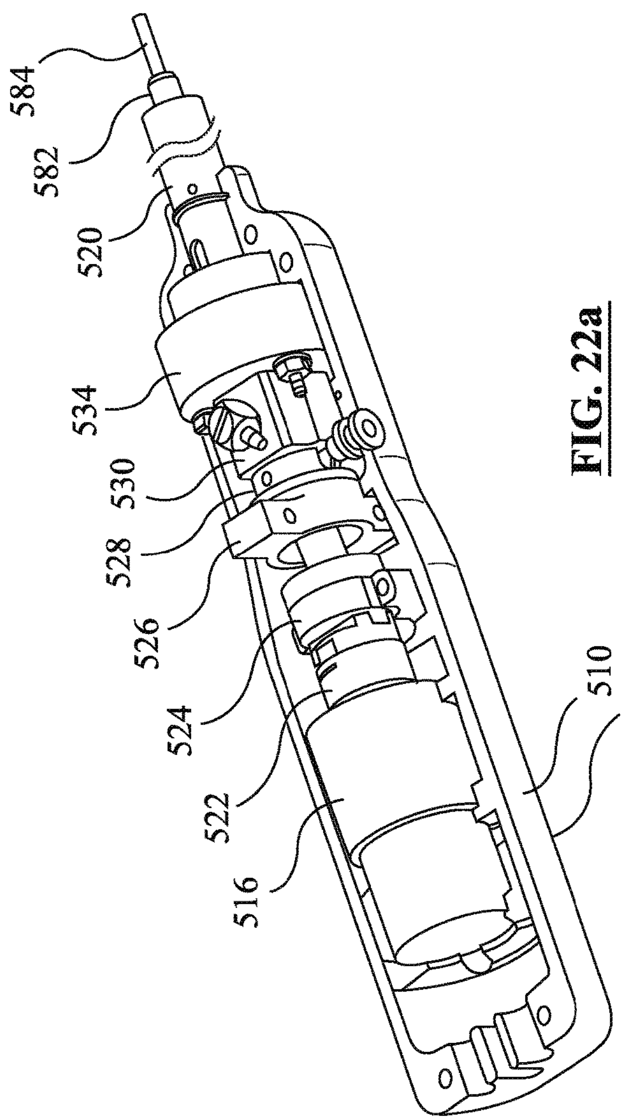


FIG. 22a

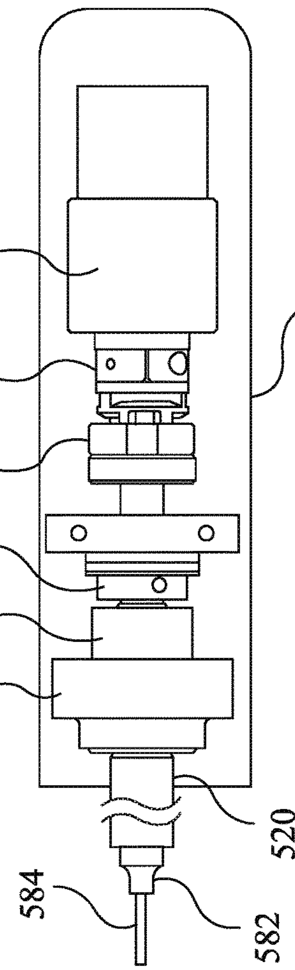


FIG. 22b

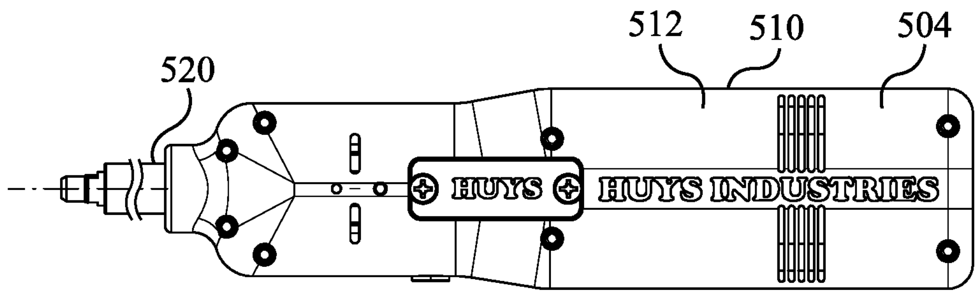


FIG. 22c

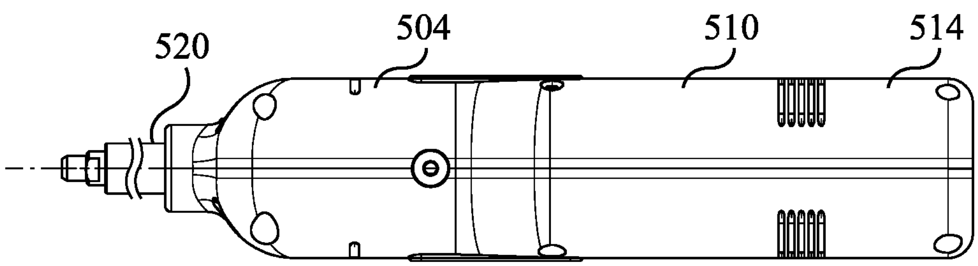


FIG. 22d

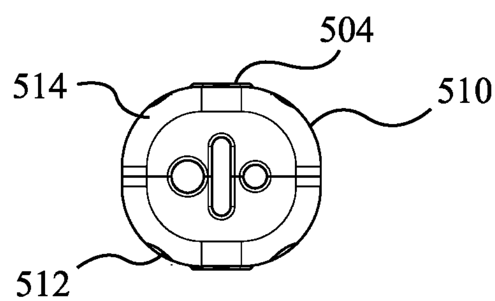


FIG. 22e

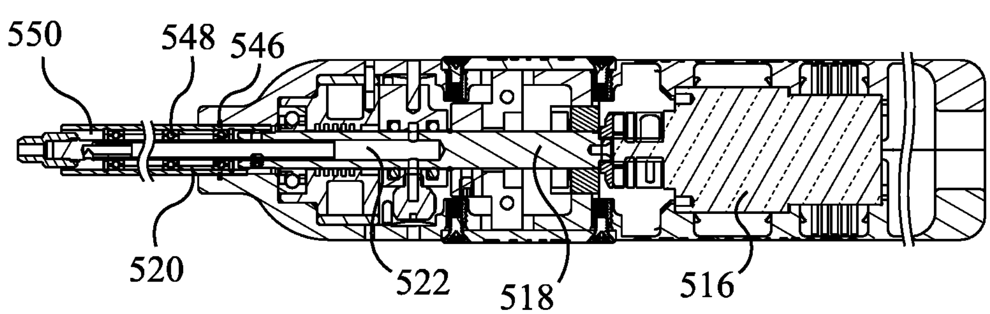


FIG. 22f

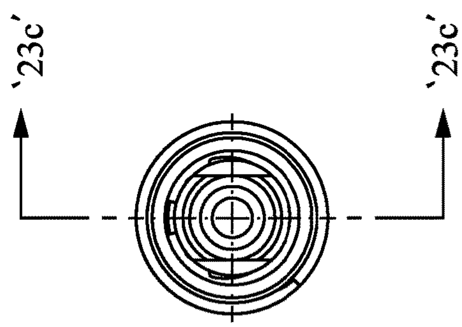


FIG. 23b

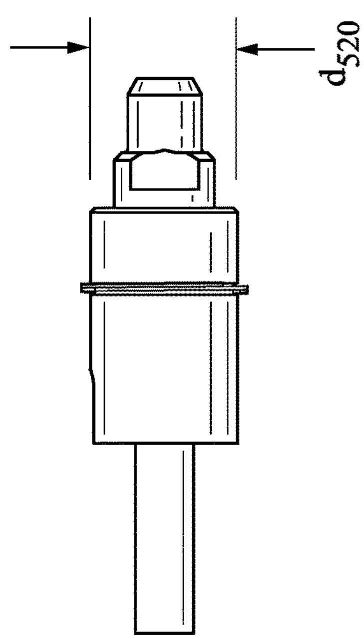


FIG. 23a

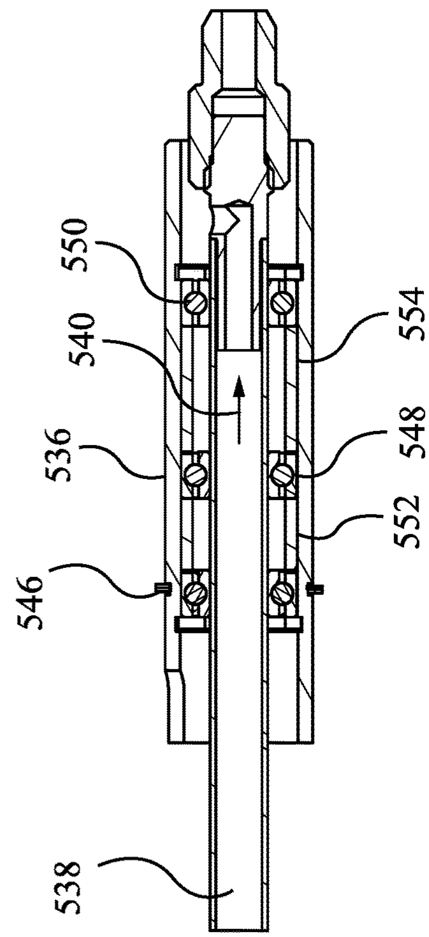


FIG. 23c

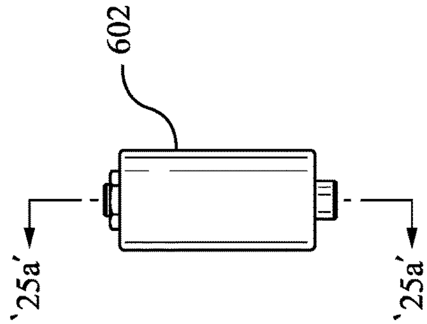


FIG. 24c

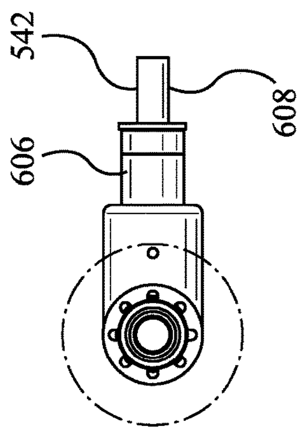


FIG. 24b

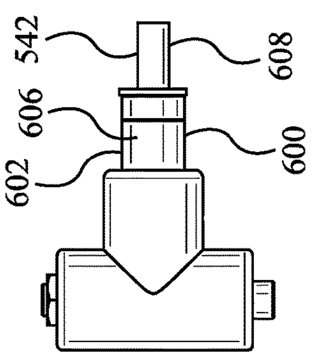


FIG. 24a

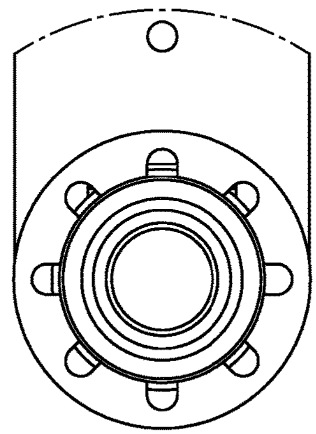


FIG. 25b

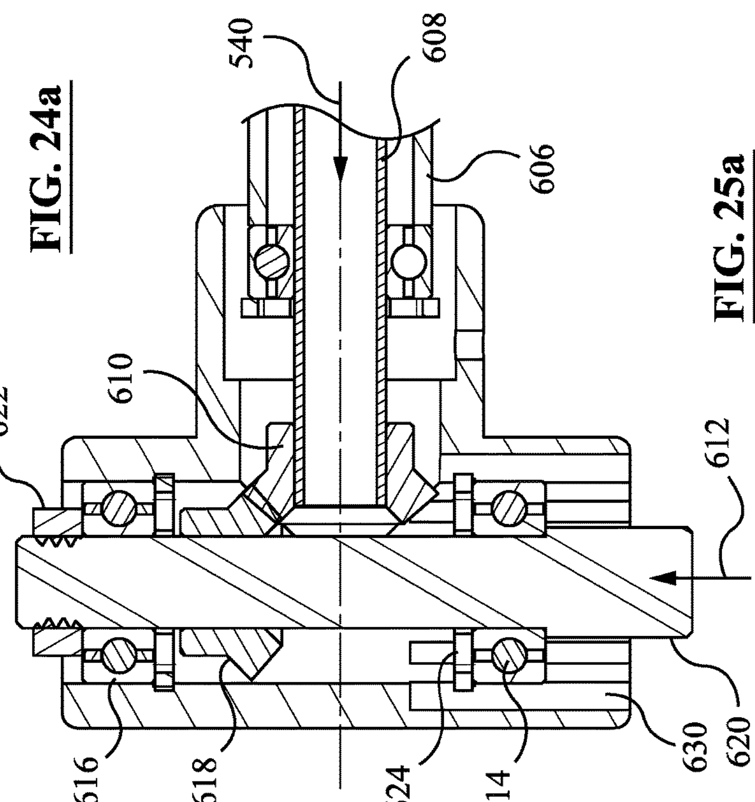


FIG. 25a

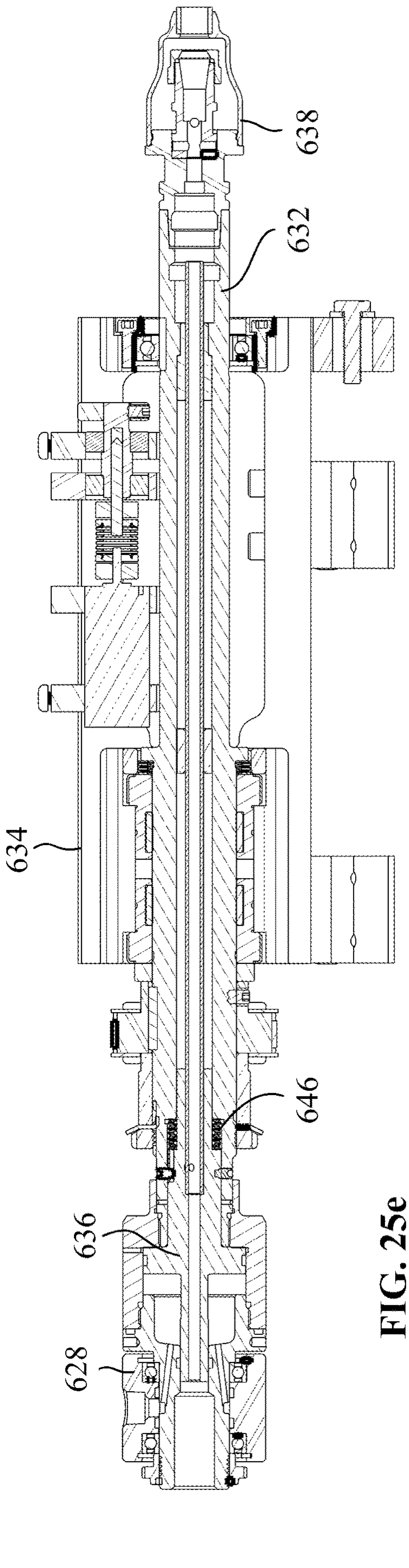


FIG. 25e

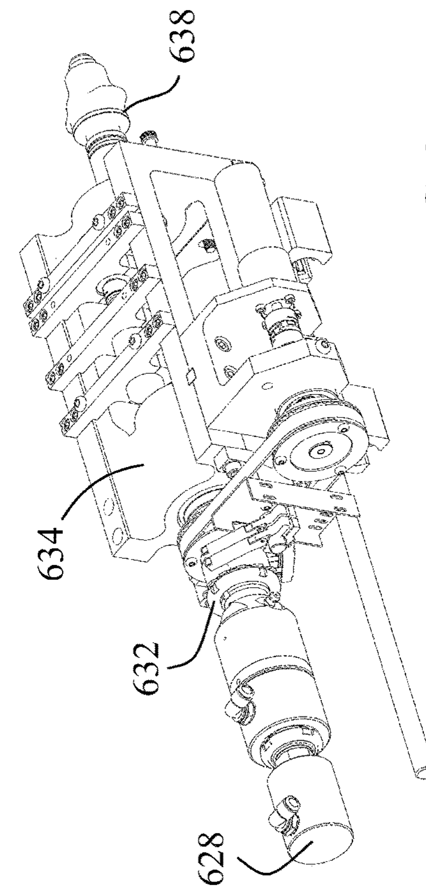


FIG. 25c

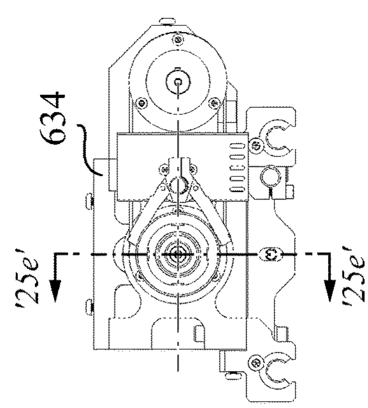


FIG. 25d

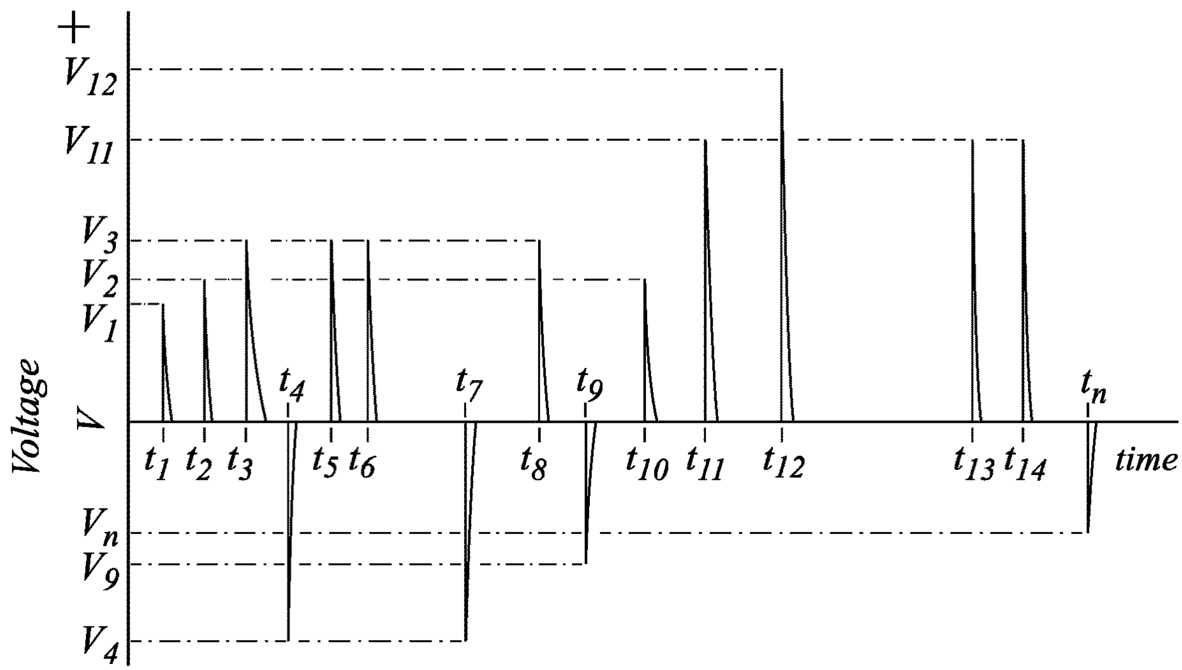


FIG. 26a

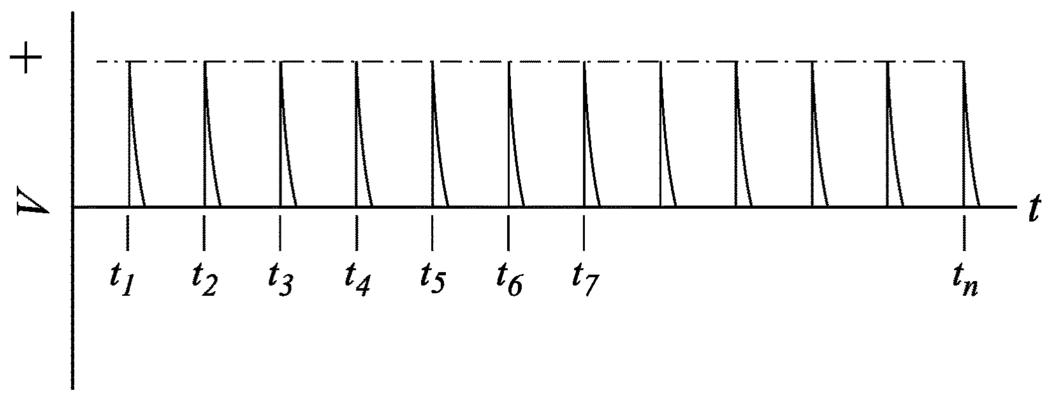


FIG. 26b

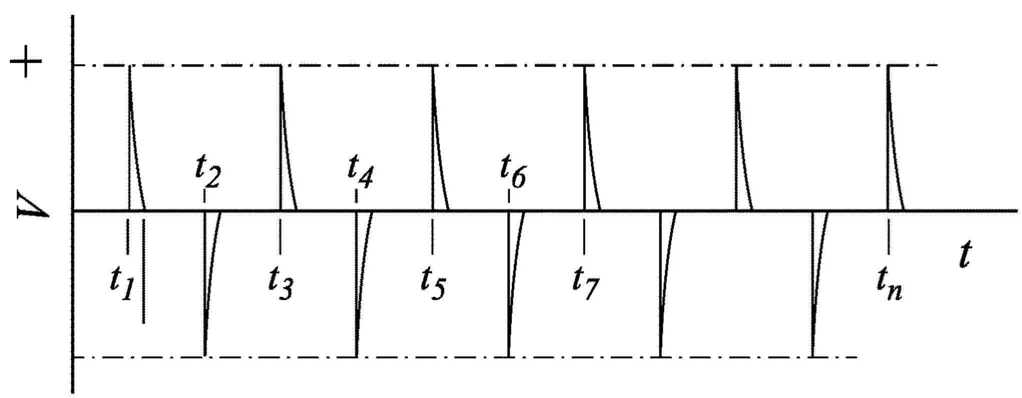


FIG. 26c

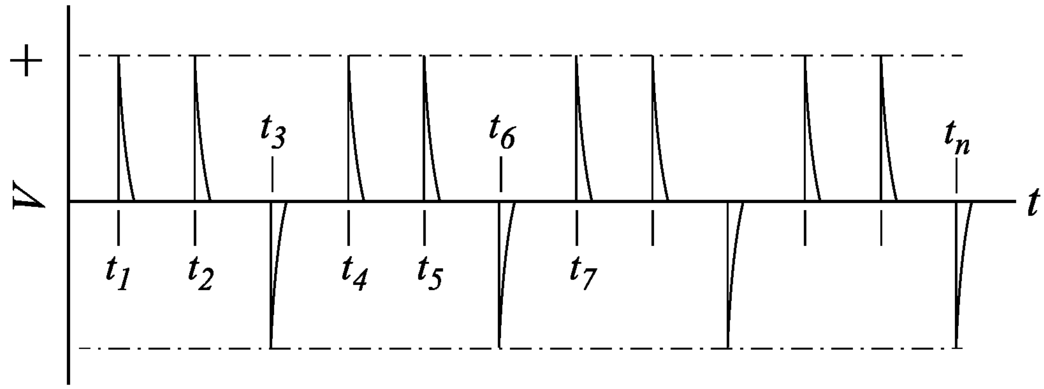


FIG. 26d

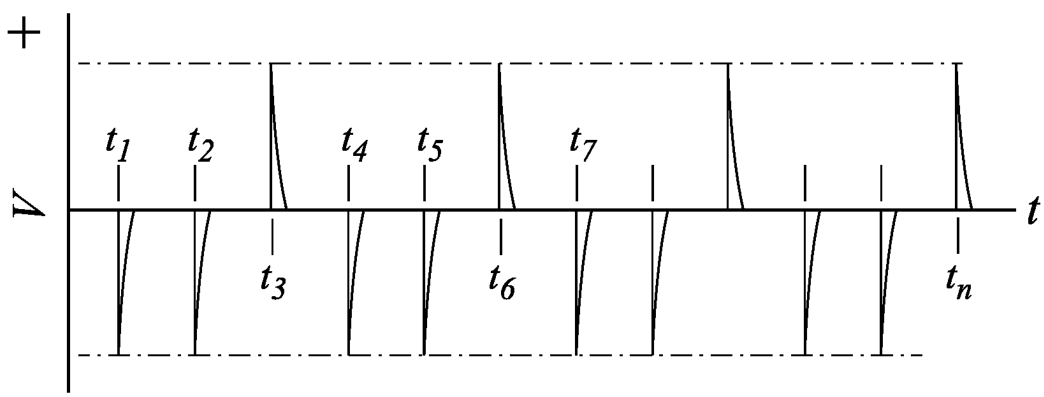


FIG. 26e

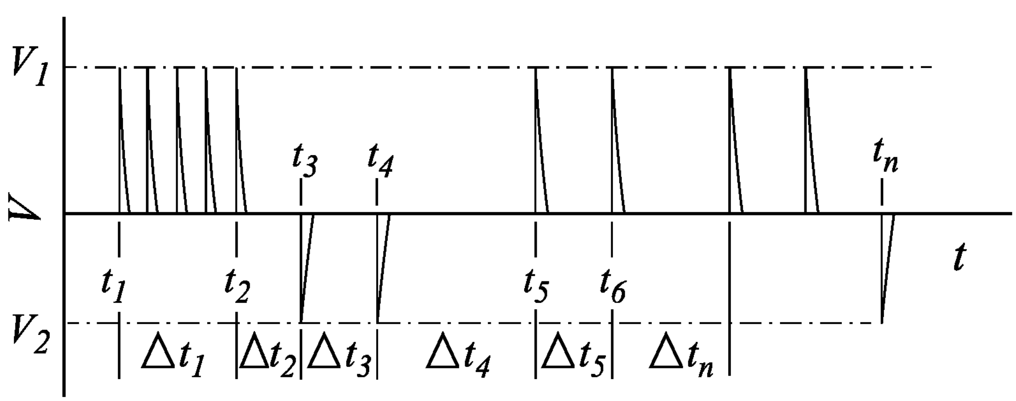


FIG. 26f

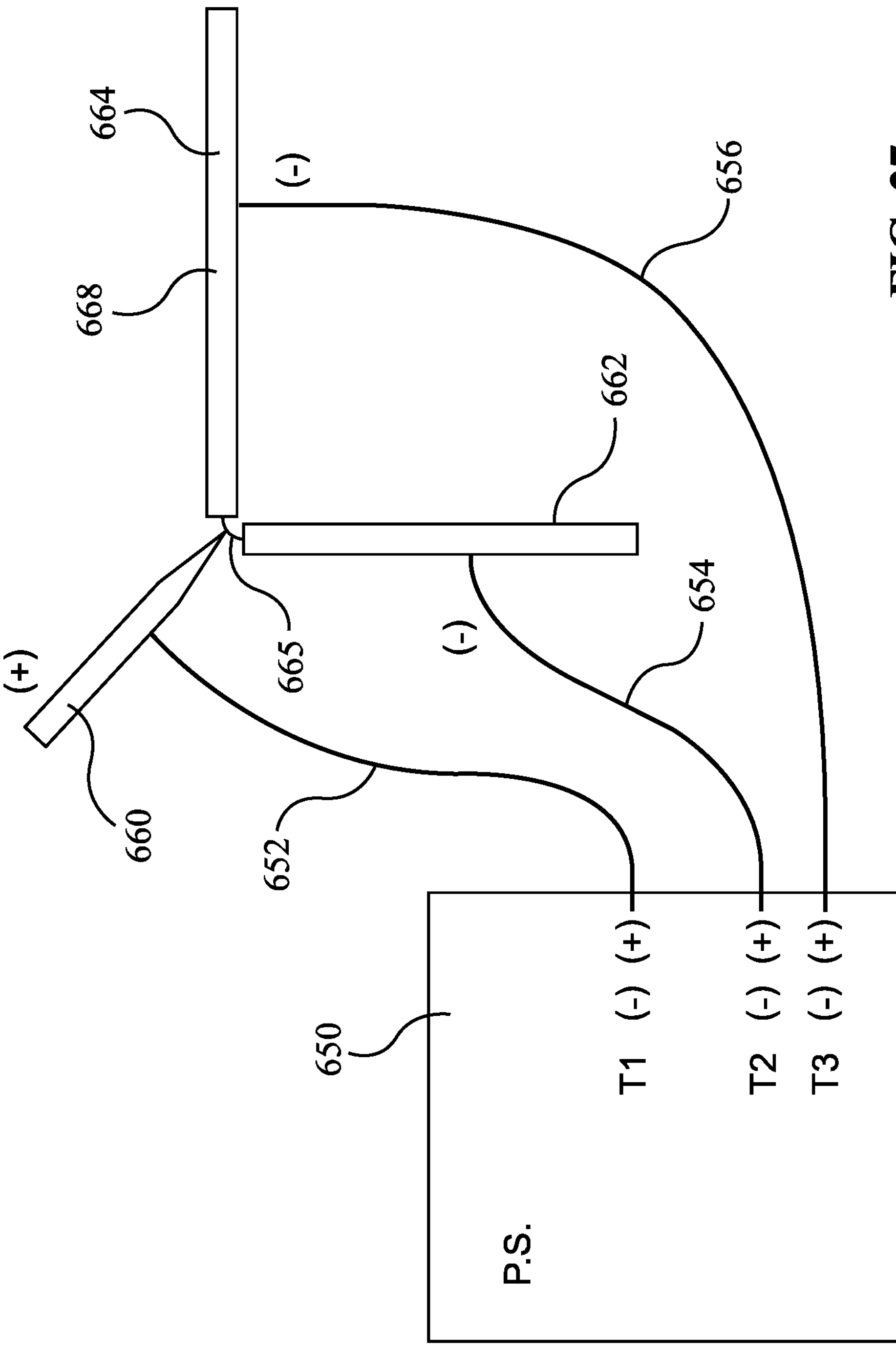


FIG. 27

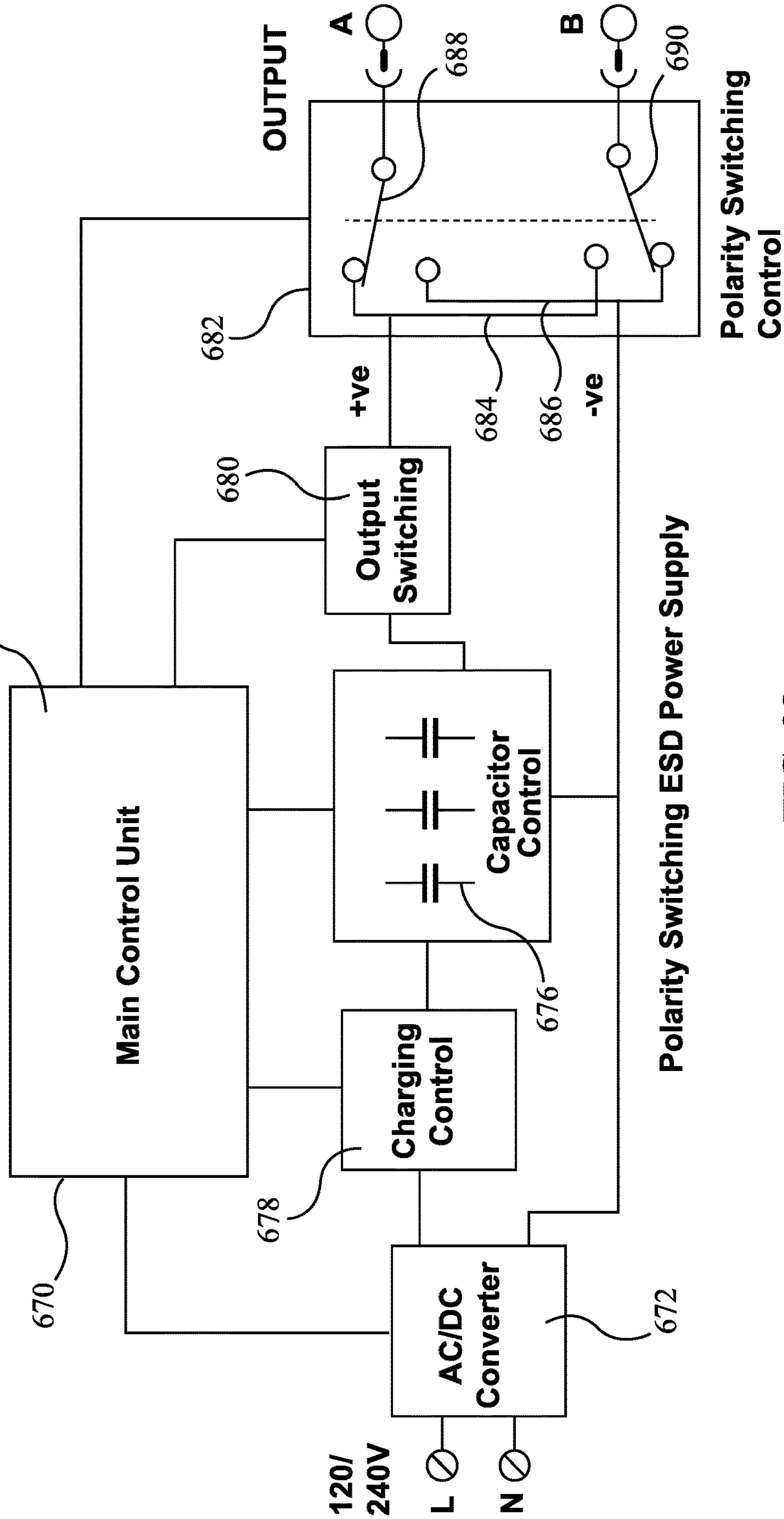

OUTPUT
A
B
688
690
682
Polarity Switching Control
+ve
684
686
-ve
680
Output Switching
674
Main Control Unit
Capacitor Control
676
Charging Control
678
670
AC/DC Converter
672
120/ 240V
L
N
Polarity Switching ESD Power Supply

FIG. 28a

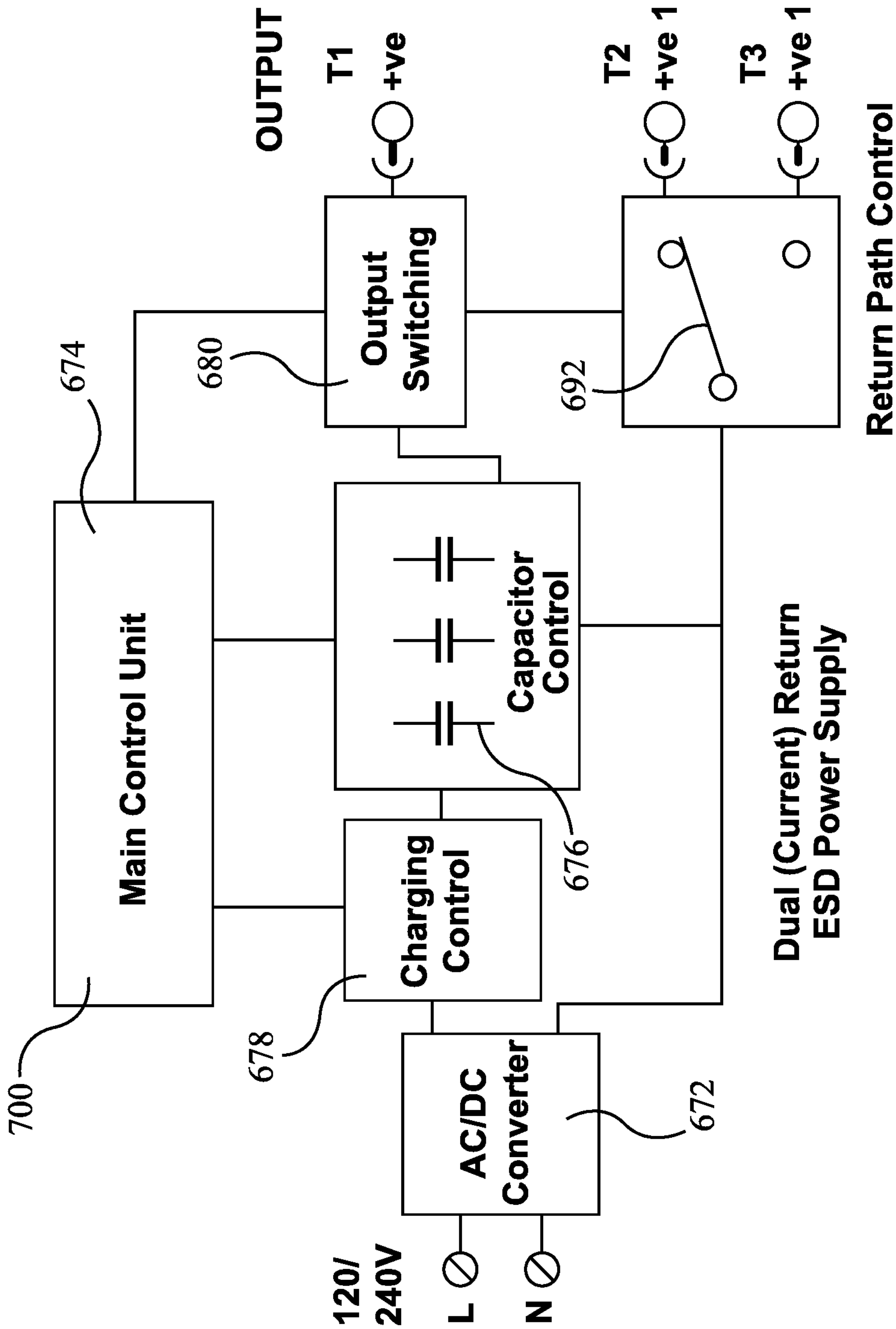
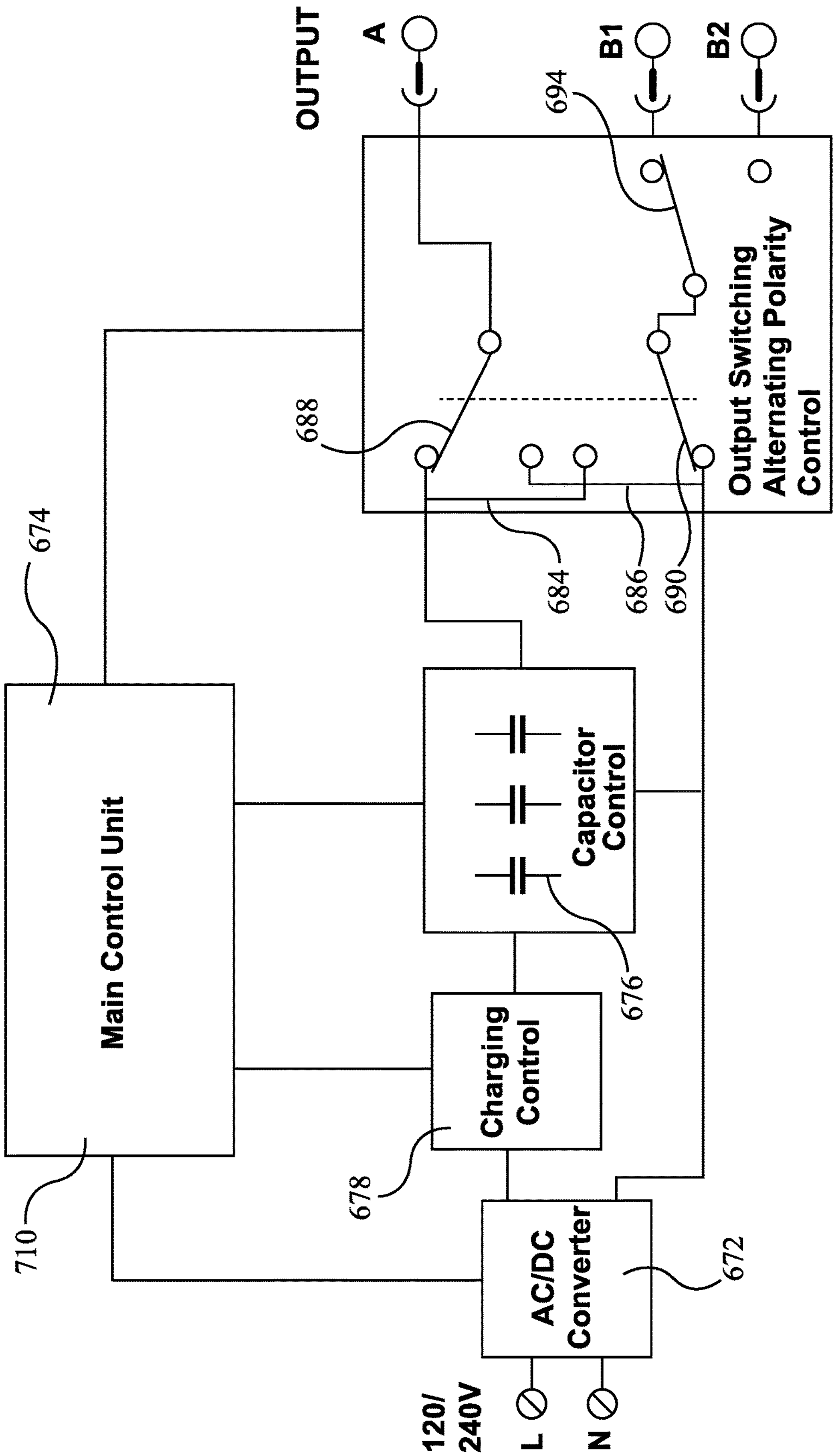

OUTPUT
T1
+ve
T2
+ve 1
T3
+ve 1
Output Switching
680
Return Path Control
692
674
Main Control Unit
700
Capacitor Control
676
Charging Control
678
AC/DC Converter
672
120/ 240V
L
N
Dual (Current) Return ESD Power Supply

FIG. 28b



Dual Return Alternating Polarity ESD Power Supply

FIG. 28c

WELDING APPLICATOR AND METHOD OF APPLICATION

This application claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 62/801,437 filed Feb. 5, 2019, the specification and drawings thereof being incorporated in their entirety herein by reference.

FIELD OF INVENTION

This application relates to tools for welding electrodes, and to a method of welding.

BACKGROUND

In a number of industries it may be helpful to be able to coat a metal substrate having certain desired qualities with a coating of a dissimilar material having other properties. One such application may be for the coating of existing parts by a deposition process. Another application may occur where it is desired to remove or repair defects in the surface of a substrate, whether as a planar surface or as part of a non-planar three-dimensional object. Another application may pertain to welding electrodes for use in a production line for the sequential assembly of parts using a large number of welding stations. Welding electrodes are generally made of copper. The electrode may have a surface coating, such as a ceramic coating, that may be intended to increase electrode life. Other objects, such as steel may be provided with a surface coating, such as nickel or chrome. For example, coatings of vanadium-carbide, tungsten-carbide, titanium-diboride, zirconium-diboride, Titanium-carbide, Cr_3C_2, and so on, might be applied to tool steels or aluminum, or other metals, as may be. Such treatments, or coatings, or repairs may occur where it is desired to change the surface, or near surface properties of an object, such as hardness, or corrosion resistance or other property.

The surface area will be coated with a layer of the electrode material when swept by the electrode. The electrode cap may be mounted to a moving device. The condition of the contact may be dependent on the relative motion of the rod of depositing electrode coating material and the electrode cap to be coated.

Sometimes the surface to be welded or treated or coated, or a defect to be repaired, or materials to be joined, may not be conveniently located. The welding location may have poor access, as on the inside of a bore, tube, or cylinder. Sometimes it may be helpful to provide shielding gas even in such a poorly accessible location. In ESD processes it may also be helpful to cause motion between the electrode at the location of application and the surface of the workpiece to which the welding or coating activity is to be applied.

Further, ESD processes tend to be Low Energy Welding processes, in which the energy of an individual welding discharge may be substantially less than 200 Joules. ESD welding is considered to be, or is grouped as, a Direct Current (DC) process, and the inadvertent application of the incorrect polarity may tend to be quickly evident.

SUMMARY

As described herein, in one aspect, there is a welding rod assembly. It has a seat, a chuck, and a welding rod, all of them being electrically conductive. The seat has a first end, a second end, and a body extending between the first and second ends. The first end of the seat is matable to a

rotational drive. The second end of the seat and the chuck are releasably matingly engageable. The chuck has a socket defined therein in which an end of the welding rod is permanently mounted. The seat has a passageway formed therein, the passageway having an inlet and an outlet. The passageway defines a fluid conduit from the first end of the seat toward the chuck.

In another aspect there is a welding rod assembly. It has a welding rod and a chuck. The welding rod is unremovably mounted in the chuck. The chuck has a releasable end fitting operable to engage a welding electrode holder seat.

In a feature of that aspect, the releasable end fitting is a threaded end fitting. In another feature, the chuck has a socket in which an end of the welding rod is mounted, and the chuck is plastically deformed after insertion of the end of the rod in the socket. In a further feature, the chuck is made of a different material from the welding rod. In a still further feature, there assembly includes the electrode holder seat, and the seat has an end fitting that mates with the end fitting of the chuck. In another feature, the respective end fittings are threaded end fittings. In still another feature, the seat has a shielding gas passageway formed therethrough.

In another aspect there is a welding coating head. It has a conductor and a shroud. The shroud extends along the conductor. The conductor has a first end and a second end. The conductor is a rotor having a torque input fitting at the first end, and a torque output fitting at the second end. The conductor has a fluid passageway defined therewithin. The fluid passageway has an inlet at the first end of the conductor, and an outlet at the second end of the conductor. The conductor is electrically conductive and has an electrical input at the first end of the conductor and an output at the second end of the conductor.

In a feature of that aspect, there is a set of bearings mounted along the shroud within which the conductor is carried, the shroud being a stator relative to the conductor. In another feature, the bearings are non-electrically conductive. In still another feature there is any one or more of (a) the shroud is electrically non-conductive; (b) the shroud has an electrically non-conductive coating; and (c) the shroud is electrically isolated from the conductor. in another feature, the coating head has a welding rod mounted to the second end thereof. In still another feature, the coating head has a welding rod holder seat mounted to the second end thereof. In a further feature, a welding rod holder is mounted to the welding rod holder seat. In still another feature, the welding rod holder is removably mounted to the seat, and the rod holder has a welding rod permanently mounted thereto. In a further feature, a welding rod holder seat is mounted to the second end thereof, and the seat includes a non-coaxial welding rod mounting. In another feature, the welding rod mounting has a transverse welding rod accommodation. In yet another feature, the welding rod mounting includes a gear drive having a transverse shaft output. In another feature, the transverse shaft output includes bevel gearing. In yet another feature, the welding head is mounted to a holding body, the holding body including any one of: (a) a mechanical transmission by which the rotor is driven; (b) an electrical stator mounted to pass electrical current into the rotor; and (c) a fluid union by which fluid is supplied to the rotor. In still another feature, the welding head is mounted to a holding body, and the holding body includes a vibration source. In yet another feature, the vibration source includes at least one of (a) a mechanical oscillator; and (b) an ultrasonic vibration source.

In another aspect, there is a method of surface treatment employing a welding head, the method of surface treatment comprising applying a direct current discharge between a welding electrode and a workpiece to be treated, and subsequently applying a direct current discharge of opposite polarity between the welding electrode and the workpiece to be treated.

In a feature of that aspect, the method includes applying a train of direct current discharges between the welding electrode and the workpiece, and the train of discharges includes both positive polarity discharges and negative polarity discharges. In another feature, there is a train of the discharges, and the number of discharges of positive polarity is different from the number of discharges of negative polarity. In still another feature, there is a train of the discharges, and the train is asymmetric as between positive discharges and negative discharges in at least one of (a) number of discharges; (b) initial voltage of discharges; (c) duration of individual discharges; and (d) total electric charge per pulse of discharge. In yet another feature, the welding head is connected to a power supply having a plurality of capacitor banks, and the method includes charging different capacitors of the capacitor banks to different voltages.

In a further feature, the welding head is connected to a power supply having at least one programmably controlled switch connected to govern discharge from a first capacitor, and the method includes operating the switch to reverse the polarity of output from the first capacitor to the welding head. In again another feature, the welding head is connected to a power supply, the power supply has a programmable controller, and the method includes operating the controller to form asymmetric positive and negative polarity discharges. In another feature, individual discharge pulses of one polarity have greater electrical charge per discharge than do individual discharge pulses of the opposite polarity. In still another feature, a synthetic AC wavetrain of discharge pulses is applied between the welding electrode and the workpiece, the synthetic AC wavetrain being made up of DC discharge pulses of varying polarity.

In yet another feature, the method proceeds while at least one of: (a) a mechanical oscillation source acts on at least one of (i) the welding electrode; (ii) the workpiece; and (c) any fixture to which the workpiece is mounted; and (b) an ultrasonic vibration source acts on (i) the welding electrode; (ii) the workpiece; and (c) any fixture to which the workpiece is mounted. In a further feature, a first vibration source acts on the welding electrode; and a separate, second vibration sources acts on any one of (i) the workpiece; and (ii) any fixture to which the workpiece is mounted. In another feature, the method includes rotating the welding electrode during welding. In still another feature, the method includes any one of (a) rotating the workpiece; and (b) shaking the workpiece, during welding. In a further feature, the method includes peening deposited weld material at times during the method when discharge is not occurring. In another feature, the method includes providing a supply of shielding gas during welding. In still another feature, the method includes initiating discharges at a frequency greater than 10 Hz. In a further feature, the method includes maintaining total energy in individual ones of the discharges to less than 20 Joules. In a still further feature, the method includes maintaining total energy in individual ones of the discharges to less than 2 Joules. In yet another feature the method includes alternating discharge current between at least two workpiece connections. In a still further feature, the method includes building a fillet of weldmetal between two workpiece parts using electro-spark deposition.

In another aspect of the invention there is an electro-spark deposition apparatus. It has a power supply having a main control unit and a welding electrode connected to the power supply. The main control unit includes a synthetic discharge signal generator programmed to produce DC discharges as output electrical signals from the power supply. Each discharge has an initial voltage, a charge, a starting time and a polarity. The synthetic discharge signal generator is programmed to release a plurality of these DC discharges in series. The set of discharges defines a synthetic AC output composed of the series of the DC discharges.

In a feature of that aspect, the apparatus includes an output polarity reversing switch and a main control unit connected to operate the output polarity reversing switch. in another feature, the apparatus includes at least three output terminals, a first of the output terminals is connected to the welding electrode, a second of the output terminals and a third of the output terminals has respective workpiece connections. In another feature, the apparatus includes an output selection switch operable alternately to connect the second and third output terminals, and a main control unit connected to operate the output selection switch. In a further feature, the power supply includes at least first and second banks of capacitors, and the main control unit is connected to govern discharge of the first and second banks of capacitors.

In another aspect, there is a method of electro-spark deposition. It includes providing a welding apparatus including a power supply, a welding electrode connected to the power supply, and a consumable welding electrode rod mounted to the welding electrode, and at least first and second workpiece connections emanating from the power supply. The workpiece connections, in operation, are of opposite polarity to the welding electrode. The method includes providing an electro-spark discharge current to the welding electrode rod while the power supply is switched to elect the first workpiece connection; switching the power supply to elect the second workpiece connection; and providing an electro-spark discharge current to the welding electrode rod while the power supply is switched to elect the second workpiece connection.

In another feature, the method includes repeatedly switching back and forth between the first and second workpiece connections. In a further feature, the method includes reversing polarity as between the welding electrode and the first and second workpiece connections from time to time. In yet another feature, the method includes changing output voltage when polarity is reversed. In still another feature, the method includes applying a first voltage during straight polarity discharge, and applying a second voltage during reverse polarity discharge. The first voltage has a greater magnitude than the second voltage. In an additional feature, the method includes applying straight polarity discharges more often than reverse polarity discharges. In a further additional feature, there is a ratio of straight polarity discharges to reverse polarity discharges, and the ratio is in the range of 2:1 to 10:1. In still another feature the method includes vibrating at least the welding electrode to peen deposited weld metal between discharges. In another feature, the method includes applying ultrasonic vibration to the workpiece.

In still yet another feature, the workpiece includes a first work piece part and a second workpiece part, and the method includes welding the first workpiece part and the second workpiece part together to form the workpiece. In another feature, the first workpiece connection is electrically connected to the first workpiece part, and the second workpiece connection is electrically connected to the second workpiece part. In a further feature, the first workpiece part is made of a first material and the second workpiece part is made of a second material. The first material is different from the second material. In a yet further feature, the method includes providing an electro-spark deposition coating of at least one of the first and second workpiece parts of a first coating alloy, followed by a further coating of a second coating alloy applied to the first coating alloy. In another feature, the method includes building a weldmetal fillet between the first and second workpiece parts. In another feature, the method is undertaken in the presence of a shielding gas.

In another aspect, there is an electro-spark deposition apparatus. It has a power supply, and an electro-spark deposition applicator. The power supply includes a main control unit operable to govern electro-spark discharge signals from the power supply. The power supply has a first output terminal connected to the electro-spark deposition applicator. The power supply has at least a second output terminal and a third output terminal. The second and third output terminals have respective workpiece connections. The first terminal is of opposite polarity to the second and third terminals.

In a feature of that aspect, the control unit is programmed to alternate connection between the second and third output terminals. In another feature, the power supply has a polarity reversing switch, and the control unit is programmed to reverse polarity as between (a) the first output terminal; and (b) the second and third output terminals. In another feature, the power supply includes at least a first capacitor bank and a second capacitor bank. In operation, the first capacitor bank is maintained at a different voltage than is the second capacitor bank. In an additional feature, the first voltage is higher than the second voltage. The first voltage is applied in a first polarity mode, and the second voltage is applied in a second, opposed, polarity mode. In another feature, the apparatus has an input power converter. The input power converter is sensitive to at least voltage amplitude and frequency of input power, and a main control unit is programmed to transform input power received by the input power converter to a DC output. In another feature, the apparatus includes a welding electrode vibrator. In another feature, the apparatus includes a workpiece vibrator.

BRIEF DESCRIPTION OF THE DRAWINGS

These aspects and features can be understood with the aid of the following illustrations of exemplary, and non-limiting, embodiments in which:

FIG. **1** is a perspective view of a welding apparatus for holding a welding electrode;

FIG. **2** shows an exploded perspective view of the apparatus of FIG. **1** with near side exterior shell removed to reveal internal details;

FIG. **3** is a right-hand side view of the apparatus of FIG. **1**;

FIG. **4** is a left-hand side view of the apparatus of FIG. **1** with near-side exterior shell removed to reveal internal details;

FIG. **5** is an end view of the apparatus of FIG. **1**;

FIG. **6** is an exploded perspective view of the forward end of the apparatus of FIG. **1**;

FIG. **7** shows an exploded view of the rotating elements of the apparatus of FIG. **1**;

FIG. **8*a*** shows a first embodiment of rotating shaft for the assembly of FIG. **7**;

FIG. **8*b*** sows a second embodiment of rotating shaft for the assembly of FIG. **7**;

FIG. **9*a*** shows a section of an alternate embodiment of welding apparatus to that of FIG. **4**, with near-side housing shell removed;

FIG. **9*b*** shows an enlarged perspective view of a detail of FIG. **9*a***;

FIG. **10*a*** is a perspective view of a further alternative embodiment of welding apparatus to that of FIG. **9*a***, with near-side half-shell cover removed;

FIG. **10*b*** is a side view of the apparatus of FIG. **10*a***;

FIG. **11*a*** is a perspective view of an alternative embodiment of the welding apparatus of FIG. **10*a***;

FIG. **11*b*** is a side view of the embodiment of FIG. **11*a***;

FIG. **12*a*** is a perspective view with near side half shell cover removed of a still further alternative embodiment to that of FIG. **9*a***; and

FIG. **12*b*** is a side view of the welding apparatus of FIG. **12*a***;

FIG. **13*a*** shows a perspective view of an alternate embodiment of coating apparatus to that of FIG. **1*a*** mounted on a multi-axis robot;

FIG. **13*b*** shows a perspective view of the alternate welding or coating apparatus of FIG. **13*a***, apart from the robot;

FIG. **14*a*** shows a left hand side view of the coating apparatus of FIG. **13*a***;

FIG. **14*b*** shows a front view of the welding or coating apparatus of FIG. **14*a***;

FIG. **14*c*** shows a right and view of the coating apparatus of FIG. **14*a***;

FIG. **14*d*** shows a rear view of the coating apparatus of FIG. **14*a***;

FIG. **14*e*** shows a top view of the coating apparatus of FIG. **14*a***;

FIG. **14*f*** shows a bottom view of the coating apparatus of FIG. **14*a***;

FIG. **15*a*** shows front view of the spindle housing of the apparatus of FIG. **14*a***;

FIG. **15*b*** shows a cross-sectional side view of the spindle housing of FIG. **15*a*** taken on the center line vertical plane of the spindle;

FIG. **16*a*** shows a perspective view from the right front of the coating head of the coating apparatus of FIG. **14*a***, with the coating head in the fully extended or lowered position as when engaging a workpiece;

FIG. **16*b*** shows a perspective view from the right front of the coating head of the coating apparatus of FIG. **14*a*** in the raised or fully retracted position, as when disengaged from the workpiece;

FIG. **16*c*** shows a perspective view of from the left front of the coating head of the coating apparatus of FIG. **14*a***;

FIG. **16*d*** shows a perspective view from the rear right of the coating head for the coating apparatus of FIG. **14*a***;

FIG. **17*a*** shows a perspective view, from the right front, of a base or slide assembly of the coating apparatus of FIG. **14*a***;

FIG. **17*b*** shows a perspective view from the left front of the assembly of FIG. **17*a***;

FIG. **18*a*** shows a tool and carriage of the coating apparatus of FIG. **14*a*** from in front and to the right;

FIG. **18*b*** shows the tool and carriage of FIG. **18*a*** from behind and to the right;

FIG. **19*a*** shows a manual release chuck for the tool of FIG. **18*a***;

FIG. **19*b*** is an exploded view of the chuck of FIG. **19*a***;

FIG. **19*c*** is a side view of the chuck of FIG. **19*a***;

FIG. **19***d* is a cross-sectional view of the chuck of FIG. **19***c* on '**19***d*-**19***d*' of FIG. **19***c* on the vertical centerline plane;

FIG. **20***a* shows a quick release chuck for the tool of FIG. **18***a;*

FIG. **20***b* is an exploded view of the chuck of FIG. **20***a;*

FIG. **20***c* is a side view of the chuck of FIG. **20***a*; and

FIG. **20***d* is a cross-sectional view of the chuck of FIG. **19***c* on '**20***d*-**20***d*' of FIG. **20***c* on the vertical centerline plane;

FIG. **21***a* is a general arrangement perspective representation of another embodiment of welding apparatus;

FIG. **21***b* is a view of the power converter of FIG. **21***a* with cover removed;

FIG. **21***c* is a schematic of a power supply alternative to the converter of FIG. **21***b;*

FIG. **22***a* is a perspective view of a welding electrode handle of the welding apparatus of FIG. **21***a* with one half of its shell housing removed;

FIG. **22***b* is a top view of the apparatus of FIG. **22***a;*

FIG. **22***c* is a front view of the welding electrode handle of the apparatus of FIG. **22***a;*

FIG. **22***d* is a back view of the welding electrode handle of FIG. **22***a;*

FIG. **22***e* is a rear end view of the welding electrode handle of FIG. **22***a;*

FIG. **22***f* is a cross-sectional view on the central longitudinal plane of the welding electrode handle apparatus of FIG. **22***a;*

FIG. **23***a* is a side view of an extension head of the electrode handle of FIG. **22***a;*

FIG. **23***b* is a cross-section of the end extension of FIG. **23***a;*

FIG. **23***c* is an end view of the end extension of FIG. **23***a;*

FIG. **24***a* is a side view of an alternate end fitting for the end extension of FIG. **23***a;*

FIG. **24***b* is a top view of the alternate end fitting of FIG. **24***a;*

FIG. **24***c* is an end view of the alternate end fitting of FIG. **24***a;*

FIG. **25***a* is an enlarged cross-sectional detail of the end fitting of FIG. **24***a* taken on section '**25***a*-**25***a*' of FIG. **24***c*; and

FIG. **25***b* is an end view of the end extension of FIG. **25***a;*

FIG. **25***c* is a perspective view of an apparatus such as that of FIG. **22***b* in a mounting head such as that of FIG. **15***b*, suitable for mounting to an automated carrier;

FIG. **25***d* is an end view of the mounting head of FIG. **25***c;*

FIG. **25***e* is a cross-sectional view comparable to FIG. **15***b* of the apparatus of FIG. **25***c* taken on section '**25***e*-**25***e*' of FIG. **25***d;*

FIG. **26***a* is a chart of Voltage v. time for the welding apparatus of FIG. **21***a;*

FIG. **26***b* is an example of a wave train produced using the apparatus of FIG. **21***a;*

FIG. **26***c* shows a different wave train produced using the apparatus of FIG. **21***a;*

FIG. **26***d* shows an apparatus of unequal positive and negative pulses;

FIG. **26***e* shows a reversed wave train to that of FIG. **26***d;*

FIG. **26***f* shows an alternate programmed wave train to that of FIG. **26***a;*

FIG. **27** shows a schematic of a welding apparatus;

FIG. **28***a* is a schematic of a switching apparatus for the power supply of FIG. **21***a;*

FIG. **28***b* is an alternate switching apparatus to that of FIG. **28***a*; and

FIG. **28***c* is an alternate switching apparatus to that of FIG. **28***a*.

DETAILED DESCRIPTION

The description that follows, and the embodiments described therein, are provided by way of illustration of an example, or examples, of particular embodiments of the principles of the invention. These examples are provided for the purposes of explanation, and not of limitation, of those principles and of the invention. In the description, like parts are marked throughout the specification and the drawings with the same respective reference numerals. The drawings may be understood to be to scale and in proportion unless otherwise noted. The wording used herein is intended to include both singular and plural where such would be understood, and to include synonyms or analogous terminology to the terminology used, and to include equivalents thereof in English or in any language into which this specification many be translated, without being limited to specific words or phrases.

For the purposes of this description, a Cartesian frame of reference may be employed. In such a frame of reference, the long, or largest, dimension of an object may be considered to extend in the direction of the x-axis, being the longitudinal axis and the main axis of rotation. The height of the object is measured in the z-direction, and the lateral distance from the central vertical plane is measured in the y-direction. Unless noted otherwise, the terms "inside" and "outside", "inwardly" and "outwardly", refer to location or orientation inside the housing of the apparatus. In this specification, the commonly used engineering terms "proud", "flush" and "shy" may be used to denote items that, respectively, protrude beyond an adjacent element, are level with an adjacent element, or do not extend as far as an adjacent element, the terms corresponding conceptually to the conditions of "greater than", "equal to" and "less than". Unless otherwise noted, the assembly shown and described herein may tend to be symmetrical, or largely symmetrical, about the vertical longitudinal central plane. In this specification distinction may be made between materials that are thermal conductors and thermal insulators. In general, the thermal conductors may be thought of as materials such as metals, such as steel, stainless steel, copper sheathing, mild steel flashing, whether galvanized or otherwise, or aluminum sheeting or aluminum extrusions, painted or otherwise. The insulators may be thought of as materials such as wood, particle board, oriented strand board, composites, and plastics, whether fiber reinforced or otherwise.

The embodiments illustrated and described are non-limiting examples of principles of the invention. It is possible to make other embodiments that employ the principles of the invention and that fall within the claims. To the extent that the features of those examples are not mutually exclusive of each other, the features of the various embodiments may be mixed-and-matched, i.e., combined, in such manner as may be appropriate, without having to resort to repetitive description of those features in respect of each possible combination or permutation. The invention is not limited to the specific examples or details given by way of illustration herein, but only by a purposive interpretation of the claims to include equivalents under the doctrine of equivalents.

By way of general overview, a welding apparatus, such as may be identified as an electrode handle apparatus, or simply as a electrode handle, is shown in FIG. **1** as **20**. Handle **20** has an electrode holder, indicated generally as **22**, in which an electrode **24** is mounted. Electrode **24** has a cylindrical shape, and is relatively long and thin. Electrode **24** may be a semi-conducting material, such as titanium carbide, titanium di-boride, or such other welding rod material, as may be. The outwardly extending tip of electrode **24** is seen positioned toward an object with which electrode **24** is to interact, i.e., that is to be subject to welding.

Considering again handle **20**, there is housing, or backshell, or haft, or body, generally indicated as **30**. Housing **30** includes first and second portions **32**, **34**, which may be referred to as first and second, or left hand and right hand backshell or housing halves or housing portions. First and second housing portions **32**, **34** are held together by an array of fasteners such as may be in the nature of threaded cap screws **26** spaced thereabout. A gasket **28** may be captured between portions **32**, **34**, and compressed by the tightening of screws **26**. Both backshell halves may have porting in the nature of vents such as inlet vent array **36** and outlet vent array **38**, by which air or other gas coolant may be admitted to, and enabled to depart from, the interior of housing **30**. The backshell halves may be made of an electrically non-conductive, or electrically insulating, material. The girth of housing **30** may be suitable for being grasped or cradled in the hand of an operator. The general proportions of housing **30** are such that it may have a through dimension in the transverse or y-direction of the order of 2 inches.

As assembled, housing **30** may be generally gun-shaped, i.e., it has a main body or barrel, or longitudinal portion **40**, and a predominantly transversely projecting hand grip portion **42**. Portion **42** may have a trigger, or activator, or switch **44**. Handle **20** has a forward end, or nose, **46** from which the welding electrode protrudes or advances in operation, and a rearward end or butt, or tail **48** that extends rearwardly of grip portion **42**. The underside of forward end **46** is somewhat flared or bulbous as at **52**, such that a recess **50** is formed rearwardly thereof. In use, the operator may support apparatus **20** with one hand in recess **50**, thereby cradling apparatus **20**, while the other hand holds grip portion **42** and operates switch **44**. Switch **44** may be a variable speed switch, e.g., in which the maximum speed of rotation of the first drive is adjusted by rotation of the switch about its axis, and the on-off function, and the speed of the motor between zero and maximum selected speed is governed by how far the trigger is depressed axially. The desired speed may be set, and the switch is squeezed the full distance to that speed.

Looking at the inside of apparatus **20**, each half portion **32**, **34** forms a cavity. The halves are molded to define cavities (or two halves of one cavity) in which to receive the controller, circuitry, and rotating elements of handle **20**. There is a first rotating assembly **60**, and a second rotating assembly **62**. Each has a respective axis of rotation, which is, or may be considered to be, or substantially to be, parallel to the long direction of the main body or longitudinal portion **40**. The axis of rotation of assembly **62** is transversely offset from, and may conveniently be parallel to, the axis of rotation of assembly **60**.

Starting with assembly **60**, and proceeding from the rear of the unit to the front, a first drive in the form of a motor **64** is rearmost. The body or housing of motor **64** fits into a pre-formed moulded seat in the backshell or housing, there being a corresponding half-seat in the other housing half. The output shaft of motor **64** extends forwardly to mate with the input of a coupling or clutch **68**. Main drive shaft **70** has a first end that engages the output side of clutch **68**. Clutch **68** is an insulating coupling that electrically isolates motor **64** from shaft **70**. Clutch **68** may also compensate for any misalignment between motor **64** and shaft **70**.

Near end bearing **72** and intermediate bearing **74** are provided to carry main shaft **70**. Near end bearing **72** is located at, or close to, the clutch-connected end of main shaft **70**. Intermediate bearing **74** is located at roughly the half-way, or mid-point, of main shaft **70**, such that a first portion **76** of shaft **70** is carried between bearings **72** and **74**, and a second portion **78** is cantilevered forwardly away from bearing **74**. An electrical power pick-up **80** mounts on shaft **70** near first bearing **72**. Power pick-up ring, **80** may typically be made of copper, and is connected to the welding power cable **82**. In operation, ring **80** is held stationary. Ring **80** is externally accessible through a slot or port covered by external cover plates **84** located on the outside of housing portions **32** and **34** forward of the rearward set of air vent ports. A slip ring **86** is mounted axially forward of pick-up ring **80**. Slip ring **86** may be a carbon-lined slip ring. It bears against main shaft **70** and against pick-up ring **80**. In operation it carries electrical current from pick-up ring **80** into shaft **70**. The distal, of most forward end of shaft **70** is enlarged at a forward shoulder into a head **88** that has a mating threaded chuck. Chuck **90** and head **88** co-operate to define a seat or accommodation for the inward end of welding rod **24**, chuck **90** being releasable to permit replacement of rod **24**, and may be tightened to secure rod **24** in place, or to adjust the protruding length of rod **24**.

As noted, a second drive in the form of a second rotating assembly **62** is transversely offset from first rotating assembly **60**. It includes a motor **100**, and isolation coupling **102**, a second shaft **104** (i.e., the first shaft being main shaft **70**); a pair of near and far bearings **106**, **108**; an impeller **110**; an eccentric weight **112** and an eccentric weight retainer **114**. Motor **100** fits in a molded seat in the housing shell. In this instance, it is nested just forward of motor **64**, above coupling **68**. Shaft **104** is carried in bearings **106**, **108**, with coupling **102** being located between bearing **106** and motor **100**. Shaft **104** is keyed or splined forwardly of bearing **106**, for mating with corresponding key or spline fitting of the hub of impeller **110**, to be able to impart torque to impeller **110**, and thereby to drive impeller **110** to force air to flow through the inlet and outlet vents or ports to permit cooling of the elements inside the housing. Impeller **110** is held axially in place by a transverse roll pin or grub screw. Bearing **108** is spaced axially forward of impeller **110**, and is located between a pair of circlips.

The second drive is an oscillator used to provide a vibrational forcing function to apparatus. To that end, an imbalance, in the form of eccentric weight **112** is mounted forward of bearing **108**, and is held in place by removable retainer **114**. Eccentric weight **112** may therefore be removed and replaced or adjusted to provide a different imbalance, as may suit. Although a rotational eccentric weight is shown, a linearly reciprocating element could also be used. It does not matter that shaft **100** be precisely parallel to shaft **70**, although it is convenient for making a compact design. Shaft **100** merely needs to be such that rotation of eccentric weight **112** results in a forcing function having a varying component of force in a transverse direction relative to shaft **64**, such as to cause oscillation therein (and, ultimately, at the tip of welding rod **24**).

At the front end or nose of apparatus **20** there is a closure member or closure plate **118**, that permits access to the inside of handle **20** without having to take the two backshell halves apart. This access permits adjustment of eccentric weight **112**. Plate **118** may also have a gas manifold fitting or conduit **120** which is connected to the inert gas supply line **122** on the inside, and which delivers that gas forwardly at the forward facing outlet or shielding gas port. Cover plate

118 also provides the seat for a guide bearing and gas seal **124** that seat about the outside of the radially outwardly facing external shoulder of head **88**.

A gas shield **126** is mounted on the outside or forward face of cover plate **118**. Gas shield **126** has a broad or somewhat bulbous of bell-shaped cowling **128** that has a large end that mounts about gas seal **124**, and a smaller forward end that carries a tip member **130**. Tip member **130** has the form of a ceramic tube such as may be suitable for exposure to high temperature materials, e.g., splatter from ESD welding. When gas shield **126** is in place, shielding gas conveyed by line **122** is carried through plate **118** and discharged into the shielded chamber or duct, or passage-way, or curtain, defined within cowling **128** and tip member **130**, thereby to bathe the electrode in inert shielding gas.

At the connected end, housing **30** has three input connections, the third input being an inert gas supply line **122**. The first input is an electrode power connection, which may be an AC or DC power connection, indicated as **132**, and which may, ultimately, be connected to an ESD power source—the same power source of which the opposite pole is connected to the work piece upon which electrode **24** is to be applied. The power source may be indicated generically as a power supply, discussed below. The second input is a motor power source **134** for operation of electric motors **64** and **100** within housing **30**, in the form of a power cable which may be 120V AC 60 Hz, or 220 V AC 50 Hz, or a 12V DC source, or such other source as may be, and could be a pneumatic source. In the embodiment shown, it may be a 24 V DC source, and motor **64** may be a 24 V DC Pittman variable speed motor having forward and reverse directions controllable by the On-Off switch. Motor **64** may be termed a low speed brush DC motor. It may have an operating speed range of 0 to 1800 r.p.m. In one application it has a rotational speed of about 300 r.p.m., which would generally be considered a relatively slow speed. Motor **100** may be an high speed DC, brushless DC or variable frequency AC, variable speed servo motor. Ability to adjust speed and imbalance of the eccentric may permit the operator to choose a setting suited to the materials to be welded and to be applied. It may have an operating speed range up to 3000 r.p.m., and may typically be run in the range of 500 to 1500 r.p.m. About 1000 r.p.m. is a speed that has been used.

FIGS. **8***a* and **8***b* show alternate embodiments of main shaft **70**. In FIG. **8***a*, main shaft **70** is substantially cylindrical from coupling **68** to the rearward face of the shoulder of head **88**. By contrast, in the alternate embodiment of FIG. **8***b*, main shaft **140** is tapered to narrow in section longitudinally forward of the mid-pint bearing, thus making shaft **140** somewhat less stiff than shaft **70**.

The main power cable, namely that of electrode power connection **132**, is secured at a terminal lug inside housing **30**. The shielding gas conduit **122** may have the form of a hollow pipe that is formed to run along the inside proximal margin of housing **30**. Coolant conduit **122** may be used to conduct an inert gas, such as argon, to electrode rod **24**, and may be used for the alternate purpose of providing an inert gas shielding to the coating process. Conduit **122** may be made of a non-electrically conductive material such as a plastic tube. That portion of conduit **122** lying within housing **30** may be made of a metal, such as copper, aluminum, stainless steel, mild steel, or such other metal as may be suitable.

As noted, also enclosed within housing **30** is a vibration assembly, or an oscillator, or shaker, or motion exciter, namely assembly **62**. The resultant vibration has an amplitude having a component in the transverse direction of electrode rod **24**. In use, an operator grasps housing **32**, and uses electrode **24** much like a pencil to paint or coat the workpiece object—provided one is accustomed to painting or writing while the pencil is oscillating. Of course, the pencil has two degrees of motion imposed upon it, namely rotation about the axis of the rod (by motor **64**) and transverse deflection (by motor **100** and eccentric weight **118**). The rotation of eccentric **118** causes apparatus **20** to vibrate, which, in turn, causes electrode **24** rapidly and repeatedly to make and break contact with the work piece. With each oscillation there is a new spark and deposition of the material of electrode rod **24** onto the workpiece.

Vibration assembly **62** provides a forcing function input to the drive transmission of rod **26**, namely assembly **100**, transmitting an input impulse, or wave-train of impulses. The force and displacement transmissibility of that transmission of the mechanical motion of the forcing function input to electrode **24** is dependent upon the natural frequency of the vibrational degree of freedom of interest. Although the axis of the cylindrical rod of electrode **24** is shown as being parallel to the long axis of apparatus **20**, this need not necessarily be so. In another embodiment, electrode **24** may have the form of a rod having an axis at least partially transverse to the main body of housing **30**.

The apparatus shown and described herein may be employed for processes that may be termed "Low Energy Welding". That is, where there may be 1 kJ of energy used in the heating of a resistance spot weld, in an intermittent electrical discharge weld the amount of energy used in heating at each contact of the electrode to the workpiece may be of the order of 1 Joule. The heating has very short time duration, is highly localised, and results in the deposition of only a very small amount of material. While the welding is true welding in terms of the fusion of materials through melting, the small energy input may tend to reduce or substantially eliminate any heat affected zone.

The handle apparatus drives the consumable electrode **24** to vibrate in a first degree of freedom of motion relative to the metallic surface being coated or treated in the process. The force or displacement is generated by attaching an eccentric circular metal load to a spinning motor. The positioning of the eccentric weight determines the pounding or contact force when the contacts are made. The frequency of vibration is controlled with the speed of the motor to which the eccentric weight is mounted. The longitudinal movement of the consumable electrode in a direction that includes a component of motion, and usually a predominant component of motion, normal to the surface to be treated, allows the periodic contacts to be made with the metallic surface of the workpiece. This occurs while that workpiece surface is being driven in a second degree of freedom of motion. The combination of motions, and the vibration-driven urge to make and break contact, may result in a relatively stable or consistent sequence of electro-sparks (when the contacts open) and depositions of coating material (when the open contacts approach) that take place in the process. The vibrating motion is, or may include, motion normal to the surface being coated.

It is known to use an ultrasonic horn to impose vibration on a welding rod. Here, however, an apparatus is provided that may provide secondary vibration (i.e., the first or primary motion forcing function is rotation about the axis, caused by motor **64**; the second forcing function is due to motor **100** and eccentric **118**), without using an ultrasonic vibration source. A variable speed motor and weight combination may tend to be a relatively low cost and robust approach to this issue.

In FIGS. **9***a* and **9***b* there is a welding apparatus **150** that is substantially similar to that of FIG. **4**. However, in addition to the features of apparatus **20**, welding apparatus **150** has a first drive **152** that includes a speed reducer, in this case a gear reducer, **154**, mounted at the front end of motor **156** to drop down the output rotational speed. Further, apparatus **150** includes a feedback control apparatus **158** that includes a digital encoder **160** mounted to observe shaft speed. Encoder **160** is mounted between coupling **162** and near end bearing **72**. In this embodiment, it may be noted that the inert gas delivery cowling **164** gives onto an output duct or tube **166** that has a mitred end **168** rather than a square cut end.

When the welding rod is depositing material on the work surface, the material at the tip is heated to a molten state. The deposited material cools rapidly. There may be a tendency for the tip of the rod to stick and jump. This may tend also to cause variation in the rotational speed of the welding rod between substantially instantaneous high load no-load conditions. The use of a feedback function may permit the variation of rotational speed to tend to be reduced. The clock pulse frequency of the digital sensor and encoder may be very where the frequency of events it is polling is much lower, i.e., of the order of Hz. The frequency of rotation of the rod may be of the order of 300 rpm, i.e., 5 Hz.

In another feature, liquid cooling is provided to electrode holder **22**, in the form of a liquid cooling jacket **170**, supplied from supply and return lines **172**, **174** attached at corresponding supply and return fittings **176**. The provision of a liquid coolant system in this way may permit more consistent control of electrode rod temperature during operation.

In this embodiment, apparatus **150** may also include an impeller **110**. The housing has a shroud **178** to discourage output air from impeller **110** from recirculating back into the input, such that flow may start by being drawn in at the vents or ports in the housing on one side of the impeller and be pushed out the vents or ports on the other side.

Some embodiments may not have all of the features of FIGS. **9***a* and **9***b*. By way of example, in FIGS. **10***a* and **10***b*, welding apparatus **180** is substantially the same as, or similar to, welding apparatus **150**, except that it does not employ a liquid cooling manifold, or liquid cooling jacket, and the motor **184** of the first drive **182** does not have a gear reducer. It does, however, have an output shaft speed feedback control system that includes a digital encoder as at **186** that is used to govern the variable speed motor **184**.

Similarly, In FIGS. **11***a* and **11***b* there is a simpler embodiment of welding apparatus **200** that has a first drive **202** including a variable speed motor **204**, but does not have a second drive, i.e., there is no rotating oscillator. Apparatus **200** does, however, have an output feedback control **206** that includes digital encoder **208** that is used to maintain a relatively smooth output speed of rotation of the welding rod. In the further alternate embodiment of FIGS. **12***a* and **12***b*, there is a welding apparatus **220** that has a first drive **222** that has a liquid cooling jacket **224** to cool the electrode holder, as described above. The embodiments of FIGS. **10***a* and **10***b*, **11***a* and **11***b*, and **12***a* and **12***b* are intended to show that the various features may be employed alone or in combination.

The first drive may have an AC or a DC motor. In one embodiment it may be a 12 VDC Pittman Brush Motor with a no-load speed of 6500 rpm and a low speed of 500 rpm. Another embodiment of this motor may have a no-load speed of 6200 rpm, and a low speed of 300 rpm. It may be a brushless DC motor. It may be a 12 VDC servo motor with a speed range of 200-827 rpm; it may be a 24 VDC servo motor with a speed range of 500-1481 rpm. In still another embodiment, the first drive motor may be a DC servo motor having a no load speed of 7200 rpm and a low speed of 2200 rpm. That motor may have a gear reducer on the output side. As the various motors are, or may be, of different sizes, adapters **190**, **192** may be used at one or both of the front and rear end to fit the motor into the mounting cavity defined in the molded housing, thereby permitting any of the motor embodiments to be used depending on, e.g., cost and availability. There may be a feed-back control system that includes a digital sensor, or digital encoder, mounted to observe output shaft speed. Inferentially, the monitoring of motor current, which may also be controlled, may typically also be a measure of output shaft torque. Output shaft torque may tend to fluctuate, e.g., when rod contact with the workpiece is broken, or when the rod starts to stick. The second drive, i.e., of the oscillator, may also be either an AC drive or a DC drive.

In the embodiments of FIGS. **13***a* through **20***d*, there is an automated low energy welding (LEW) applicator unit identified as a welding or coating apparatus **300**. It interfaces with, i.e., mounts on and operates in cooperation with, a robot **250** and its power supply. The apparatus automatically applies LEW coatings and repairs and welds as programmed. That is, the robot is programmed to present the coating apparatus to a workpiece according to a pre-programmed path, where the path may follow the surface of the workpiece, whether that surface is flat or has a curvature. It may follow a particular path, as where the operation is to lay down a particular configuration of welding material, whether to follow a crack or defect in making a repair, or in building up a low energy weld of several passes, and so on.

Robot **250** is a multiple degree of freedom robot, such as may be purchased commercially, as, for example, from ABB. In the example, the robot has a base **252** such as may be mounted to stationary structure (e.g., a concrete floor, or other suitable pedestal), as may be. The workpiece is then positioned in a known location relative to robot **250**. Robot **250** may include a laser sensing system to establish the relative location of the workpiece. The workpiece may itself be mounted on a pedestal or bed and may be stationary. Alternatively, the workpiece may be moving, as along an assembly line, whether continuously or intermittently, according to either a pre-determined path, or according to a path that can be sensed by the robot such that the relative position, orientation, and motion of the workpiece are known in the sense of the robot having the ability to correlate the path of tool operation to the workpiece. For ease of explanation, the workpiece is stationary during welding or coating, unless otherwise noted.

Robot **250** has a first degree of freedom, namely freedom of rotation about the vertical axis between the robot first member, or shoulder **254** and base **252**. Robot **250** has a second degree of freedom of motion, namely angular rotation of upper arm **256** about the shoulder joint **258**. Robot **250** has a third degree of freedom of motion, namely angular rotation of the forearm **260** relative to upper arm **256** about elbow joint **262**. Elbow joint **262** may have an axis parallel to the pivot axis of shoulder joint **258**. As such, motion of upper arm **256** and forearm **260** can place wrist **270** in a wide selection of positions in the radially extending plane perpendicular to the shoulder joint and elbow joint axes. Forearm **260** may have a further rotational degree of freedom about its long axis. Wrist **270** has a hand, or finger, or knuckle that has a further rotational degree of freedom about the pivot axis of wrist joint **272**. Apparatus **300** has a mounting interface, or base, or plate **304** that mounts to a knuckle of robot **250**, the knuckle being carried on a spindle extending from wrist **270**. That knuckle may also have a rotational degree of freedom about its own longitudinal spindle axis such that plate **304** may be rotated. Plate **304** itself is mounted to a resilient intermediate member, or resilient suspension member **306**. These multiple degrees of freedom permit the welding head, i.e., apparatus **300**, to be oriented in a wide variety of locations and orientations to engage the workpiece.

Apparatus **300** has a head or carriage **310** that mounts on a base, or slide, or slides **320**. The base, or slide, or slides **320** are mounted to the resilient intermediate member, or suspension member, **306**. Resilient suspension member **306** may include springs or springs and dampers. In one embodiment it may have the form of a substantially stiff polymeric plate, which may be made of a Nylon™ or UHMW polymer. In other embodiments member **306** may be made of a metal, such as steel, and may be electrically insulated from the body of robot **250**. While apparently rigid, such a plate is not rigid in the manner of a thick steel plate, and, being polymeric, has much higher anelasticity. It is, by the nature of the material, a damper or moderator of high frequency vibration. The plate is substantially rectangular, or, rather, has a substantially rectangular mounting fitting connection footprint at which the slide or slides, or base **320** is attached, e.g., on the front side of resilient support member **306**. Resilient support member **306** also has a fastener footprint corresponding to the fastener footprint of mounting face plate **304**, which, in this case, may be a circular mounting place of a diameter falling within the rectangular shape of resilient suspension member **306**. As shown plate **304** has a diameter that is about half the width of the generally rectangular, or four-cornered shape of resilient suspension member **306**. The fastener fittings at the corners of the plate, identified at **308** may be provided with resilient bushings or gaskets. Suspension member **306** may be non-electrically conductive. Suspension member **306** may tend to attenuate relatively high frequency vibration, such that vibration in apparatus **300** may tend to be isolated to some extent from robot **250**. Apparatus **300** is connected through resilient suspension member **306** and mounting plate **304**. When mounting plate **304** is installed on the spindle emanating from wrist **370**, slides **320** travel with, and have their position, motion, and orientation dictated by the position of wrist **370**.

Carriage **310** is mounted on guideways, slides, or rails **312**, **314** so that it has a degree of freedom of motion, in this instance in linear translation, along those guideways. In the orientation shown in FIG. **13***a*, this would yield motion in vertical translation. Clearly, the direction of motion, whether vertical, horizontal, lateral, or some combination of components thereof, will be dictated by the orientation of wrist **270**, and is variable according to the programming of robot **250**. Apparatus **300** includes a drive **316** that causes carriage **310** to move along rails **312**, **314**. Drive **316** may be a screw drive, e.g., an Acme screw, or it may be a pneumatic or hydraulic drive. In this instance, a pneumatic drive may be convenient. There is also a brake, **318**, which, in the embodiment illustrated is an air brake. It could be a magnetic or friction brake. Axial-direction drive **316** mounted is midway between rails **314**, **314**, such that all three of items **312**, **314** and **316** are roughly co-planar. While this need not necessarily be so, the use of a co-planar, or approximately co-planar mounting may tend to reduce or avoid secondary eccentric, out-of-plane forces that may not aid in the smooth operation of apparatus **300**. A symmetrical co-planar mounting may tend to be simple, and convenient. There is also an adjustable over-travel limit abutment, **308** rigidly mounted to carriage **310**. The adjustment may have the form of a lag bolt, or threaded rod that is adjusted to set a limit on upward travel of carriage **310** relative to the base or slide or slides.

Apparatus **300** has a spindle assembly **330**. In the most general case, spindle assembly may have a tool engagement interface **322** to which a variety of tools may be attached—drills, arbors, mills, and so on—in addition to the welding or coating equipment described herein. Too engagement interface **322** may include a socket for receiving the shank of the tool. Spindle assembly **330** is mounted in main bearings **324**, **326**, and a forward pilot bearing **328**. Main bearings **324**, **326** may be angular contact bearings. Pilot bearing **328** may have clearance to accommodate vibration as induced in the apparatus. A timing gear, or pulley, or sheave **332** is mounted to drive the spindle shaft **334** which extends longitudinally through the various bearings. Shaft **334** is electrically conductive. Slip rings **336** are mounted to the end of shaft **334** and receive power from brushes **338** (which are connected to a power supply), to the power input to shaft **334**. Dampers, in the form of damper springs **340**, **342** may be mounted axially outside, i.e., bracketing, main bearings **324**, **326**. A main spindle drive **344** is mounted to carriage **310** and includes a motor **346** and transmission **348**. The motor can be a servo motor as described in the embodiments above. In this instance, transmission **348** has the form of a timing belt. It is driven by the output pinion or pulley or sheave **350** of motor **346**, and carries motion to the input gear, or pulley, or sheave **332** of spindle shaft **334**. The timing belt is non-electrically conductive. As may be understood, motor **346** operates at a given motor speed, and is comparable to any of the main drives described above in terms of speed of rotation. As before, the motor speed may be digitally controlled, as by use of an encoder **352** and associated feedback loop. This arrangement may be substantially the same as the encoder examples above.

Carriage **310** has first and second laterally extending structural members or frames, such as identified as a head frame **280**, a tail frame **282**, and left and right hand side frames **284**, **286**. Head frame **280** is substantially or predominantly cylindrical, and has an axial extent greater than the axial spacing of main bearings **324**, **326**. The main bearings **324**, **326** of spindle shaft **334** of spindle assembly **330** are mounted in headframe **280**. Pilot bearing **328** is likewise mounted in tail frame **282**. Side frames **284**, **286** form a pair of spaced apart axially extending beams whose lateral spacing and orientation is governed by head frame **280** and tail frame **282**. Similarly, side frames **284**, **286** provide the structural rigidity and define the stiffness of carriage **310** in respect of maintaining the spacing and orientation of headframe **280** and tail frame **282** relative to each other. Upper and lower slide followers **294**, **296** are mounted to the near margins of side frames **284**, **286** to engage the slide rails, **312**, **314**. The axial drive is mid-way between rails **312**, **314**, and that all three.

Apparatus **300** also includes an offset oscillator or vibration source **360**, which, to avoid repetition of explanation, may be the same as, or substantially the same as above. That is, there is a vibrator motor **364**, which drives through a clutch **366**, thereby turning an output shaft **368** carried in bearings **372**, **374**. An eccentric **376** is mounted to the far end of driven output shaft **368**. Motor **364** is secured in position by mounting frames **290** that attach to the outside (i.e., furthest away from wrist **270**) margins of side frames **284**, **286**. Similarly, mounting members, or frames, **292**, also mounted to the outside margins of side frames **284**, **286**, secure oscillator bearings **372**, **374** in position. Again, this arrangement may be the same as, or substantially the same as, the oscillator arrangements noted above. The oscillator need not be mounted on an axis parallel to the spindle axis. It is, however, convenient, and relatively compact, for these axes to be parallel.

Inasmuch as spindle shaft **334** is hollow, conducts shielding gas. To that end, a gas supply manifold **380** is provided in the form of a rotating union shielding gas fitting mounted to the end of shaft **334** distant from the workpiece. At the opposite end of shaft **334** there is an O-ring seal **382** to prevent diversion of shielding gas. The flow of shielding gas is governed by a solenoid controlled valve **378** having on and off positions or conditions.

Carriage **310** is constrained to travel within a permitted range of travel. To that end there are left and right hand linear slides that have respective left and right hand fixed stops **384**. Carriage **310** has upper and lower indexing fittings **386**, **388**, having the form of slide followers **294**, **296** that mate with the linear slide. The upper and lower fittings alternately bottom, or run into, or abut, on fixed stop **384** at the upper and lower limits of travel.

A cooling jacket **390** may be mounted to carriage **310**. That mounting may be to the distal face (i.e., the face oriented toward the workpiece) of tail frame **282**. Cooling jacket **390** may include upper and lower seals **392** that engage the outside of spindle shaft **334**. Cooling jacket **390** includes an annular cooling manifold **394** formed therein for the circulation of liquid coolant, and has liquid coolant inlet and outlet fittings **396**, **398**.

The application tool **400** is mounted at the distal end of spindle shaft **334**. Application tool **400** may, in general, be any kind of tool that could be mounted to the end of a robot arm or milling machine, whether a welding or coating applicator or some other tool such as a drill, an end mill, an arbor, a grinder, or such as may be.

In the embodiment of FIG. **19***a*-**19***d*, application tool **400** is a manually changed welding deposition of coating tool **410**. Tool **410** may be referred to generally as a "chuck assembly", or merely a chuck. Tool **410** includes a spindle or chuck (i.e., the actual chuck fitting itself, as opposed to the overall chuck assembly), indicated as **414**. Chuck **414** has a root or stub shaft **416** that fits closely within spindle shaft **334**, and engages the inside of a seal, such as an O-ring seal or gas seal **382** mounted inside hollow spindle shaft **334**. The outside of shaft **416** includes a securement engagement fitting **420**. In the embodiment shown, securement engagement fitting **420** has the form of an external circumferential slot or groove or channel that is engaged by a mating locking fitting of spindle assembly **330**. Fitting **420** will include a torque-receiving element. The locking fitting **422** (FIG. **15***b*) may be a circlip or circumferential spring clip, or screw fitting, or clamp that seats in a mating fitting of spindle shaft **334**. The locking fitting (or fittings) secure chuck **414** so that torque is transmitted from shaft **334** to chuck **414**. In the embodiment shown, fitting **422** is a lateral key. Fitting **420** has a geometry, such as a spline, channel, or socket that mates with the key.

Chuck **414** has a radially outwardly extending flange **424**. The radially outward margin of flange **424** is profiled to mate with the wider portion of a shielding gas cowling or shell, or housing, or skirt **426** that defines within it a chamber **426**. Cowling **426** is open at its distal end, and provides a seat for an end member that may have the form of a sleeve or cuff **428** that is hollow in the center and provides an outlet passage for the shielding gas. Cuff **428** may be made of a ceramic material. The ceramic material may tend to be suitable for high temperature use in a corrosive environment. In some instances cuff **428** may be mitred.

Chuck **414** has a through bore **430**. The distal end of through bore **430** is widened and tapered to receive the shank of a collet **432**, which has a matching taper. Collet **432** has an internal longitudinal passageway **434** that ends in an accommodation **436** for a welding electrode rod **440** such as may be used in operation with apparatus **300**. Rod **440** will be mad of such welding material as suitable for the welding or coating operation, be it a steel alloy, a nickel or nickel alloy, an aluminum alloy, copper or a copper alloy, molybdenum or a molybdenum allow, titanium or a titanium alloy, or a carbide of boride ceramic composition, such as titanium carbide or titanium di-boride, or such other material noted hereinabove, or as may be. As used in the claims herein, the term "alloy" includes both a composition of more than one material, and also of only one material. That is, "nickel alloy" includes coatings that are only nickel, for example. An internally threaded end piece of collet nut **438** engages the outside threads of the distal end of chuck **414** as at **442**, causing the chamfered inside end of collet nut **438** to urge collet **432** into the tapered mouth of bore **430**, thus compressing collet **432** and clamping rod **440** in position. Chuck **414** and collet **432** are electrically conductive, and so carry electrical current from spindle shaft **334** into rod **440**. Chuck **414** has a passageway, or conduit, or gas vent **444** by which shielding gas conveyed along the inside of shaft **334** is permitted to flow to chamber **426**, and thence outwardly through cuff **428** which is mounted concentrically about rod **440**.

In the embodiment of FIG. **20***a*-**20***d*, application tool **400** is a manually changed welding deposition coating tool **450**. Tool **450** may be referred to as a "quick release chuck assembly", or merely a chuck. In this case, rather than having chuck **410** that seats directly in the end of spindle shaft **334**, a chuck **452** is seated in an intermediate fitting, or adapter, **448**, such as may be known as an "HSK Holder". Chuck **452** is threaded or otherwise secured into the output end of adapter **448**, and the near end of adapter **448** fits to spindle shaft **334** by a releasable securement fitting. Adapter **448** is changed by moving robot **250** to present tool **400** to a tool changer that is designed to interact automatically with adapter **448**. The HSK holder and the automatic tool changing equipment are available from commercial suppliers.

From that point, tool **450** is the same, or similar to tool **410**, although the gas cowling or gas shield **460** may be more compact and predominantly cylindrically sided, as at **462**, giving a smaller internal chamber **464** since the upstream mounting is to the outside threaded portion of the HSK holder, as at **466**, rather than to the radially larger flange of chuck **412** described above. Tool **450**, as assembled, is shown in FIG. **20***d*.

In all cases, apparatus **300** and robot **250** are provided with appropriate electrical connectors (as at **470**), pneumatic, and fluid connections, piping, and other ancillary fittings to supply electrical power, whether AC or DC, compressed air, and hydraulic or cooling fluid, such as may be required. These ancillary fittings are understood to be conventional. Apparatus **300** is also provided with sensors such as an inclinometer **472**, as well as vibration, motion, and force sensors. The feedback from these sensors allows apparatus **300** to adjust as welding is progressing, e.g., as welding rod **440** is being consumed, it may automatically adjust the axial position of carriage **310** to advance toward the workpiece.

The use of an automatic tool changer and a programmable robot permits apparatus **300** to be used to lay down a welding pass, or a coating pass of material, be it steel, nickel, molybdenum, or such other material, and then, without releasing the workpiece from its accurately known position and orientation, to dress the surface with another tool, be it a drill bit, an end mill, an arbor, or a grinder, as may be appropriate to yield a finished, machined part. Alternatively, it permits a new welding rod to be installed without significant delay. That is, where more than one tool **450** is provided, one can be kept in readiness with a new welding rod while another tool **450** is being used. When the rod is eaten away in use, tools **450** can be swapped out automatically. I.e., the head control automatically stops the progress of the head, retracts the tool to the upwardly withdrawn position, (i.e., the "datum gauge line), and then the machine swaps out the head. The new head has a rod that has been engaged to a pre-set length, set to a gauge length. The programmed machine is given the gauge length as a known parameter, and so returns apparatus **300** to the workpiece with the new welding rod installed, at the correct position and height to continue from the point of interruption (i.e., the previous stopping point). With the new holder tool in place, with the new electrode, the machine moves to the last position and restarts the LEW process where it left off (or at such other location or for such other task as it may have been programmed to perform after swapping one tool for another).

The new replacement rod can be installed in the out-of-service tool **450** while the other tool **450** is working. This replacement can also be an automated process. Furthermore, it is possible to use different welding rod compositions with successive tools. For example, a first pass or treatment may lay down a coating of nickel on copper. A subsequent pass may lay down a coating of titanium carbide on the nickel. Similarly, an initial pass or treatment may coat steel with molybdenum, while another coating, perhaps on a different area, lays down a pass of titanium or a ceramic composition. In these examples, since they do not involve a manual tool change, apparatus **300** may be installed within a controlled environment, which may be flushed with shielding gas, or which may involve exposure to high temperatures or to corrosive or otherwise harmful processing substances.

As noted above, "Low Energy Welding" tends to involve spark deposition of welding or coating material in which the energy of deposit is of the order of 1 J per spark. The spark deposition, or "Low Energy Welding" approach may tend to yield a very small heat affected zone. The coating thickness may be in the range of 0.001" (or less) up to 0.100". Up to now, Low Energy Welding has been a hand-held process, often dependent upon the skill and intuition of the operator. Apparatus **300** may be suitable for mounting on an existing CNC machine tool or robot that controls the coating path or welding as would be done with an end mill, while retaining the rotation, vibration, and peening capabilities of the above-described hand-held units. Using force, current, and motion feedback, the apparatus adjusts electrode stick-out according to the controlled, programmed pattern of direction, angle, and force. The frequency of the electrical supply, the speed of electrode rotation, and the frequency of vibration are all adjustable (or fixed) as welding occurs. Spindle speed is known, because it is monitored, and can be adjusted in real-time, thereby tending to permit a more consistent processing of a workpiece, and promoting consistency of processing from workpiece to workpiece.

The coating head itself, i.e., the apparatus on carriage **310**, is a sliding coating head. It is driven by a linear servo motor that governs axial position. The spindle can be driven by different motor systems, depending on the coating speed required. Encoder feedback is provided to permit spindle rotational speed to be monitored and adjusted (e.g., stabilized) during processing. The system may provide the quick-change capability described above; a shielding gas flushing or flooding capability; vibration of the spindle nose; provision of electrical power to the spindle through the main slip-ring connection; and water cooling of the hot end of the spindle. As described, in the embodiments above both the electro-spark deposition (ESD) electrical power and the shielding gas are supplied at the rear or distant end of shaft **334** and carried axially along shaft **334** to the electrode.

Considering the embodiment of FIG. **21***a*, there is a welding apparatus indicated generally as **500**. It includes a power source module **502** and a welding electrode handle apparatus, or applicator, **504**. Power source module **502** may be a power converter that itself receives power from another power supply that is ultimately connected to a source of main power, e.g., 120 V, 60 Hz or 220 V 50 Hz. The power converter may be receiving DC power from the intermediate power supply at some voltage, such as 150 V DC, and converting it to a different DC output. Alternatively, power source **502** may be a power supply such as power supply **480** seen in FIG. **21***c*. During operation, the power supply **480** provides the welding electrode with current. As seen in FIG. **21***c*, power supply **480** has an input interface in the form of an input AC/DC power converter **482** which converts line voltage to voltage usable within the power supply. The input power may be alternating current, e.g., 120 V, 60 Hz or 240 V, 50 Hz; or it may be a DC supply voltage up to about 150 VDC. Input power converter **482** may be a two-terminal input having a first input L, for line voltage, and a second terminal N for neutral or ground. Power supply **480** also has a main control unit **484**. Main control unit **484** may also be termed, or may include, a central processing unit which may have the form of a circuit board and ancillary components. Main control unit **484** is programmed to determine the nature of the input power signal received at converter **482**, and to convert it accordingly into rectified DC at an appropriate voltage for charging the capacitors of the capacitor bank (or banks) **486**. Capacitor banks **486** may include a single set of capacitors, two sets of capacitors, or more sets of capacitors. Main control unit **484** is also controls the charging of the capacitors of capacitor banks **486**, and monitors their stored voltage levels, setting those voltage levels according to the voltage required for the programmed output pulses. This may be done by controlling the positive voltage output from input power converter **482** using a charging control **488** connected in series between input power converter **482** and capacitor banks **486**. Main control unit **484** also controls a main discharge switch, indicated as output switching **490** which is connected in series between the positive side of capacitor banks **486** and the output terminal 'A'. The other terminal 'B' is connected to the common ground, or neutral, of the converter output and the capacitor banks. The total charge of each discharge is determined by the magnitude of the voltage stored in the capacitors, and the number of capacitors switched to discharge through output 'A' when the switches are closed to conduct charge.

In any case, whether power source module **502** is a power converter or a power supply, it may be considered to be, generically "a power supply", as it may receive power from electrical mains, whether directly or indirectly, and convert it into a welding power discharge carried to a long-nosed applicator **504** through wiring such as may be symbolised by a cable or wiring harness **506**, and which may include a grounding cable, or opposite polarity output, indicated generally as **508**, that is electrically connected to the workpiece, or to the jig or fixture, or work table to which the workpiece is mounted during the welding activity. Alternatively, power supply **502** may receive power from another power supply, and may function rather as a power interface box between a commercially available power supply and applicator **504**. To this end, power supply **502** has detection circuitry to identify whether the input source is AC or DC, and the voltage magnitude. Power supply module **502** may be adjusted to yield the desired output voltage, current, and train of electrical output signal used in the welding process.

The welding electrode output handle, or handle assembly or applicator **504**, has a main body **510**. Main body **510** has an external skeleton, or containment enclosures, or housing, or shell that includes mating first and second portions or halves, or back shells **512**, **514**. Main body **510** contains a motor **516** that receive power from power supply **502**. Motor **516** may be a brushless DC motor, and may be geared, i.e., the motor may include reduction gearing between the motor and the output shaft. Apparatus **500** includes a rod, or rod assembly that protrudes forwardly of body **510**. It may have an extension, or mandrel, or arm or rod **520**.

Between motor **516** and rod **520** there is a non-electrically conducting transmission, or coupling **522** that carries torque and rotational displacement from motor **516** to an output shaft **518**. Output shaft **518** is electrically conductive. It is carried in a pair of first and second, or proximal and distal, or rear and front, bearings **524**. Forward of rear bearing **522**, output shaft **518** passes through, or is mounted within, a set of brushes **526** and a slip rings **528** that receive the welding electrical feed from power supply **502**. That electrical charge, current, and voltage supply, however it may be termed, is then carried through slip rings **528** and into output shaft **518**.

Output shaft **518** is further carried through a shielding gas union **530** by which shielding gas, such as argon, is introduced into the passageway defined by the accommodation or central bore **532** formed in the forward end of shaft **518**. Forward of gas union **530**, the foremost end of shaft **518** is carried through a coolant union **534**. Coolant union **534** can also be termed a cooling jacket. It has an internal cooling gallery through which externally supplied coolant is circulated, and by which, ultimately, heat is transferred away from the inner end of rod **520**.

Rod **520** is actually an assembly, or head, which may be termed a "straight coating head", that is mounted to the holding body defined by main body **510**. It could be mounted to a robot, as described in the embodiments above. The coating head, namely rod assembly **520** collectively, includes the outer shaft or tube or rod **536**, which could be of square or hexagonal, or other shaped section, but is conveniently of round cylindrical section. Outer rod **536** is a stator. Outer rod **536** is electrically isolated, or has a surface treatment that is non-electrically conductive, such that if outer rod **536** should inadvertently come into contact with other objects during operation, no current will flow.

Outer rod **536** carries within it, or envelops, an inner shaft or tube or rod **538**. That is, outer rod **536** forms, or defines, a protector, or shroud, or cowling, or shield surrounding inner rod **538** that is nested within outer rod **536**. Inner rod **538** is the conductor that carries the welding discharge current. Inner rod **538** may have an outside diameter of the order of 5 mm (about V). In contrast to conventional conductors, inner rod **538** is hollow, so that it may also be employed to conduct shielding gas to the welding interface. That is, it performs three functions: it provides a rotating drive shaft to spin the electrode; it provides an electrical conduit to carry electrical current to the electrode; and it provides a fluid communication path by which to deliver shielding gas to the weld site. Moreover, the flow of shielding gas within the conductor may tend also to provide cooling to the conductor, and to rod assembly **520** more generally. That is, the rapidly flowing shielding gas provides forced convention heat transfer from inner rod **538**.

Inner shaft or tube or rod **538** has a first or proximal end **542** that seats in the accommodation defined by central bore **532**. Inner rod **538** is, or includes, a duct or passageway or conduit **540** through which to transport shielding gas received from shaft **518** to the second or distal end **544** of inner rod **538** located at the far (i.e., forwardmost) end of rod **520**. Inner rod **538** is a hollow transmission shaft, or transmission member that is mounted in a torque transfer relationship (e.g., with a cotter pin or spline or keyway) to output shaft **518**, such that it is a rotor driven rotationally by shaft **518**. Inner rod **538** is electrically conductive. Inner rod **538** is carried in inner, mid-length and forward end bearings **546**, **548** and **550** that keep it spaced (i.e., radially or annularly spaced) relative to outer rod **536**. Bearings **546**, **548** and **550** are electrically insulating. Bearings **546**, **548** and **550** may be made of ceramic material. Rod **520** may include first and second bearing spacer tubes **552** and **554**, those tubes having a shape to fit within outer rod **536** and to block the passage of bearings **546**, **548** and **550**, such that, on installation, inner bearing **546** locates against an inner end stop **556**; proximal spacer tube **552** seats next to bearing **546**; middle bearing **548** seats against the forward end of spacer tube **552**; the rearward end of spacer tube **554** seats against middle bearing **548**; and forwardmost bearing **550** seats again the forwardmost end of spacer tube **554**, i.e., the parts are axially stacked. The string of parts is locked in place by a circlip **556** that seats in a groove in the inside wall of outer rod **538**.

The forewardmost end of inner rod **538** defines an accommodation, or socket, into which a welding electrode adapter fitting, or seat **560** is mounted. This may be a treaded connection, or a permanent connection made by welding or plastic deformation of the material, e.g., as by crimping or swaging. It is an electrically conductive interface through which welding rod current passes. Seat **560** has a body **562** that has a first end **564** and a second end **566**. First end **564** is the rear or inward end that engages the forward end of inner rod **536**. Second end **566** is the front or forward, or outward end that engages a welding rod assembly **580**. In the example, second end **566** has a welding rod engagement interface, or welding rod engagement fitting, however it may be called, which may be threaded as at **568**.

Seat **560** also has two ports, inlet **572**, outlet **574**, and a passageway, path or conduit **576** extending between inlet **572** and outlet **574**. In the example, passageway **576** is formed of a blind bore extending forwardly in the body of seat **560** from inlet **572** which is located in the rear end of seat **560**, and an intersecting cross-bore that is let radially into seat **572** through the circumferential sidewall at the base, or forwardmost end, of the blind bore. When seat **560** is matingly engaged with inner rod **536**, passageway **540** conducts shielding gas to inlet **572**, whence it is carried through passageway **576** to outlet **574**. Outlet **574**, in turn, gives onto the mouth, or bell, or end of the inside cavity of outer rod **536**. In the example shown, first end **564** of seat **560** is a male fitting that fits within the female fitting defined by the peripheral wall of the cylindrical section of inner rod **538**. This could be reversed, such that inner rod **538** has the male fitting, and seat **560** has the female fitting.

Welding rod assembly **580** includes a welding rod holder **582** and a welding rod **584**. Welding rod holder **582** may also be termed a chuck. The body of welding rod holder **582** has a first end **586**, which is the rearward end; and a second end **588** which is the forward end. It is arbitrary which end is called the first end, and which end is called the second end. First end **586** mates with second end **566** of seat **560**. Second end **588** mates with welding rod **584**. In the example, the body of rod holder **582** as a bore extending inward from the forward end, and a larger, countersunk bore extending inward from the rearward end. The two bores intersect in the mid-body portion of holder **582**.

The first engagement fitting **592** at first end **586** is a non-permanent fitting. It allows holder **582** to be matingly engaged and disengaged from seat **560**. In the embodiment shown, fitting **592** is a threaded fitting. In the embodiment, it is a female thread that mates with the corresponding male thread of seat **560**.

The second engagement fitting **594** receives one end of welding rod **584**. Fitting **594** and welding rod **584** could be matingly threaded. However, a permanent mechanical connection can be made by inserting the end of welding rod **584** in the seat, accommodation, or socket defined by the bore formed in the forward end, and then plastically deforming the forward end of the chuck to squeeze onto the welding rod. Where a permanent connection is made, the chuck is disposed once the welding rod has been consumed. Given the single-use nature of such a chuck, rod holder **582** is removably mounted to seat **560**. As may be noted, the outside diameter of the chuck body is less than the inside diameter of the mouth of outer rod **536**, such that shielding gas carried through passageway **576** may tend to flow predominantly axially in the annular gap between them, somewhat in the manner of a nozzle or shower head, with the shielding gas flowing about the welding rod.

In the alternate embodiment of FIGS. **24***a*-**24***c* and of FIGS. **25***a* and **25***b*, there is an apparatus **600** that is generally as described above. However, in place of seat **560** and welding rod holder **582**, apparatus **600** has a generally T-shaped housing or seat **602** that mates with the end of a rod assembly **604** that is otherwise the same as rod assembly **520**. In this case, the stem of the T of seat **602** is internally threaded, and the end of the external rod **606** or rod assembly **604** is correspondingly externally threaded. The end of internal rod **608** carries a bevel gear **610**. The T-shaped housing of seat **602** is either non-electrically conductive, or has a non-electrically conductive coating such that it does not provide and electrical current path if inadvertently place in contact with an electrically conductive object.

The cross-bar portion of the T-shape has a through bore **612**, into which a front bearing **614** and a back or rear bearing **616** are mounted. Bearings **614** and **616** are non-electrically conductive. Welding rod **620** is carried through the bearings. A second bevel gear **618** is mounted to the rearward portion of welding rod **620**. An end nut **622** on rod **620** works in cooperation with second bevel gear **618** to fix the longitudinal position of rod **620** relative to back bearing **616**. The driving and driven bevel gears are electrically conductive. The interior of the chamber define inside the T-shape of the housing is in fluid communication with the outlet end of hollow shaft of inner rotating shaft, or rod, **608**. Front bearing **614** similarly cannot move rearwardly beyond an internal stop defined by circlip **624**. Welding rod **620** also carries a positioning ring **626** that is crimped on the rod prior to insertion. Once inserted, nut **622** is tightened. Positioning ring **626** is a consumable element that is discarded with the remainder of rod **620** after it has been used. The T-shaped housing has grooves or splines that form axial gas passageways **630** by which shielding gas vents to ambient about the exposed portion of welding rod **620**.

In operation, inner rod **608** drives first bevel gear **610**, which turns the second, or driven bevel gear **618**, which turns welding rod **620**. While this occurs, shielding gas is carried from inner rod **608** through the internal passageway defined within the T-shaped body of seat **602**, and out the slot or slots of passageway, or passageways **630** to bathe welding rod **608**, and the welding interface, during welding.

Although the embodiments of FIGS. **22***a*-**22***f* and **24***a*-**24***e* are described in the context of a hand held application, i.e., with handle assembly or applicator **504**, such an assembly can be mounted to an applicator or base that is mounted to a robot arm, and carried along an application path, or schedule, according to a pre-programmed function, or in response to surface and tool location sensors, in accordance with any of the embodiments described above. An example of such an application is seen in FIGS. **25***c*, **25***d* and **25***e*, in which there is a spinning mandrel or cylinder assembly **632**, substantially as seen in FIG. **15***b*, but with the shielding gas transmitting applicator end tool as seen in FIG. **22***f* or **25***a*, possibly carried in a quick change tool, similar to, or largely the same as, tool **400** seen in FIG. **19***a*-**19***d*. Assembly **632** is mounted in head **634** which is substantially similar to, or the same as, carriage **310**. As may also be seen, head **634** has a quick-release, in this instance a pneumatic quick release **636** that includes a one-way acting pneumatic piston and passive return springs **646**. When air pressure is applied through sealed pneumatic union **628**, the tool e.g., **638** is released, and a new tool can be installed automatically. This may permit automatic tool changes when the consumable welding rods have been reduced in length and require replacement, largely without interruption or intervention, relatively expeditiously.

Similarly, although the embodiments of FIGS. **22***a*-**22***f* and **24***a*-**24***e* are described in the context of a hand-held applicator **504**, whether the welding head is mounted to a hand held holding body such as hand-held applicator **504** of FIG. **21***a*, or to an automated holding body, in the form of a robot as described above, they can also be provided with a source of vibration, be is an oscillating imbalance or an ultrasonic horn, as described in the other embodiments included herein.

Referring back to power supply **502**, the power supply includes capacitor banks that are charged and discharge during the welding process. In the context of low energy welding, as noted above, the energy of each discharge may be kept to a low figure, of the order of 1 Joule, or a few Joules. This is very small as compared to conventional welding.

It is known to use DC voltage for welding. It is also known to use AC voltage, with sinusoidal wave forms. In this regard, the inventors refer to the minaprem website at http://www.difference.minaprem.com/joining/difference-between-dcen-and-dcep-polarities-in-arc-welding/

The content of this website, including its definitions of welding terminology, are incorporated herein by reference. In particular, the applicant incorporates the definitions:

"In arc welding, the base plates are connected with one terminal of the power source, while the electrode is connected with other terminal. When sufficient potential difference is applied, electrons flow from negative terminal to the positive terminal in the external circuit. Avalanche of flow of electrons constitutes the electric arc, which is prime source of heat. Polarity in arc welding decides the direction in electron flow between base plates and electrodes. Arc welding power sources provide either AC or DC power; however, depending on the connection made, DC power can provide two polarities—straight polarity and reverse polarity."

Direct Current Straight Polarity (DCSP) or Direct Current Electrode Negative (DCEN)—The electrode is connected with the negative terminal of the power source and base metals are connected with the positive terminal. So with DCSP polarity, electrons emit from electrode and flow towards base plates.

Direct Current Reverse Polarity (DCRP) or Direct Current Electrode Positive (DCEP)—The base metals are connected with the negative terminal of the power source and electrode is connected with the positive terminal. Thus with DCRP polarity, electrons emit from base plates and flow towards electrode.

As further noted on the Minaprem website:

"Arc welding power sources can supply either AC or DC or both forms of current. In case of DC polarity, current flows only in one direction; whereas, in case of AC, current flow direction reverses in every cycle (number of cycles per second depends on the frequency of supply). In arc welding, base metals are connected with one terminal and the electrode is connected with other terminal. Under presence of sufficient potential difference, continuous flow of electrons between them through a small gap constitutes the arc (prime source of heat in arc welding).

The Low Energy Welding power supply **502** has been developed based on the principle of micro-arc capacitor discharge welding control. It is a combination of arc and capacitor discharge welding controls. Power supply **502** includes a programmable control. It is programmable to control the initial discharge voltage, and the quantity of stored charge to be released during the individual discharges. The programmable control includes switching, be it one switch or many switches, to permit the polarity of the DC discharge through the workpiece and welding rod to be reversed.

One terminal, the "first terminal" of the power supply is connected to a consumable electrode held in an applicator running in a rotating or vibrating motion, or both, as in the various embodiments described above. In one example, the "first terminal" would be a cable in wiring harness **506**. For the purposes of discussion, that first terminal will be designated, at least nominally, as the negative terminal. The second terminal, or positive terminal (at least nominally) of power supply **502** is connected to the workpiece via the connection of a low impedance copper cable. In one example, that terminal would be a cable in wiring harness **508**. A complete electrical circuit is made when the electrode is brought near the workpiece, and current begins to flow.

A high current of very short duration will be produced along the circuit when a contact is made between the electrode and the workpiece. The high current pulses melt a portion of the electrode and the molten material is transferred to the workpiece during the moments when contact is made. In the intermittent contact of ESD welding, this may tend to be a relatively slow process. However slow it may be, eventually substantial amount of electrode material is transferred and accumulates on the surface of the workpiece to form a metallurgical layer.

In principle, as noted, there are two major categories of power supplies, AC and DC. In conventional AC, the output polarity of the first and second terminals changes sinusoidally from positive to negative, and back to positive and so on in a preset manner according to the frequency of the supplied power. In DC, the polarities of the output terminals are fixed throughout the process.

In arc welding, as noted above, DC power has two modes of operations: straight polarity (DCEN) and reverse polarity (DCEP). In straight polarity DC mode, the electrode is connected to the negative terminal while the workpiece is connected to the positive terminal of power supply **502**. Electrons flow from the electrode to the workpiece when weld current conducts. On the other hand, electrons flow from the workpiece to the electrode in the DC reverse polarity (DCEP) mode of operation.

The choice of polarity of the welding circuit is important for the arc welding of different materials, and different shapes as that choice may tend to produce different results. There are advantages and disadvantages associated with these two modes of DC polarity of operation. With electrons flow from the electrode to the workpiece, in general the DCEN mode has the following features:

Electrons emit from electrode to workpiece
More heat to be generated on the workpiece surface
Better melting on workpiece surface
Better welding penetration
Better for high welding point metals
Might have higher distortion, stress and wider HAZ
No arc cleaning effects on workpiece surface
No good for thin metals
Lower rate of electrode deposition

In the DCEP mode of operation, electrons flow from the workpiece to the electrode giving the following features:

Electrons emit from workpiece to electrode
More heat to be generated on the electrode
Less weld melting and penetration on workpiece surface
Suitable for thin metal
Not suitable for high melting point metals
Arc cleaning or oxide cleaning on workpiece surface
Higher electrode deposition rate.

Power supply **502** herein may be used for ESD welding. The power supply control and power circuit has been modified to provide the following modes of operations:

1. Direct Current Straight Polarity (DCEN)
2. Direct Current Reverse Polarity (DCEP), and
3. Programmable Alternating Current

That is, apparatus **500** (or any of the other embodiments of apparatus shown and described herein), may be operated in DCEN mode, or in DCEP mode. In addition, or in the alternative, however, it may be operated in a programmable alternating current, or alternating polarity mode. In this AC mode, or alternating polarity mode, of operation described herein, a mix of DCEN and DCEP arc welding phenomenon may occur during the welding process. The choice of DCEN, DCEP or AC modes for Low Energy Welding depends on a number of welding conditions and factors.

Power supply **502** includes a circuit board **640** having a programmable logic device, which may be termed the "Main Control Unit". The logic device is used to provide the control functions of power source **502**. The selection of the DIP switches of power source **502** sets the operating mode. Also a thyristor module group **642** has also been developed at the output. It provides the varying polarity of the power outputs to the electrode and workpiece in real time during operation. ESD processes heretofore have employed Direct Current Reverse Polarity, which has been observed up to now to produce better results than Direct Current Straight Polarity.

The controller of the power supply is programmed to provide welding discharge pulses across the terminals of the power supply, and thus to provide a method of welding. The method of welding may be a Low Energy Welding method, in which each specific welding pulse has an energy discharge of less than 200 Joules. In these methods, the welding discharge per pulse may be less than 20 joules. In some methods it may be less than 2 joules per pulse of discharge. Methods of Low Energy Welding described herein may be intermittent methods, rather than continuous methods, where the time duration of welding is short, and the surface location of welding is rapidly changing. Where this occurs, the energy discharge per unit of surface area is small. This may tend to yield a small, or very small, heat affected zone. The method of discharge is made intermittent by the rotation of the welding rod, or by the vibration of the welding rod, or both. Welding may also be made intermittent by moving the workpiece, e.g., be shaking or rotating the work table, jig, or fixture to which the workpiece is mounted, independent of vibration of the welding electrode.

Moreover, the programmable control permits the magnitude of the voltage at the commencement of discharge also to be controlled, and for the magnitude of voltage to be varied between successive discharges, or groups of discharges, and to be varied as between discharges of differing polarity. That is, where one or more discharges may be made with positive polarity, and the initial voltage of those pulses may be varied from pulse to pulse, additionally, the power supply may be programmed to reverse the polarity of the pulses from time to time during welding. By varying the polarity, a series of pulses in one direction, be it positive polarity, may be used to obtain greater penetration, while a subsequent pulse or pulse of opposite polarity may be used to improve the cleaning effect such as may tend to reduce oxidation effects in the welding process. This reversal of polarity may be used in combination with the use of a shielding gas.

Where the polarity of one or more pulses of discharge in a train of discharges has been reversed, the controller is programmable to select individual discharges or multiple discharges in the reversed polarity. The voltage of the reversed polarity pulses need not be symmetric. That is, the controller is programmable to vary the voltage of one or more of the pulses of that reversed polarity, and to vary the magnitude of the initial voltage of the reverse polarity from the magnitude of the initial voltage of the original polarity. That is, unlike a conventional sinusoidal AC signal, the mean voltage of the series of pulses may not be zero, and so therefore the train of pulses need not have a symmetric positive and negative peak voltages relative to a zero voltage mean. The mean voltage of the train of pulses may be positive or negative. The supplied voltage may not be sinusoidal. Where more than one bank of capacitors **644** is used, the peak amplitude of voltage of one polarity, be it a positive voltage, may be different from the peak polarity of the other polarity, be it a negative voltage. Similarly, the total charge of the individual pulses may vary. It may be that a discharge pulse of greater (or lesser) energy, and therefore of greater (or lesser) heating, may be desired in one direction to achieve a level of penetration, whereas discharge pulses of lesstotal charge, and therefore of less energy, may be used for cleaning. In one embodiment the peak or initial voltage of the positive polarity pulses may be greater than the peak or initial voltage of the negative polarity pulses. Power supply **502** may include a plurality of sets of capacitors that may be charged to the same or different voltage levels according to the discharge voltages of the programmed welding process.

In this welding method, the output signal may be termed "digitally controlled AC". It may also be termed "reversing polarity digital controlled direct current welding", in the sense that each individual pulse or discharge is a direct current discharge. However, the controller is programmed to reverse the polarity of the direct current at times during the method. Hence the process is a direct current process in which the polarity is alternated digitally. Where a train of such discharges occurs, and the discharge pulses are of varying polarity, wither of equal number or otherwise, the resulting output may be termed synthetic AC. The "AC" is an assembled, (or synthesised and therefore synthetic), train of DC discharges of varying polarity that alternates between positive and negative, although it is not necessary that it alternate at each successive discharge. There may be more than one successive discharge of one polarity between alternations of polarity.

The quantity of charge of each discharge, and the initial voltage of each discharge may also be varied by digital control. Each bank of capacitors may include several individual capacitors that can be switch to discharge in any given pulse. Where a pulse of greater total energy is sought, a greater number of capacitors of that bank may be switched to discharge at the same time. As noted, the number of discharges of one polarity need not be equal to the number of discharges of the opposite polarity. They may be different in number. They may have different total charge. They may be different in initial voltage magnitude. They may be different in total energy of discharge. In particular, there may be more DCSP discharge pulses than DCRP pulses. The DCSP pulses may have a higher initial voltage. The DCSP pulse may have a greater energy input than the DCRP discharge pulses. FIG. **26***a* is provided as an illustration of a programmable, or programmed, series of pulses, or series of discharges, or, collectively, a wave train, or discharge wave-train, of discharge voltage v. time. In these drawings, each pulse is understood to be of short time duration, having the form of a spike, starting at the initial programmed voltage, and then decaying rapidly. The time duration of the pulse is not controlled directly, but tends to be a function of the total quantity of charge of the pulse and the resistance to the flow of the pulse. The voltage of the pulse is shown as being roughly triangular, with the amplitude (i.e., voltage) of the pulse decreasing. The pulses commence at t_1, t_2, t_3, etc., to t_n. In the example of FIG. **26***a*, each pulse in the wavetrain is variable. The pulses may have the same or different voltages V_1, V_2, V_3, etc., to V_n, from preceding or following pulses. Total charge may be different in each pulse. The pulses may have the same or opposite polarity. The pulses may be spaced by the same or different time between pulses. In that sense, the illustration of FIG. **26***a* is intended to be generic. It is also to be noted that referring to the series of pulses as a "wave train" may be understood in the sense of the pulses forming a continuing sequence of events. However, the term "wave train" tends to imply, or to be interpreted as, a series of waves of some measure of uniformity, or standard repeating periodicity (i.e., set frequency or set wavelength) such as produced by an oscillator, which may be an analog wave generator. In the case of FIG. **26***a*, however, each individual pulse in the series may be unique in terms of voltage, polarity, total charge discharged, and time-wise spacing from its neighbours. At the most basic level, it is a program for the creation of individual discharges, where the output includes several such individually programmed discharges spaced in time from each other. While it may be referred to as a wave train, and, as indicated above, as an Alternating Current wave train, it is more accurately thought of as a synthetic AC wavetrain that is made up of several programmed DC pulses strung together.

In one embodiment, the method may include discharging a series of DCSP discharge pulses followed by a DCRP discharge, or a plurality of DCRP discharges, followed by one or more further DCSP discharge pulses. The energy per discharge may be less than 2 Joules. In the example of FIG. **26***b*, the apparatus is programmed to produce a series of discharges of the same voltage, with discharges at times t_1, t_2, t_3, etc., to t_n. The pulse spacing is equal, yielding a wave train resembling a rectified square wave. In FIG. **26***c*, the same wave form is shown, with alternating pulse having opposite polarity. This is a synthetic AC wave form produced by adding alternative polarity DC pulses. In FIG. **26***d*, the wave-form is asymmetric, having more positive pulses than negative pulses. The magnitude of the voltage discharge may vary from pulse to pulse. In a power supply with a single set of capacitor banks, the magnitude of the voltage may be equal in both directions. In this example, the spacing of the pulses is equal from pulse to pulse. The example of FIG. **26***e* is the opposite of FIG. **26***d*, being a wave form having more negative pulses than positive pulses. In FIG. **26***f*, the wave form may have a single magnitude of straight voltage, and a single magnitude of reverse voltage. The pulse duration, the number of pulses per time, and the pulse spacing may be individually variable, as shown. The variation in period between discharge, or sets of discharges, may also include “dead time”, i.e., time periods or intervals, or interruptions, or pauses, when electrical discharge is suspended, notwithstanding that either the workpiece or the welding electrode applicator may continue to be subject to mechanical vibration such as to cause peening of the weld-metal coating or fillet, at may be. The frequency of peening, such as driven by a mechanical oscillator, may be in the range of 300 to 6000 rpm, or, roughly 5 Hz to 100 Hz, such as might be applied by an oscillator mounted to, or with, the welding electrode holder. Alternatively, peening may occur when vibration is provided at an ultrasonic frequency, applied either to the workpiece or to a jig in which it is mounted, or applied at the welding electrode holder. The peening vibration may also occur at the same time as electrical discharge. The initial voltage at the start of each discharge pulse may typically be in the range of 30 V-150 V DC, although lower voltages may be used depending on the material to be coated or welded. That is, welding TiC to Cu, as in welding caps, may be best suited to one voltage, perhaps a lower voltage such as about 30 V, whereas welding Aluminum, Magnesium, Nickel, or molybdenum may be suited to a different, perhaps higher, voltage, such as greater than 50 V. Similarly, the energy per discharge, the rate of discharges per unit time, and therefore the heat to be transferred, may vary according to the workpiece material and the coating material to be deposited.

While the use of a digitally-generated reversing DC sequence of pulses (or, alternatively, a “synthetic AC” wavetrain) may be applicable in a variety of ESD, or low energy welding, generally, FIG. **28***a* shows a two-pole apparatus to which reference may be made when considering the use of reversing or alternating ESD processes. In FIG. **27** there is a power supply, P.S., identified as **650**. It receives line voltage, or such other source electrical power as may be, and converts it to a suitable output form. That conversion may involve rectification to a DC signal, and accumulation of charge on capacitor banks. Power supply **650** has three output terminals T**1**, T**2** and T**3**, respectively. T**1** is connected to the welding handle, and ultimately to the welding electrode, identified notionally as handle-and-electrode-assembly **660** by a conductor such as indicated as cable **652**. The handle-and-electrode assembly **660** may be any of the welding electrode apparatus described herein.

In this instance, ESD, or low energy welding, or spark deposition welding may be used to join, or to form a weld between, two different parts. That is, it may be desired to attach two parts **662** and **664** of a desired final assembly indicated as **668**. The weld fillet is indicated as **665**. FIG. **27** may be exaggerated for the purpose of simplification and explanation. Parts **662** and **664** may be thin parts, perhaps less than 1/16" thick, perhaps less than 0.060" thick. They may be placed in very close contact and held in jigs for that purpose. Parts **662** and **664** may not necessarily be made of the same material, and they may not be made of materials that are easily joined under other circumstances. Alternatively, they may be the same material. In any case, they may be materials for which a large heat affected zone is not desired. As can be seen, T**2** is electrically connected to part **662** and T**3** is electrically connected to part **664**, as indicated by cables **654**, **656** respectively.

ESD, or low energy welding, may be commenced by applying a voltage discharge across T**1** and either of T**2** or T**3**. The welding rod and handle assembly **660** may be very finely guided along the site at which a weld filet is desired between parts **662** and **664** by an automated welding electrode holder, carriage, or robot. Alternatively, the handle may be held and moved manually.

Where the parts are of dissimilar materials, welding may start by coating the exposed surface of one or both of parts **662**, **664** with an intermediate composition, such as nickel, or a nickel-rich alloy, to build up a stratum to which other materials may be welded more easily. Whether for similar metals or for dissimilar metals, ESD, or low energy welding, may be used to build up a coated layer. It may then be used to build up a fillet of weld metal between the parent metal of the two parts. This may take several passes, or coating sessions, as may be. The process may occur under an inert atmosphere, or in the presence of a supplied flow of shielding gas using suitable apparatus as described above. It may occur using a hand-held apparatus or a robot mounted welding electrode. When completed, the resultant weld may have only a small heat affected zone, or no appreciable heat affected zone. The weld may be very close to near net size, and may not require grinding or other surface finishing. Alternatively, the surface may be polished after welding, as suitable.

During operation, the power supply provides the welding electrode with current. As seen in the schematic drawing of FIG. **28***a*, power supply **650** may be a polarity switching electro-spark discharge power supply such as indicated at **670**. Power supply **670** includes an input interface in the form of an input power converter **672** which converts line voltage to voltage usable within the power supply. The input power may be alternating current, e.g., 120 V, 60 Hz or 240 V, 50 Hz; or it may be a DC supply voltage, such as 150 V from another power supply to which power supply **650**, **670**, **700** or **710** may be connected as a power interface box, or converter. Input power converter **672** may be a two-terminal input having a first input L, for line voltage, and a second terminal N for neutral or ground. Power supply **670** also has a main control unit **674**. Main control unit **674** may also be termed, or may include, a central processing unit which may have the form of a circuit board and ancillary components. Main control unit **674** is programmed to determine the nature of the input power signal received at converter **672**, and to convert it accordingly into rectified DC at an appropriate voltage for charging the capacitors of the capacitor bank (or banks) **676**. Capacitor banks **676** may include a single set of capacitors, two sets of capacitors, or more sets of capacitors. Main control unit **674** is also controls the charging of the capacitors of capacitor banks **676**, and monitors their stored voltage levels, setting those voltage levels according to the voltage required for the programmed output pulses. This may be done by controlling the positive voltage output from input power converter **672** using a charging control **678** connected in series between input power converter **672** and capacitor banks **676**. Main control unit **676** also controls a main discharge switch, indicated as output switching **680** which is connected in series between the positive side of capacitor banks **676** and the input positive terminal of a polarity switching control unit **682**. During discharge output switching **680** is held in the closed position or condition, i.e., in which it conducts electricity by main control unit **674**.

Polarity switching control **682** has two internal pairs of terminals **684**, **686**, the first being positive, the other being negative, neutral, or ground. Polarity switching control **682** also has two internal throws, or switches, **688**, **690** that are slaved, i.e., linked, together. Control unit **674** is operates switches **688**, **690**, connecting them alternately to the first and second discharge power outlet terminals, seen as "A" and "B". In the normal, or straight polarity context, terminal pair **684** is connected through switch **688** to terminal "A". Similarly, the other side of terminal pair **686** is connected through switch **690** to terminal "B". In this configuration a "positive" charge pulse will be sent to the welding electrode. Alternatively, in the opposite position, main control unit **674** sets the switches such that the positive side, of terminal pair **684**, is connected through switch **690** to terminal "B", and the negative, neutral, or ground side, of terminal pair **688**, is connected through switch **688** to terminal "A", thus reversing the discharge polarity.

In the embodiment of FIG. **28***b*, power supply **700** may be taken as being the same as power supply **650**, other than that it does not have an alternating polarity output, but rather a fixed positive (or negative, depending on which output is attached to the welding electrode) output at T**1** (the positive output). On the negative, neutral, or ground side, there is a switch **692** that is movable between T**2** and T**3**, such that discharge, and therefore deposition of welding rod material as a weld metal coating, will vary between parts **662** and **664**, for example. Over time, as there is weld metal accumulation that bridges the join between parts **662** and **664**, it will not matter which side provides the negative, neutral or ground.

In FIG. **28***c*, power supply **710** combines the output switching features of power supply **670** and power supply **700** to provide both switching between alternate negative, neutral, or ground polarity for the parts upon which accumulation of deposited weld metal is to occur, and, in addition it also has the ability to reverse polarity between the welding electrode and the workpieces. That is, main control unit **674** operates to control the switching of alternating polarity switches **688** and **690**, and to control the switching of alternate output control switch **694** which moves between alternate outputs "B**1**" and "B**2**".

In operation, the output switching of the embodiments, or examples, of FIG. **26***a*-**26***f*, FIG. **27**, and FIGS. **28***a* to **28***c* is controlled by main control unit **674**. Although the synthetic DC electrical signals, or electrical pulses, however they may be called, may not have the same period or pulse duration, they may have an average rate of discharge, or an accumulated number of signals per elapsed unit of time. For example, there may be 10 to 10,000 signals, or discharges, over a period of 1 second. In some embodiments this rate may be in the range of 1500 discharges per second to 5000 discharges per second. This can be termed a frequency range of 10 Hz to 10 kHz, except that the individual pulses are not cyclic, but rather are discrete, programmed, DC discharges, as noted above. The programming may call for signal pulses of equal spacing, but it need not necessarily do so. As will also be noted, the rate of emission of discharge pulses is not amenable to manual control of the internal reversing polarity and alternate output terminal switches, but rather is an automated, programmed, electronically controlled process. The operator may program the power supply by adjusting the discharge voltage levels, and the overall energy discharge per unit time (effectively, the pulse voltage, total charge, and the number of pulses per second) to govern the overall heat input into the workpiece interface (e.g., to avoid over-heating). However, once having set those external input parameters, the main control unit is programmed electronically to implement the selections made by the operator.

The operator may also select whether straight polarity is to be employed, and to what extent. Alternatively, the deposition apparatus may sense the rate of consumption of the welding electrode, and, when that rate of consumption has fallen relative to the initial rate by a datum amount, such as 1/5 or ¼ (i.e., to 4/5 or ¾ of the original rate), to initiate a cleaning cycle using straight polarity. The cleaning cycle may include a series, or burst, of straight polarity pulses, or it may be implemented by alternating between forward or straight (i.e., cleaning) and reverse (i.e., deposition) pulses. The number of straight pulses may be different from, (i.e., not equal to), the number of reverse pulses. For example, the ratio of cleaning pulses to deposition pulses may be in the range of 1:1 to 1:10.

In summary, as described above, there is an electro-spark deposition apparatus. It has a power supply having a main control unit and a welding electrode connected to the power supply. The main control unit includes a synthetic discharge signal generator programmed to produce DC discharges as output electrical signals from the power supply. Each discharge has an initial voltage, a charge, a starting time and a polarity. The synthetic discharge signal generator is programmed to release a plurality of these DC discharges in series. The set of discharges defines a synthetic AC output of the series of the DC discharges.

The apparatus includes an output polarity reversing switch. The main control unit is connected to operate the output polarity reversing switch. The apparatus may have three output terminals: a first of the output terminal is connected to the welding electrode, a second of the output terminals and a third of the output terminals have respective workpiece connections, such as cables, connected to the workpiece, or workpieces, or jigs in which they are mounted. The apparatus has an output selection switch operable alternately to connect the second and third output terminals. The main control unit is connected to operate the output selection switch. The power supply may have first and second banks of capacitors (and possibly more banks of capacitors). The main control unit is connected to govern discharge of the first and second banks of capacitors.

There is a method of electro-spark deposition that includes providing a welding apparatus having a power supply, a welding electrode connected to the power supply, a consumable welding electrode rod mounted to it, and first and second workpiece connections emanating from the power supply. The workpiece connections, in operation, are of opposite polarity to the welding electrode. The method includes providing an electro-spark discharge current to the welding electrode rod while the power supply is switched to elect the first workpiece connection; switching the power supply to elect the second workpiece connection; and providing an electro-spark discharge current to the welding electrode rod while the power supply is switched to elect the second workpiece connection.

The method includes repeatedly switching back and forth between the first and second workpiece connections. It may include reversing polarity as between the welding electrode and the first and second workpiece connections. The method may include changing output voltage when polarity is reversed to apply a first voltage during straight polarity discharge, and a second voltage during reverse polarity discharge. The first voltage may be of greater magnitude than the second. Reverse polarity discharges may be applied more often than straight polarity discharges. The ratio of reverse polarity discharges to straight polarity discharges may be in the range of 2:1 to 10:1. The method may include vibrating at least the welding electrode to peen deposited weld metal between discharges. It may also include applying ultrasonic vibration to the workpiece.

As in FIG. **27**, the workpiece may include a first work piece part and a second workpiece part. The method includes welding the first and second workpiece parts together to form the workpiece. The first workpiece connection, i.e., the terminal cable, or electrical conductor strap, is electrically connected to the first workpiece part, and the second workpiece connection, or terminal cable, is electrically connected to the second workpiece part. In some examples of the method, the first workpiece part is made of a first material and the second workpiece part is made of a second material, which is different from the second material. The method may include providing an electro-spark deposition coating of at least one of the first and second workpiece parts of a first coating alloy, such as nickel or a nickel alloy, followed by a further coating of a second coating alloy applied to the first coating alloy. The method may include building a weldmetal fillet between the first and second workpiece parts. The method may be undertaken in the presence of a shielding gas.

In some embodiments, more than one welding or coating apparatus may be used at the same time, as where multiple passes are to be made, or one material is to be deposited upon another, or where a large area is to be treated.

Although the various embodiments have been illustrated and described herein, the principles of the present invention are not limited to these specific examples which are given by way of illustration, but only by a purposive reading of the claims.

We claim:

1. A method of electro-spark deposition comprising:

providing an electro-spark deposition welding apparatus including an electro-spark deposition power supply, a welding electrode holder connected to the power supply, and a consumable welding electrode rod mounted to the welding electrode holder, and at least first and second workpiece connections emanating from said electro-spark deposition power supply;

in use said first and second workpiece connections being at different locations;

said welding electrode, in operation, being of a first polarity;

said first and second workpiece connections, in operation, being of a second polarity;

said second polarity of said first and second workpiece connections, in operation, being of opposite polarity to said first polarity of said welding electrode;

providing a first electro-spark deposition discharge current to said welding electrode rod while said power supply is switched to elect said first workpiece connection;

switching said electro-spark deposition power supply to elect said second workpiece connection; and

providing a second electro-spark deposition discharge current to said welding electrode rod while said power supply is switched to elect said second workpiece connection;

said first electro-spark deposition discharge current having the same polarity relative to said welding electrode rod as does said second electro-spark deposition discharge.

2. The method of claim **1** wherein said method includes repeatedly switching back and forth between said first and second workpiece connections.

3. The method of claim **1** wherein said method includes occasionally reversing polarity as between (a) said welding electrode and (b) at least one of said first and second workpiece connections.

4. The method of claim **3** wherein said method includes applying a first voltage during a first electro-spark deposition discharge, said first electro-spark deposition discharge being a straight polarity discharge, and applying a second voltage during a second electro-spark deposition discharge, said second electro-spark deposition discharge being a reverse polarity discharge, and said first voltage having a greater magnitude than said second voltage.

5. The method of claim **3** wherein said method includes applying a different number of straight polarity discharges than reverse polarity discharges.

6. The method of claim **1** wherein said method includes maintaining total energy in individual discharges of said welding electrode rod to less than 20 Joules.

7. The method of claim **1** wherein said method includes at least one of:

(a) vibrating at least the welding electrode to peen deposited weld metal; and

(b) applying ultrasonic vibration to the workpiece.

8. The method of claim **1** wherein the workpiece includes a first work piece part and a second workpiece part, said first workpiece connection is electrically connected to said first workpiece part, and said second workpiece connection is electrically connected to said second workpiece part; and said method includes welding said first workpiece part and said second workpiece part together to form said workpiece.

9. The method of claim **8** wherein said method includes having said first workpiece part made of a first material and said second workpiece part made of a second material, said first material being different from said second material.

10. The method of electro-spark deposition of claim **8** wherein said method includes building a weldmetal fillet between said first and second workpiece parts.

11. The method of electro-spark deposition of claim **1** wherein said method includes providing a main control unit; and programming said main control unit to alternate connection of said power supply as between said first and second workpiece connections.

12. The method of electro-spark deposition of claim **1** wherein said power supply has a main control unit and a polarity reversing switch, and said method includes programming said main control unit to reverse polarity of said power supply as between (a) said electrode; and (b) the connected one of said first and second workpiece connections.

13. The method of electro-spark deposition of claim **1** in which the power supply includes at least a first capacitor bank and a second capacitor bank, and said method includes charging said first capacitor bank to a first voltage and charging said second capacitor bank to a second voltage, said first voltage of said first capacitor bank being different from said second voltage of said second capacitor bank; said first voltage of said first capacitor bank having a magnitude that is larger than the magnitude of said second voltage of said second capacitor bank; applying said first voltage to said welding electrode rod in a first polarity mode;

and applying said second voltage to said welding electrode rod in a second polarity mode; said second polarity mode being opposite to said first polarity mode.

14. The method of electro-spark deposition of claim **1**, wherein the power supply has an input power converter that is sensitive to at least voltage amplitude and frequency of input power, and a main control unit and said method includes programming the main control unit to transform input AC power received by said input power converter to a DC output and then converting said DC output to a synthetic AC output.

15. The method of electro-spark deposition of claim **1** wherein said apparatus includes at least one of (a) welding electrode vibrator; and (b) a workpiece vibrator and said method includes applying vibration correspondingly to at least one of (a) the welding electrode; and (b) the workpiece.

16. The method of electro-spark deposition of claim **1** wherein there are time periods between successive electro-spark deposition discharges, and said method includes varying said time periods.

17. The method of electro-spark deposition of claim **1** in which the power supply includes at least a first capacitor bank and a second capacitor bank, and a polarity reversing switch, and said method includes:

providing a main control unit;

charging said first capacitor bank to a different voltage than said second capacitor bank in which voltage of said first capacitor bank has a magnitude that is larger than is said voltage of said second capacitor bank;

applying said first voltage in one polarity mode, and applying said second voltage in an opposite polarity mode; and

programming said main control unit to reverse polarity as between (a) said electrode;

and (b) the connected one of said first and second workpiece connections between successive discharges from said first and second capacitor banks.

18. The method of electro-spark deposition of claim **17**, wherein the power supply has an input power converter that is sensitive to at least voltage amplitude and frequency of input power, and there is a main control unit, and said method includes programming the main control unit to transform input AC power received by said input power converter to a DC output and then converting said DC output to a synthetic AC output.

19. The method of electro-spark deposition of claim **18** wherein the welding apparatus includes at least one of (a) a welding electrode vibrator; and (b) a workpiece vibrator, and said method includes applying vibration correspondingly to at least one of (a) the welding electrode; and (b) the workpiece.

20. The method of electro-spark deposition of claim **17** wherein the welding apparatus includes at least one of (a) a welding electrode vibrator; and (b) a workpiece vibrator and said method includes applying vibration correspondingly to at least one of (a) the welding electrode; and (b) the workpiece.

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